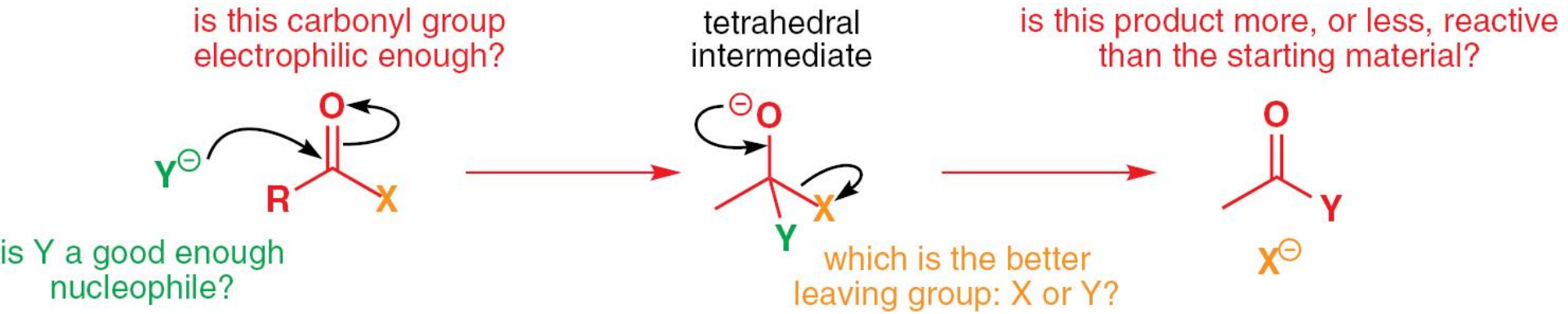
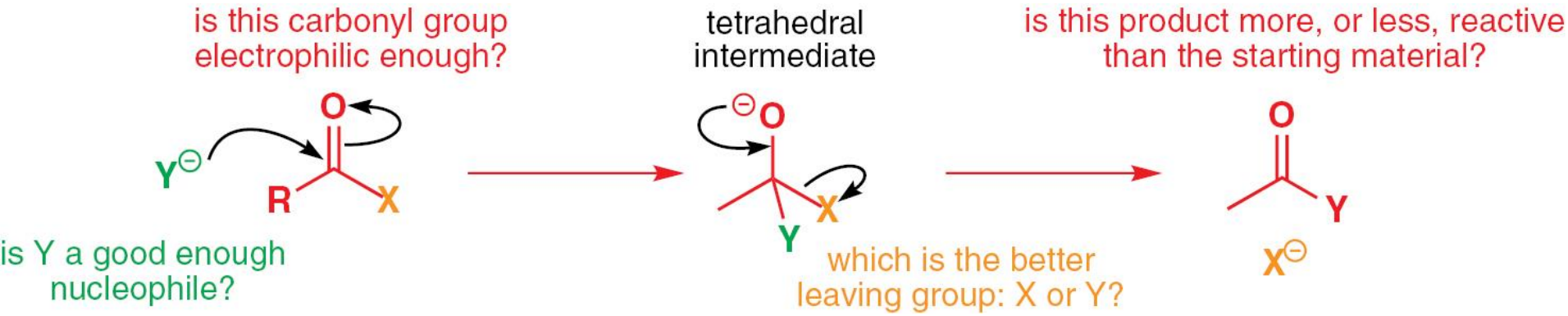
Nucleophilic *Substitution* at the Carbonyl Group

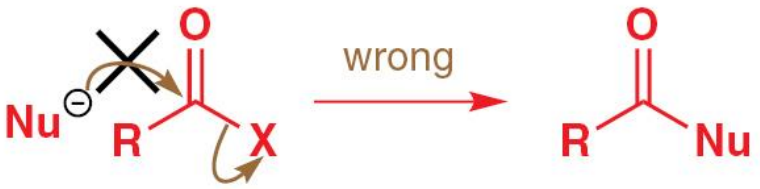
- Nucleophilic attack followed by loss of leaving group
- What makes a good nucleophile
- What makes a good leaving group
- There is always a tetrahedral intermediate
- **How to make acid derivatives**
- **Reactivity of acid derivatives**
- **How to make ketones from acids**
- **How to reduce acids to alcohols**

Nucleophilic substitution reactions

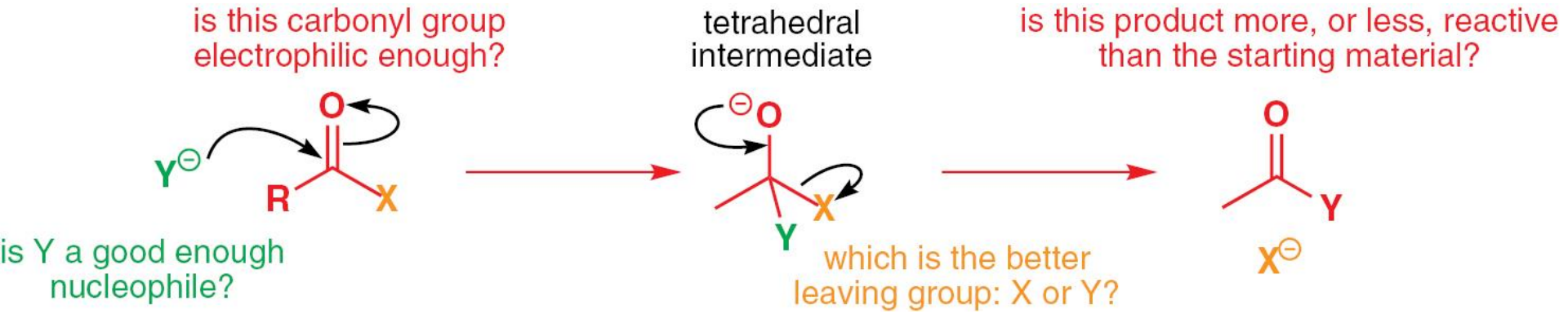


Nucleophilic substitution reactions

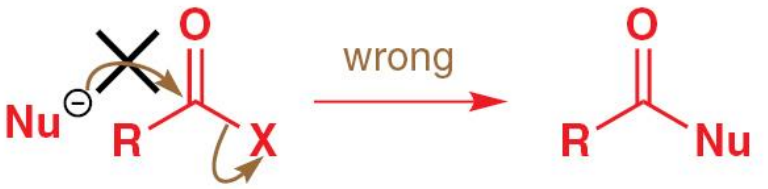




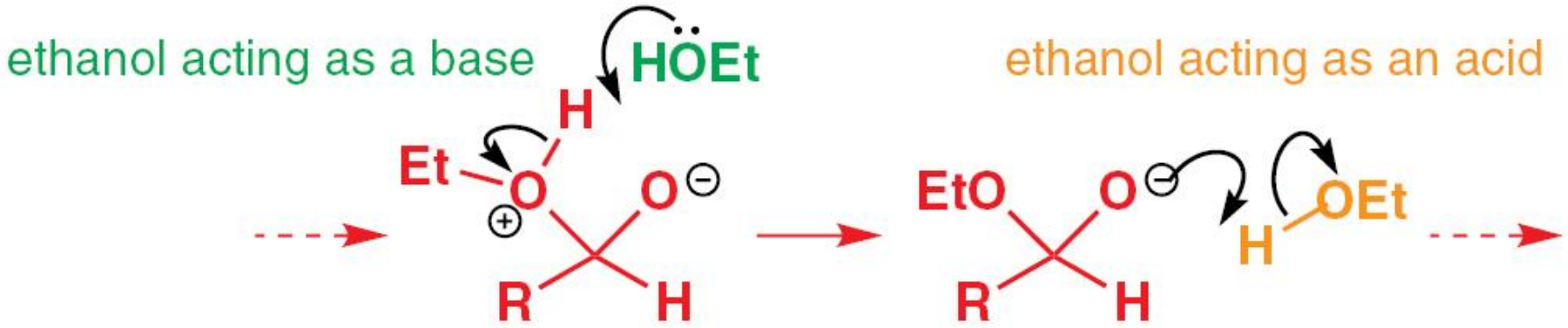
Nucleophilic substitution reactions



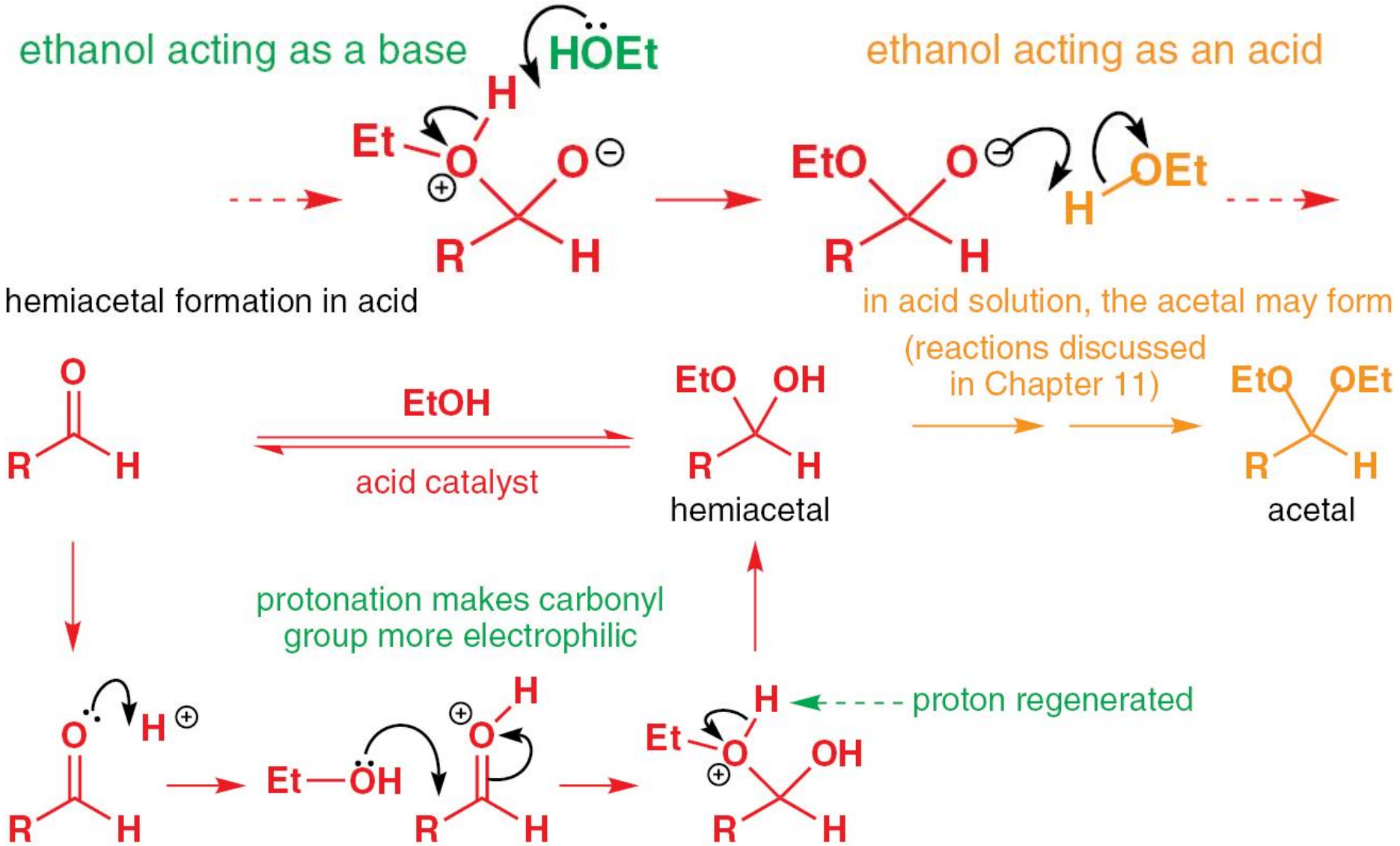




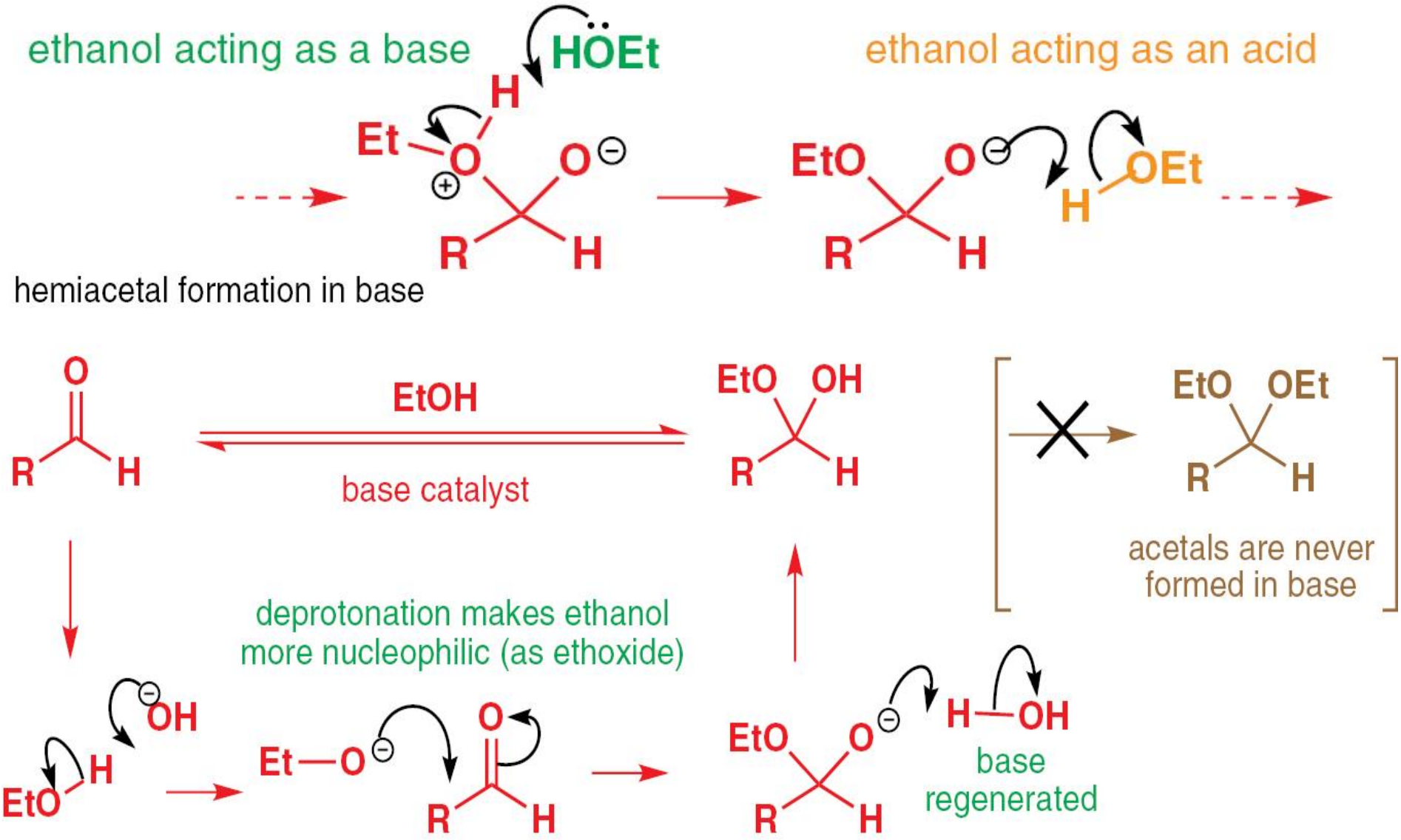
Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision



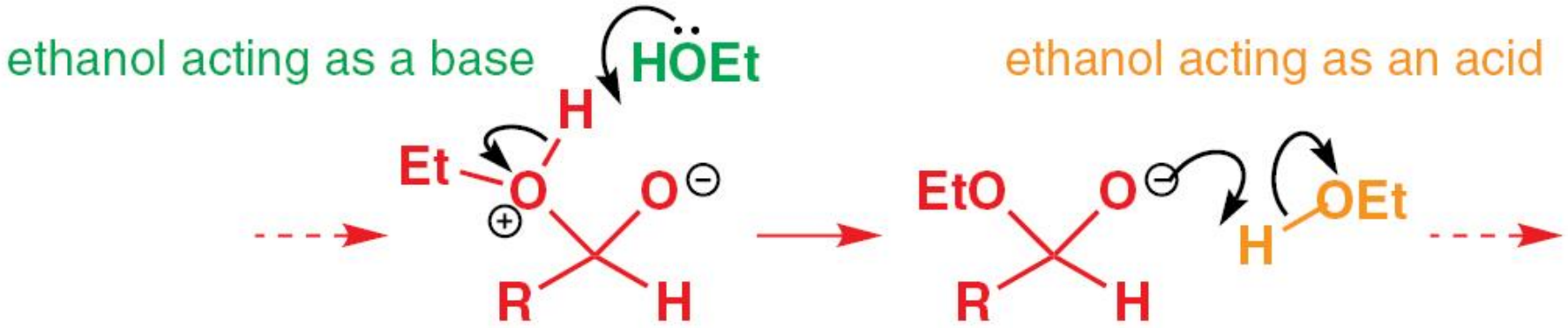
Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision



Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision

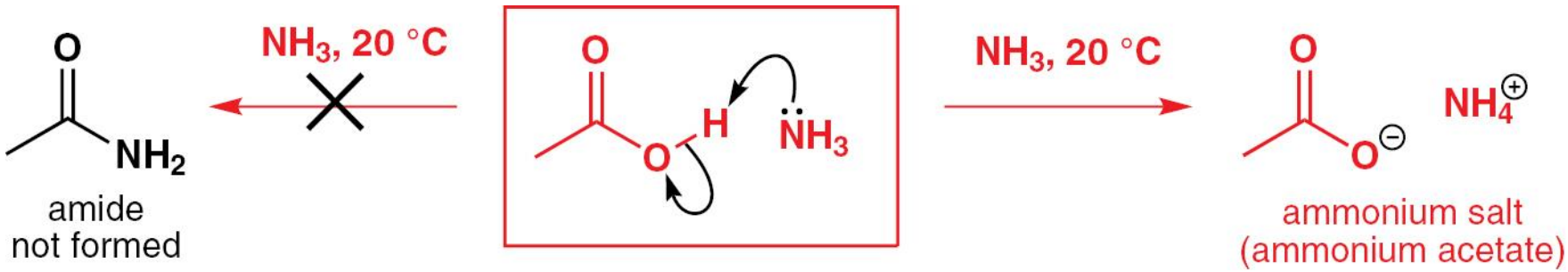


Addition of Alcohols ROH to C=O: Acid and base catalysis-Revision

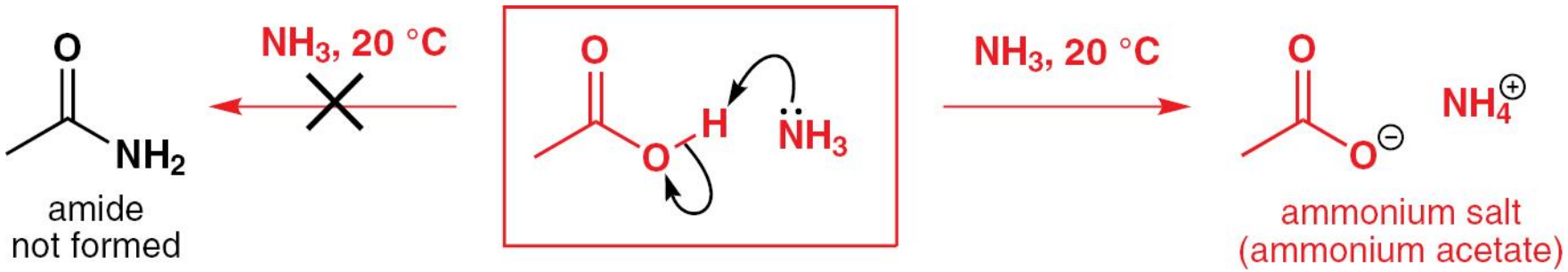


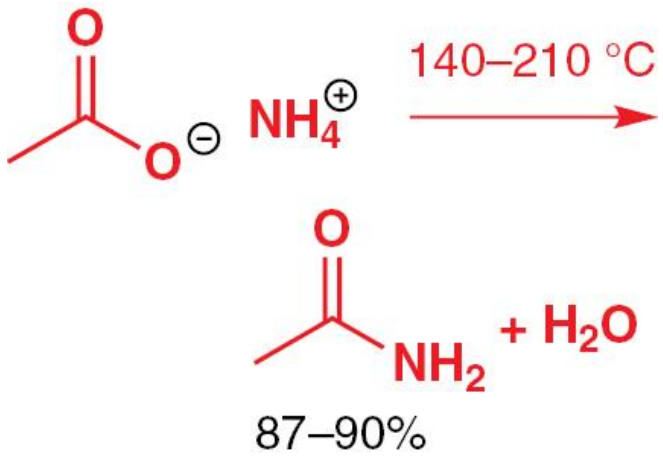
- For nucleophilic additions to carbonyl groups:
  - Acid catalysts work by making the carbonyl group more electrophilic
  - Base catalysts work by making the nucleophile more nucleophilic

Why carboxylic acids do not undergo substitution reactions under basic conditions?



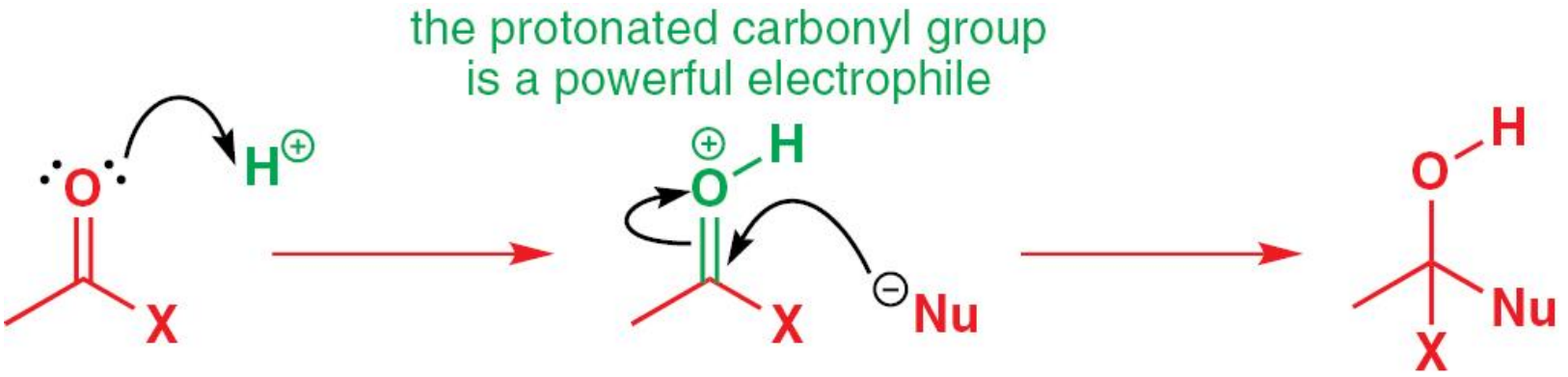
Why carboxylic acids do not undergo substitution reactions under basic conditions?





How acid catalysts increase the reactivity of the carbonyl group?

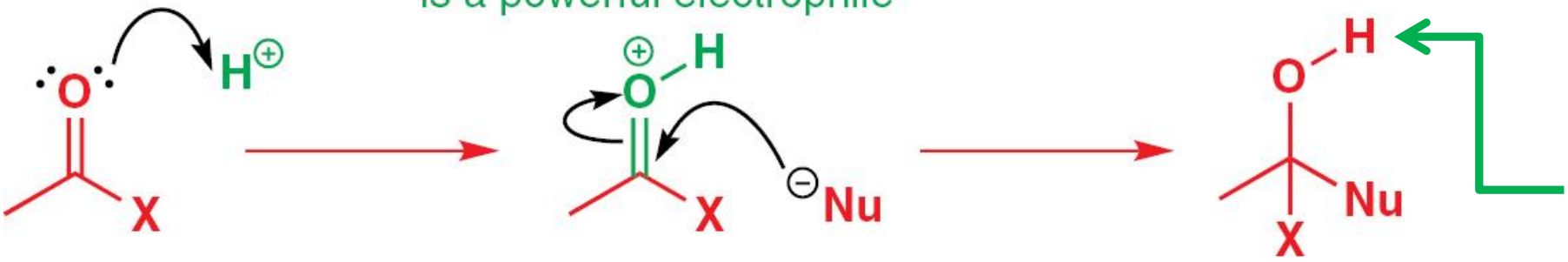
- **Acid catalysts catalyse substitution reactions of carboxylic acids**
  1 They increase the electrophilicity of the carbonyl group by protonation at *carbonyl* oxygen



How acid catalysts increase the reactivity of the carbonyl group?

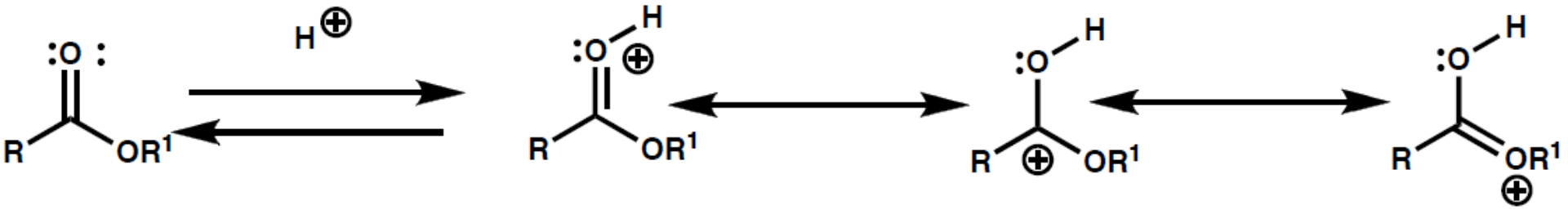
- **Acid catalysts catalyse substitution reactions of carboxylic acids**
  1. They increase the electrophilicity of the carbonyl group by protonation at *carbonyl* oxygen
  2. They lower the pK_{aH} of the leaving group by protonation there too

the protonated carbonyl group is a powerful electrophile

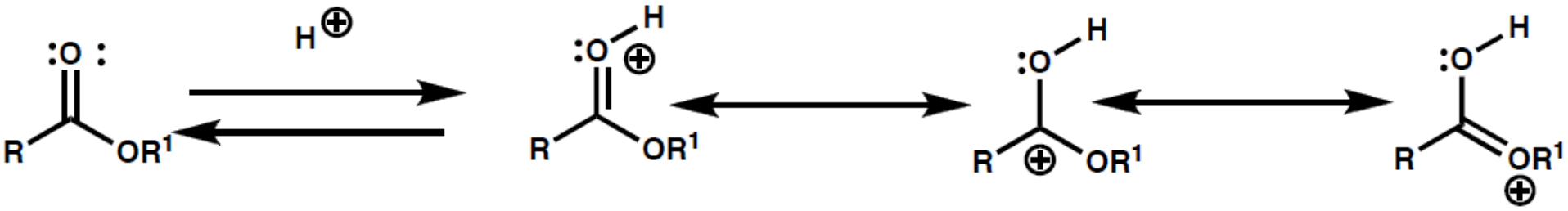


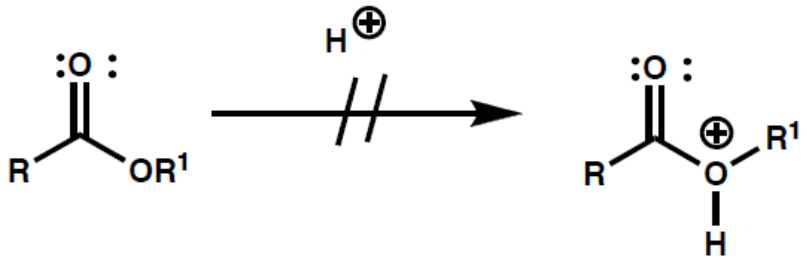
Protonation here will convert the poor leaving group OH^- to the good leaving group H_2O

Protonation occurs always on the carbonyl oxygen

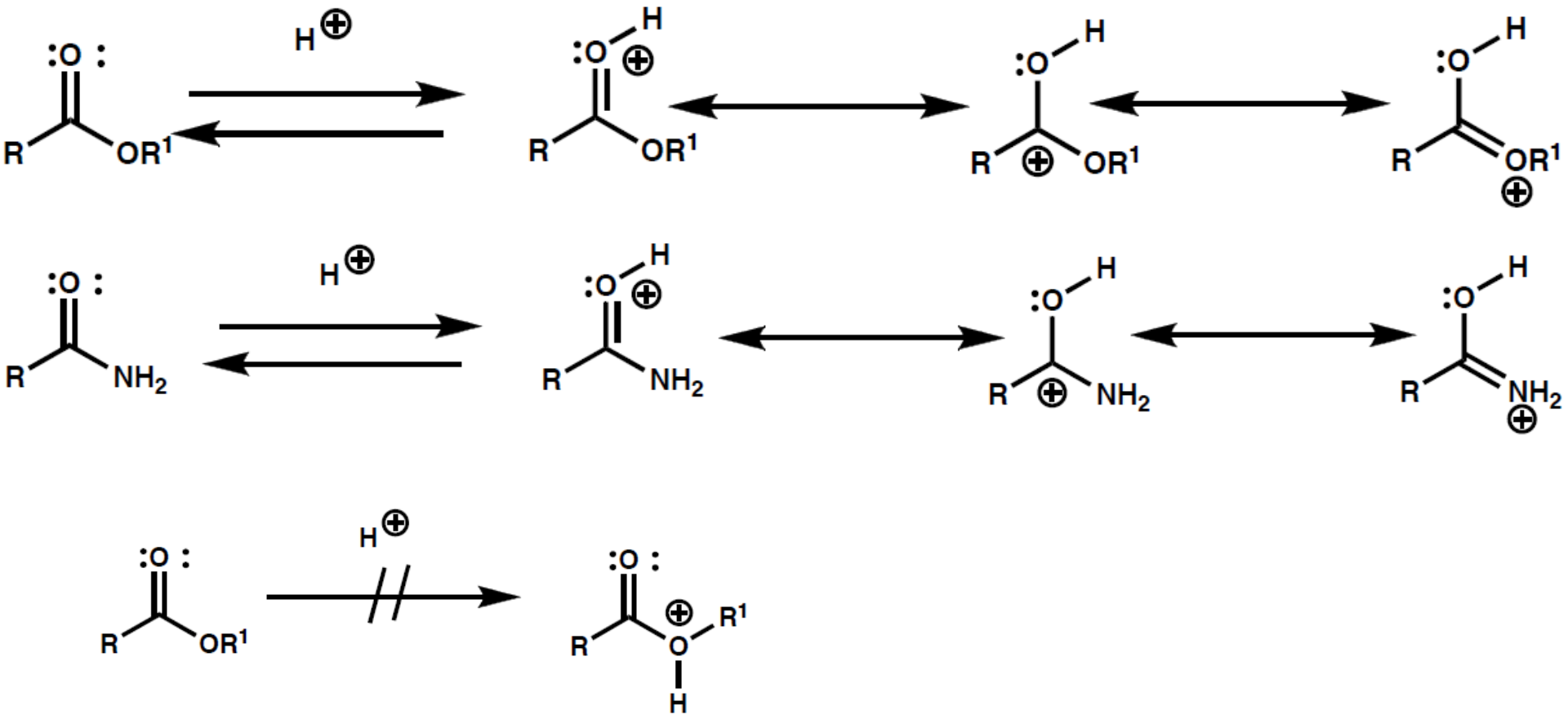


Protonation occurs always on the carbonyl oxygen



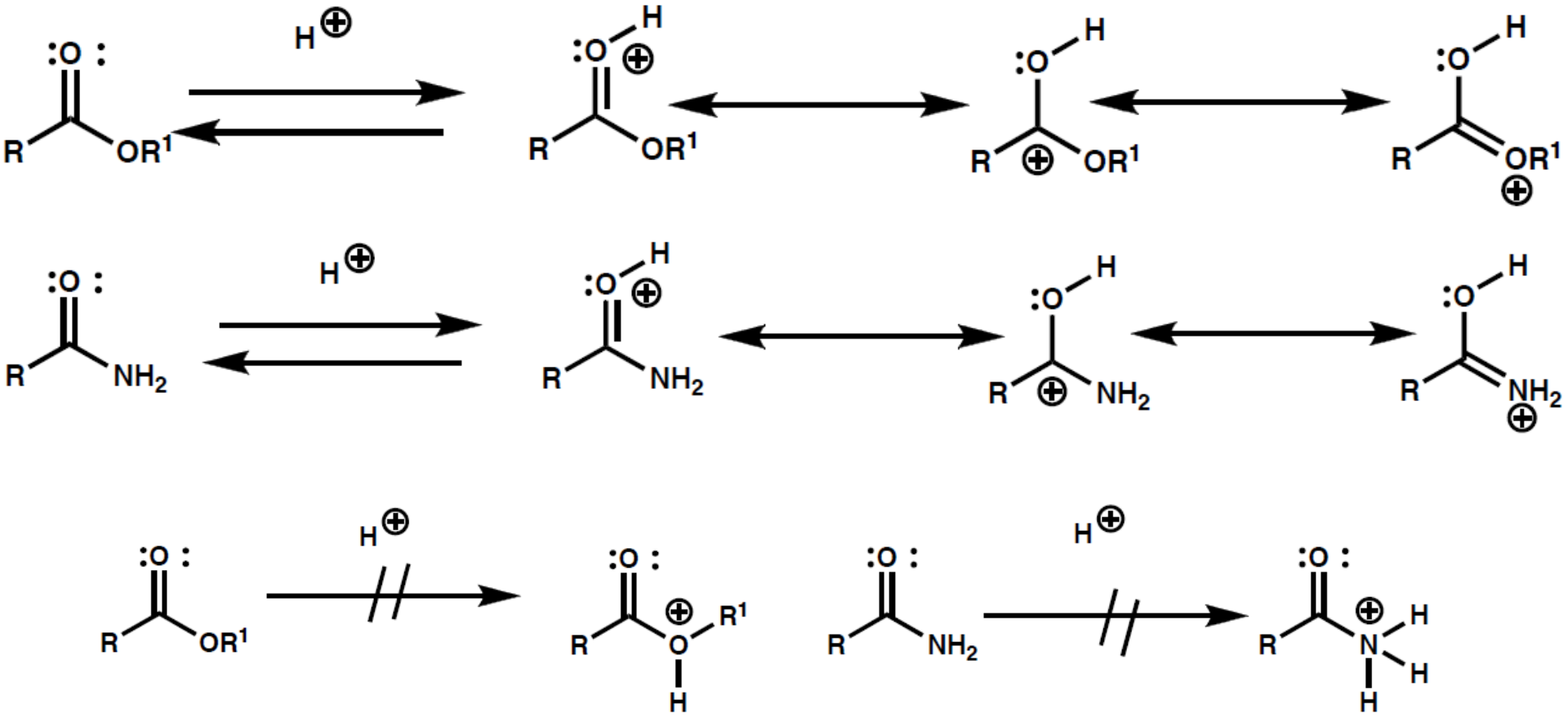


Protonation occurs always on the carbonyl oxygen



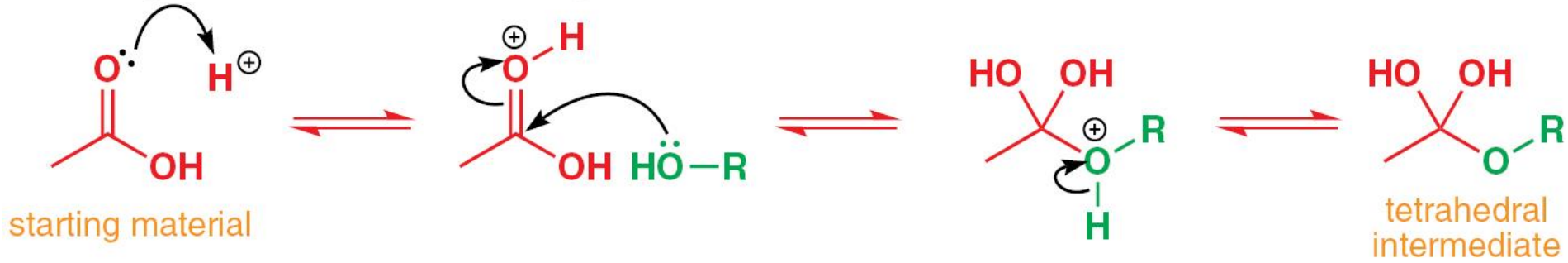
No delocalization possible

Protonation occurs always on the carbonyl oxygen



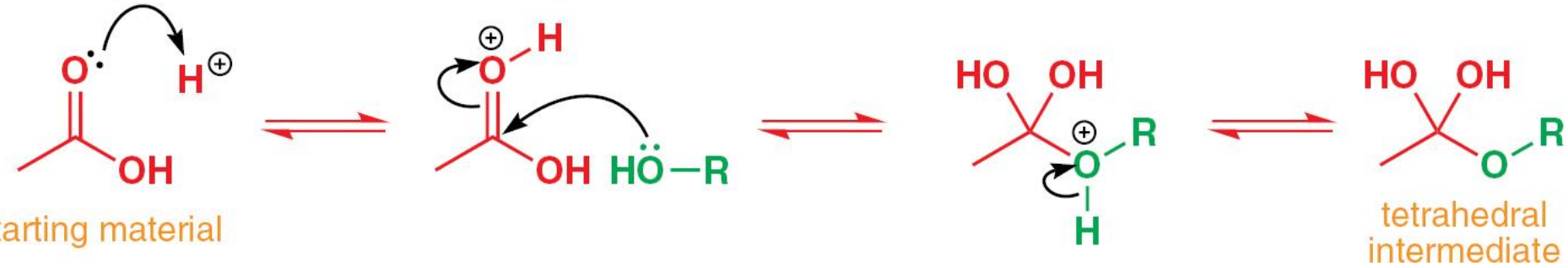
Acid-catalyzed ester formation

acid-catalysed ester formation: forming the tetrahedral intermediate

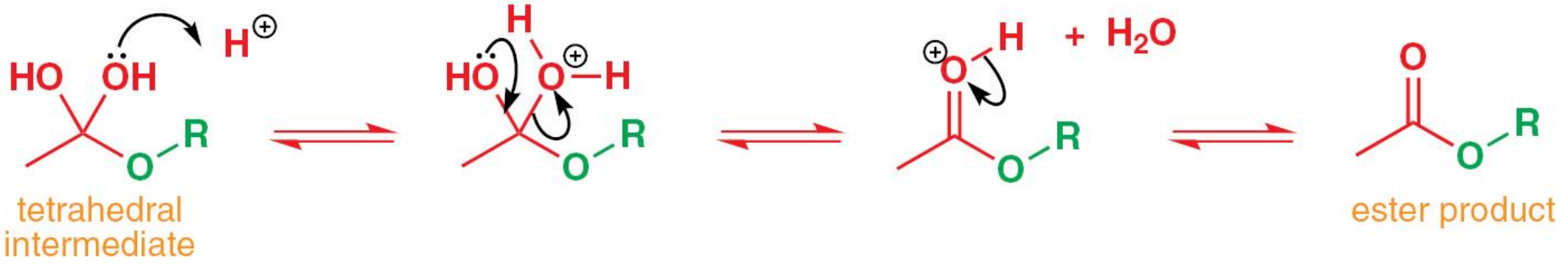


Acid-catalyzed ester formation

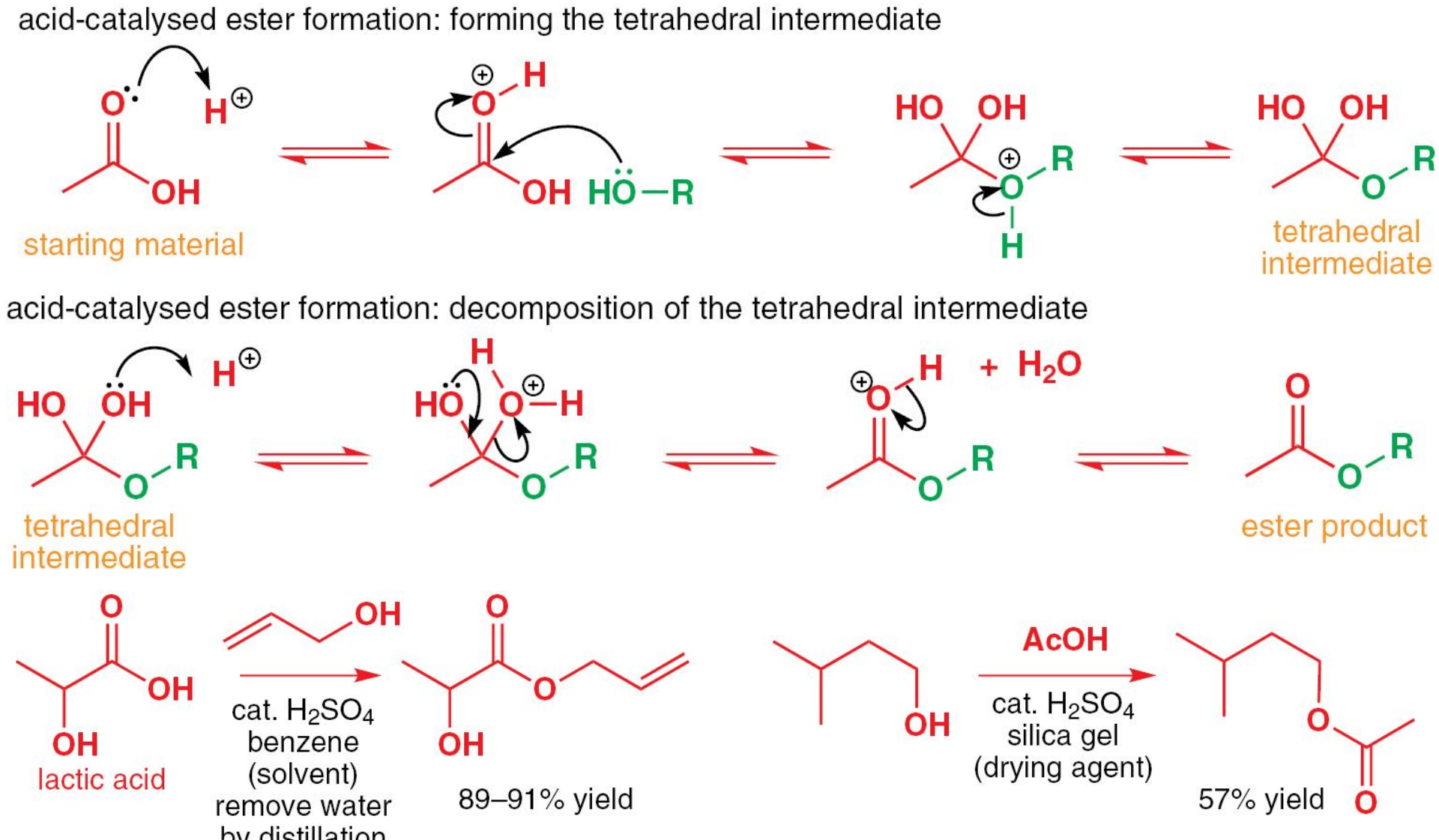
acid-catalysed ester formation: forming the tetrahedral intermediate



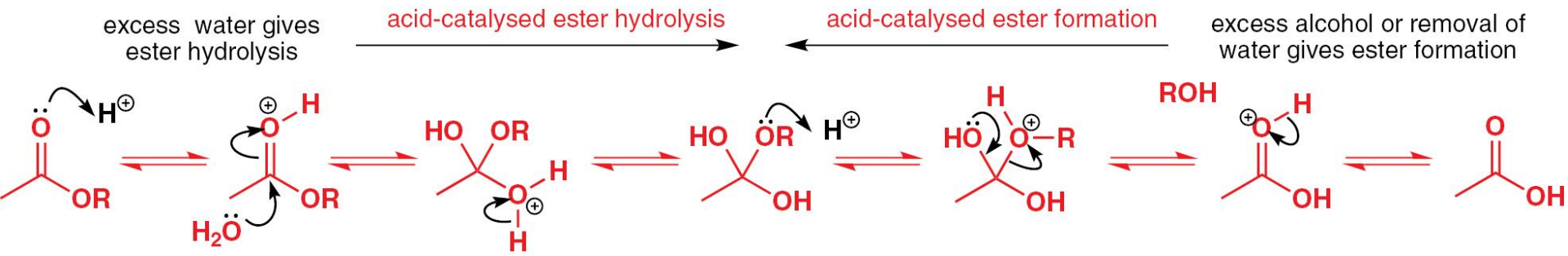
acid-catalysed ester formation: decomposition of the tetrahedral intermediate



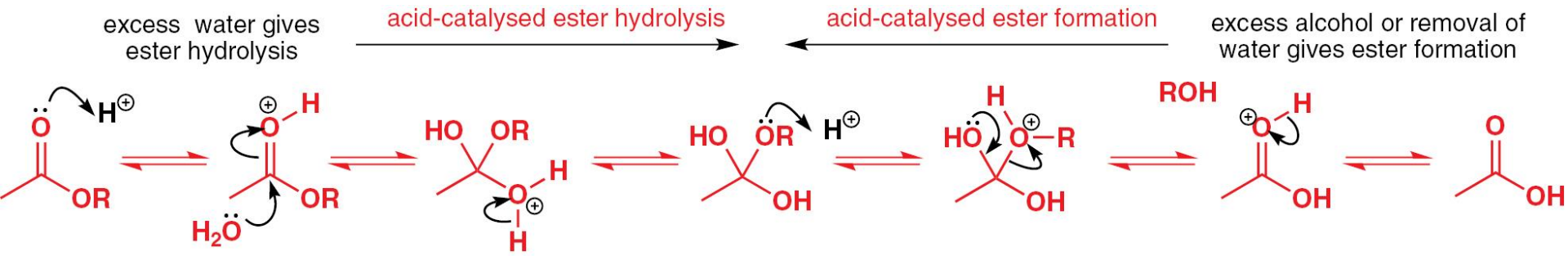
Acid-catalyzed ester formation



Acid-catalyzed ester hydrolysis and transesterification



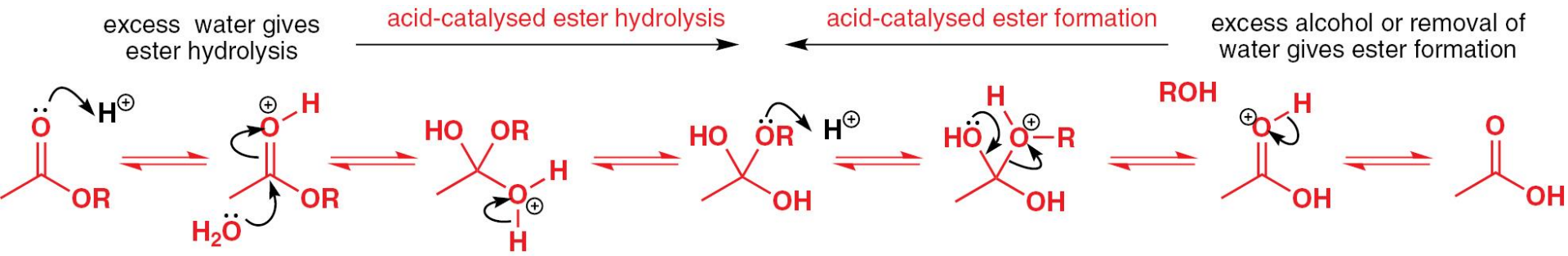
Acid-catalyzed ester hydrolysis and transesterification



Example



Acid-catalyzed ester hydrolysis and transesterification

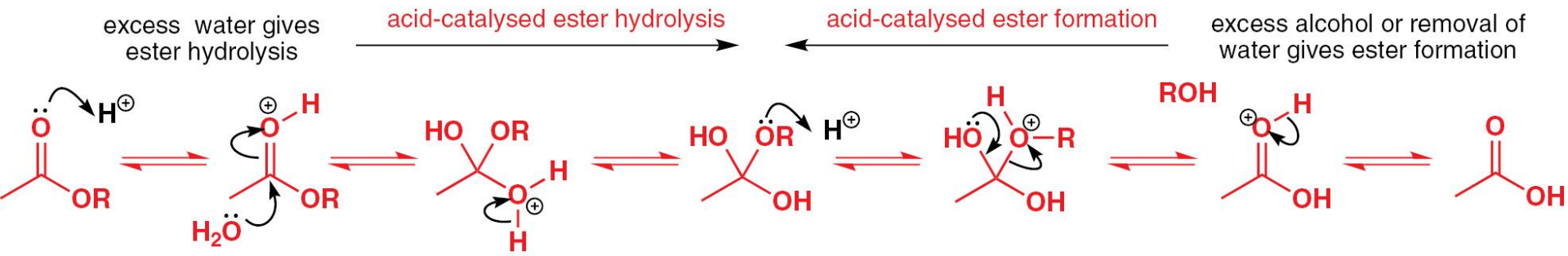


Example



Mechanism

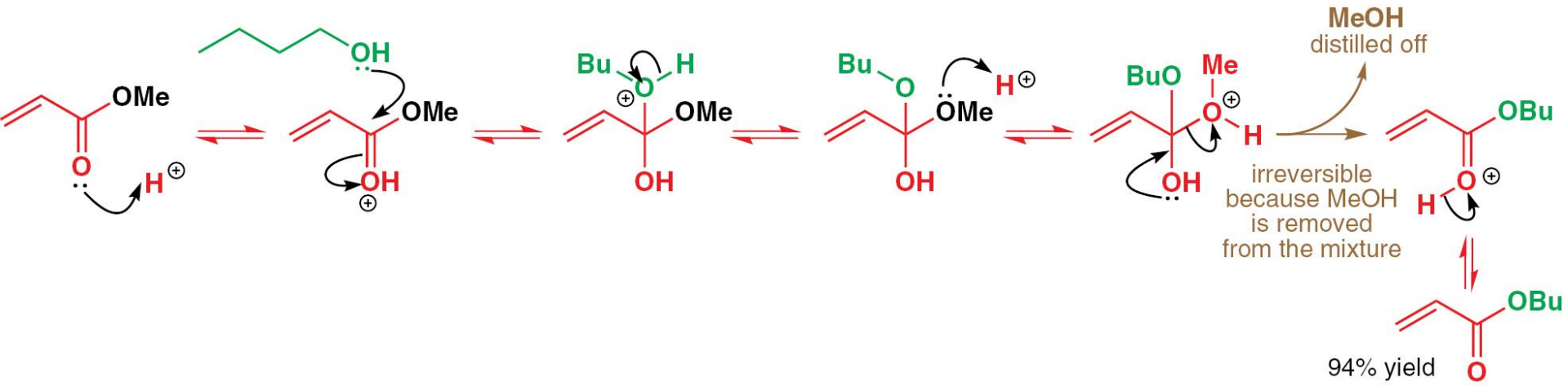
Acid-catalyzed ester hydrolysis and transesterification



Example



Mechanism



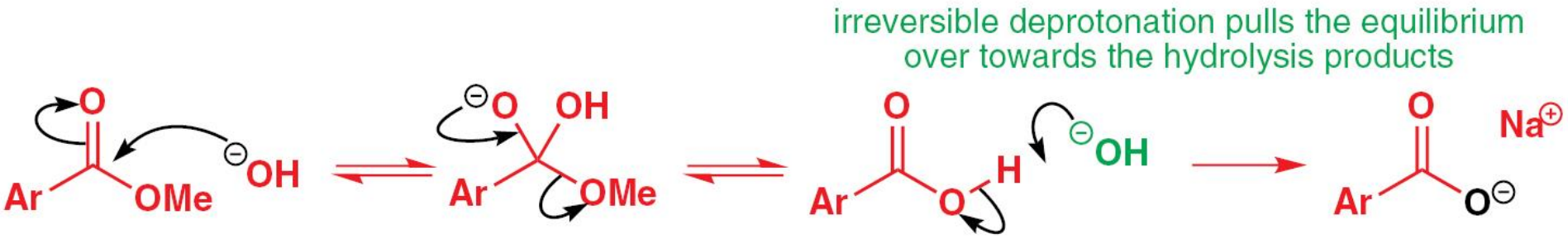
Base-catalyzed ester hydrolysis is irreversible



Base-catalyzed ester hydrolysis is irreversible

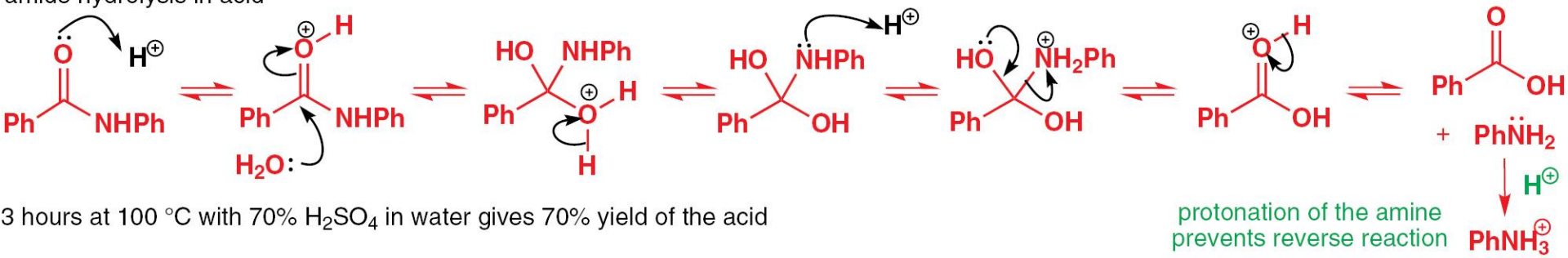


Mechanism



Acid and base-catalyzed amide hydrolysis

amide hydrolysis in acid

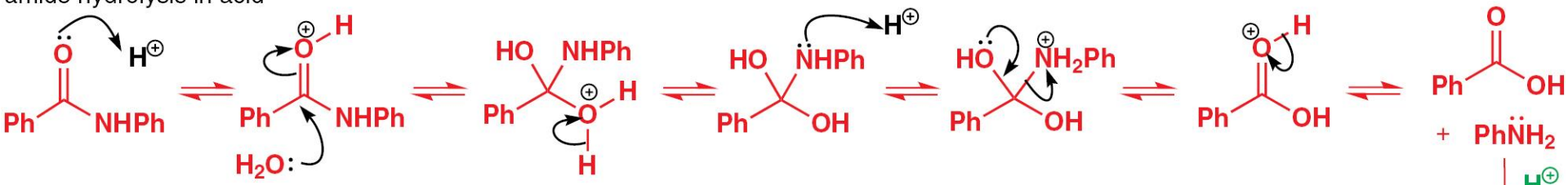


3 hours at 100 °C with 70% H_2SO_4 in water gives 70% yield of the acid

protonation of the amine prevents reverse reaction

Acid and base-catalyzed amide hydrolysis

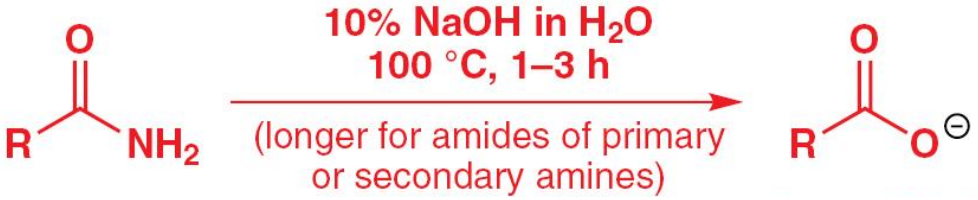
amide hydrolysis in acid



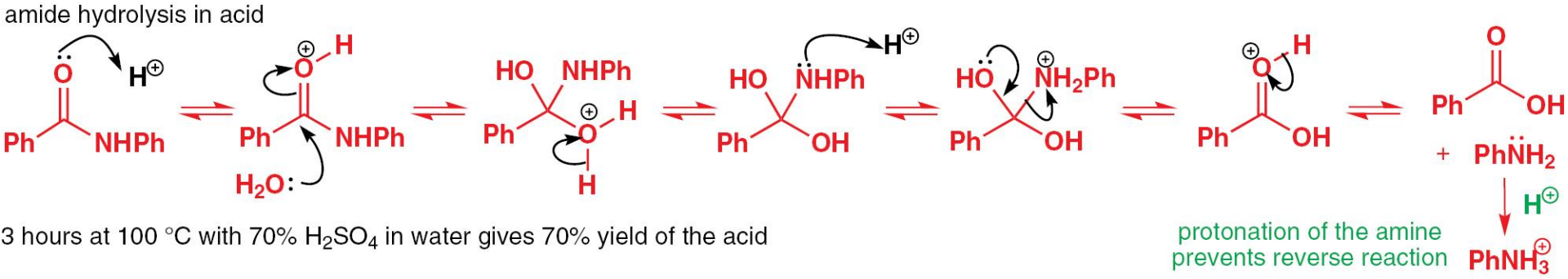
3 hours at 100 °C with 70% H_2SO_4 in water gives 70% yield of the acid

protonation of the amine prevents reverse reaction

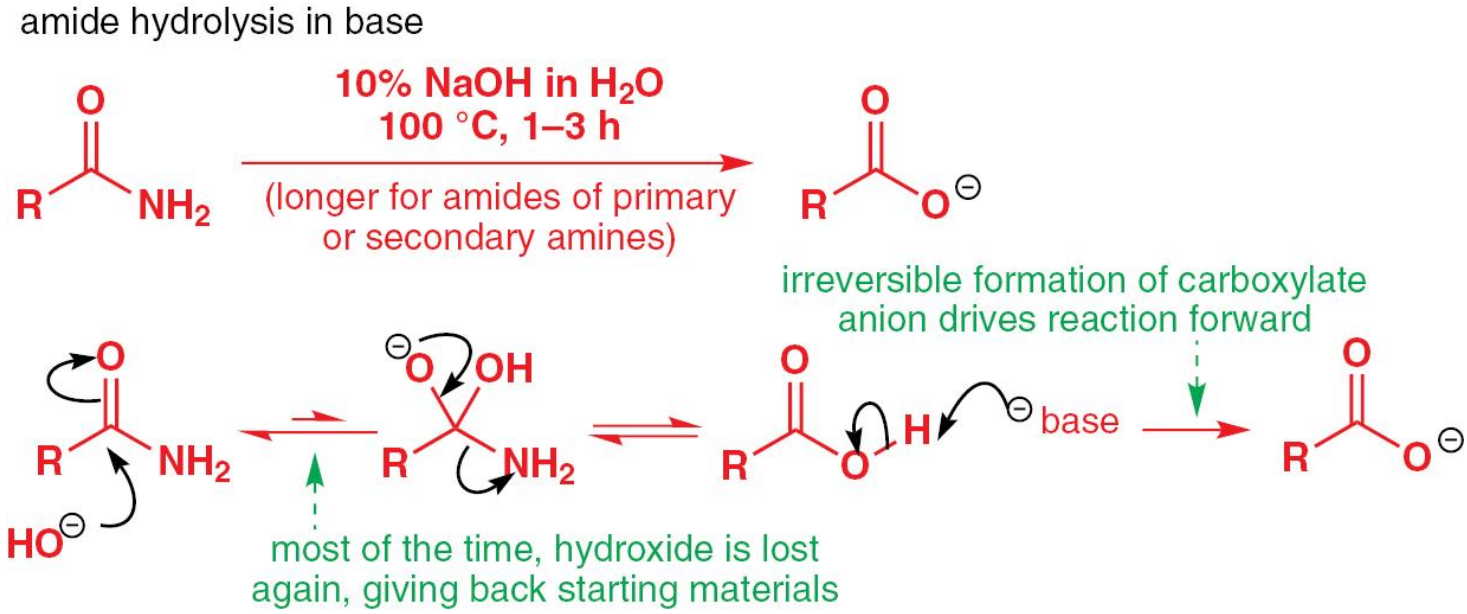
amide hydrolysis in base



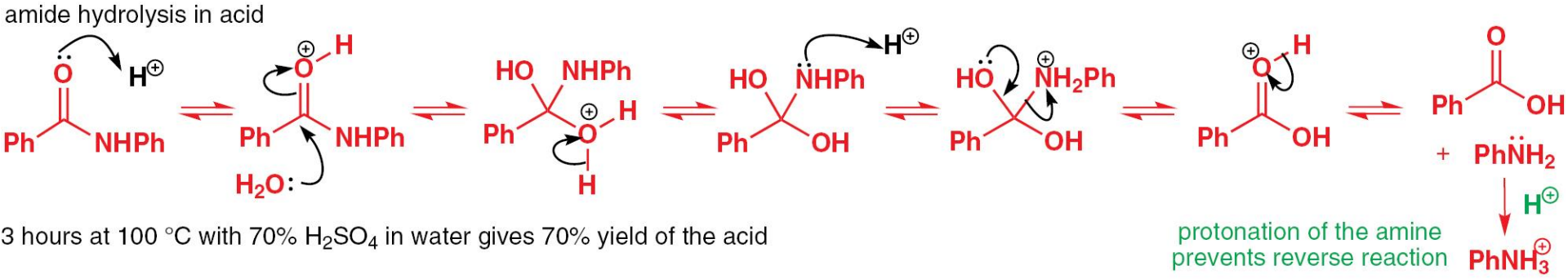
Acid and base-catalyzed amide hydrolysis



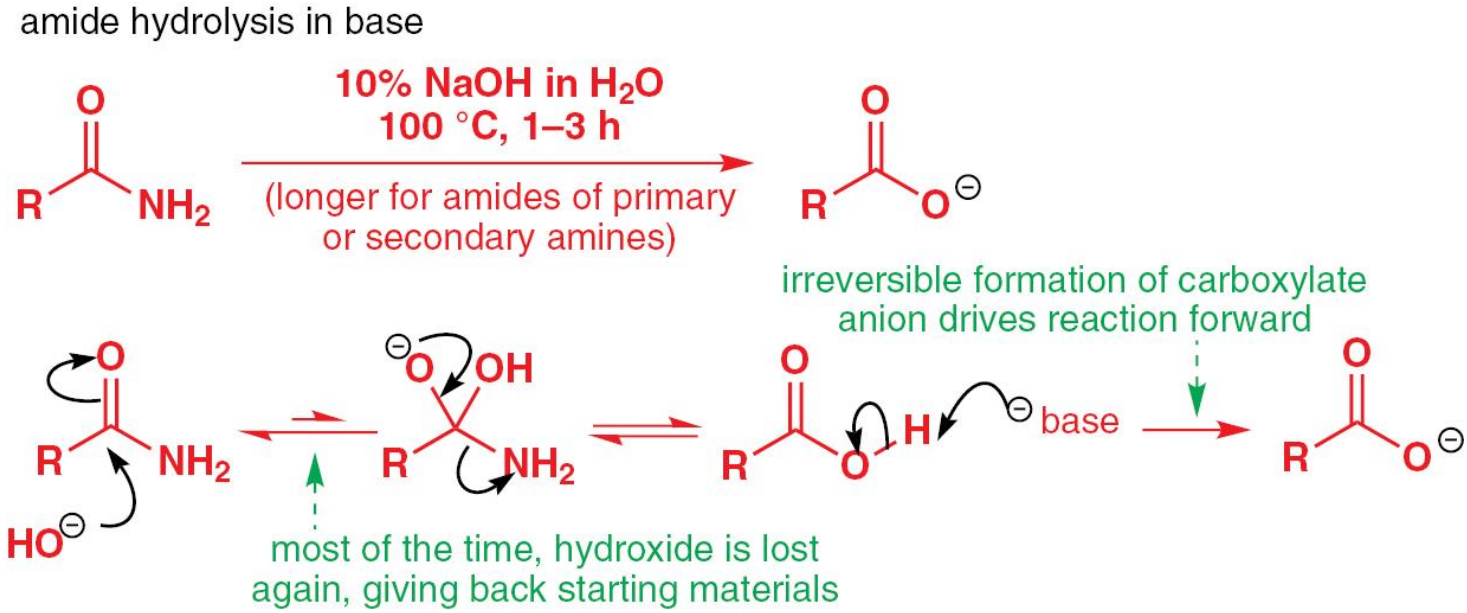
Mechanism



Acid and base-catalyzed amide hydrolysis

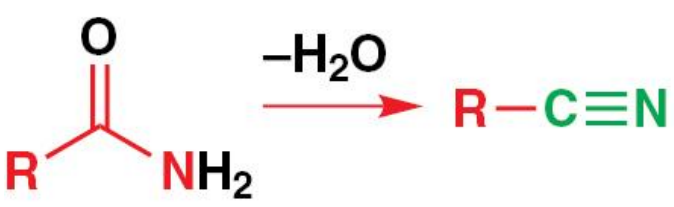


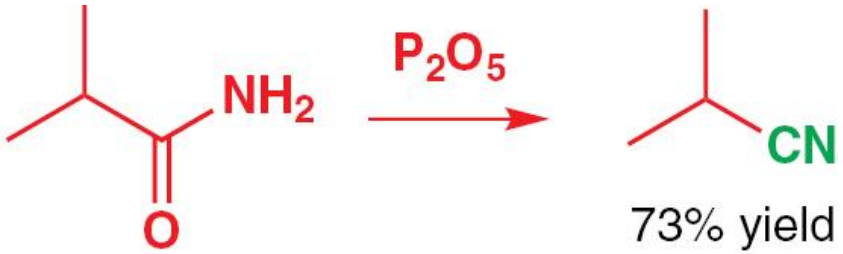
Mechanism



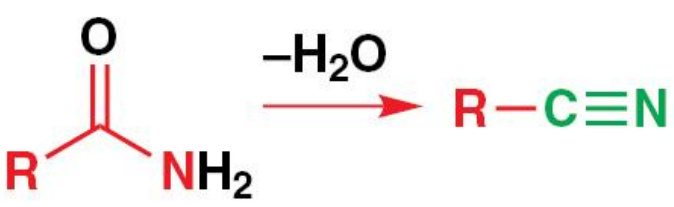
H_2O p*K*a=15.7, NH_3 p*K*a=35

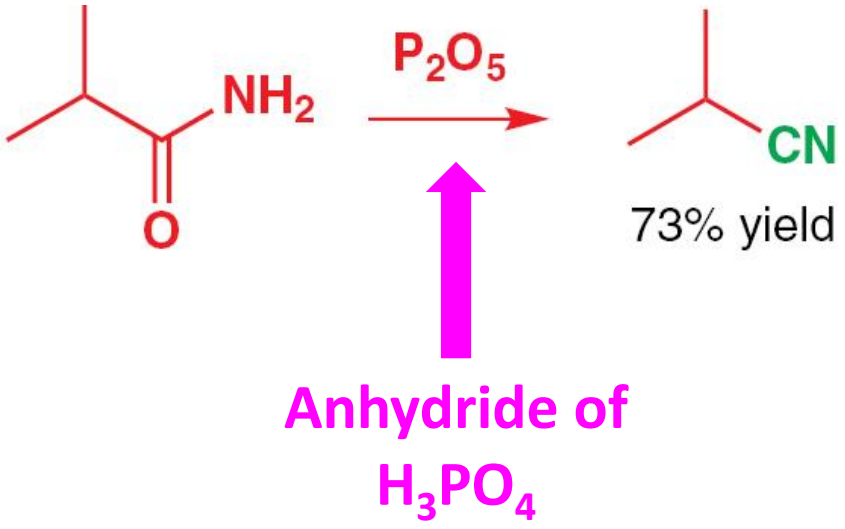
Amide dehydration-Nitrile hydrolysis



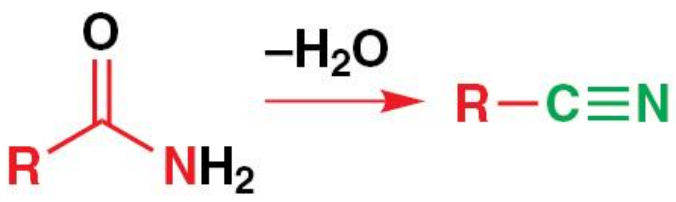


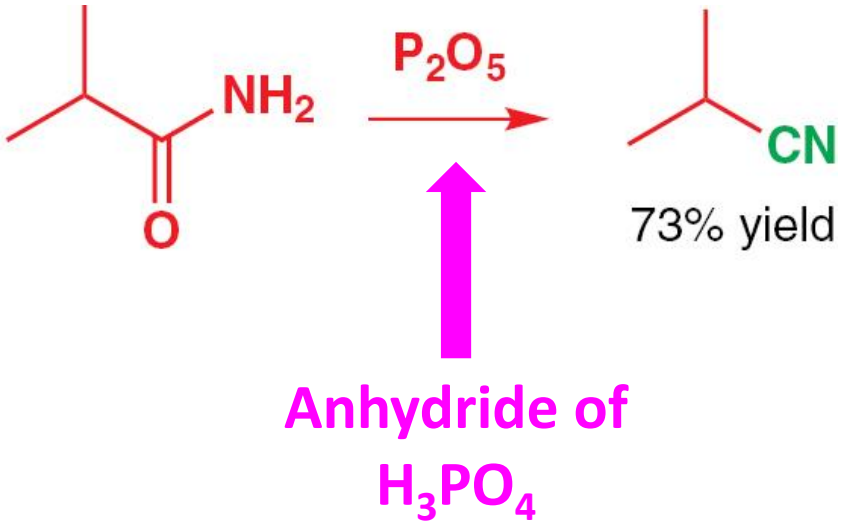
Amide dehydration-Nitrile hydrolysis

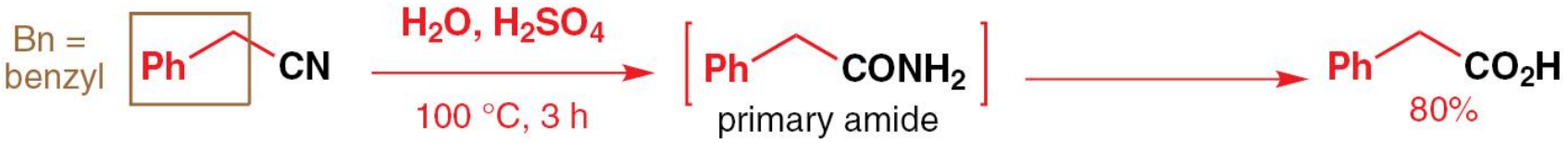




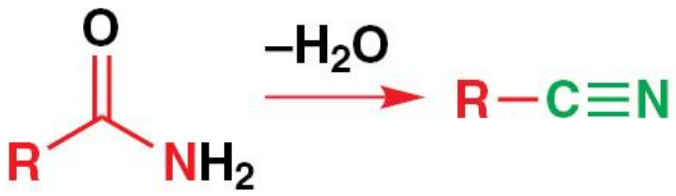
Amide dehydration-Nitrile hydrolysis

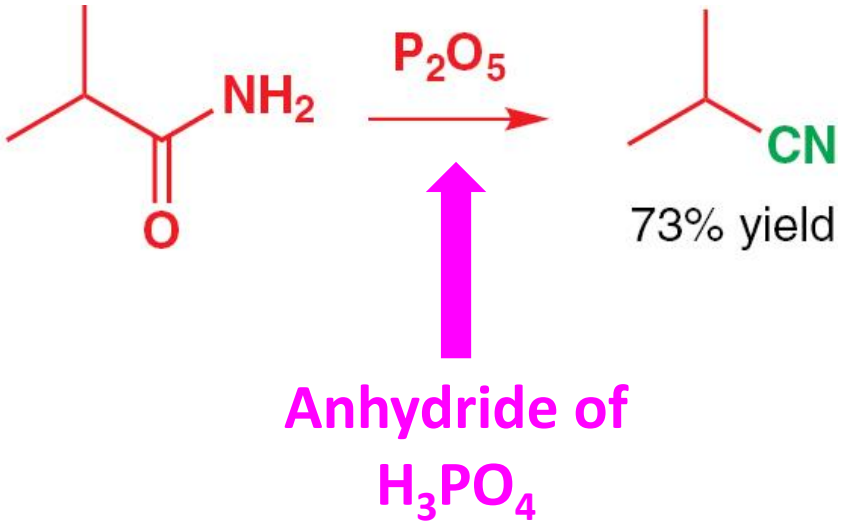


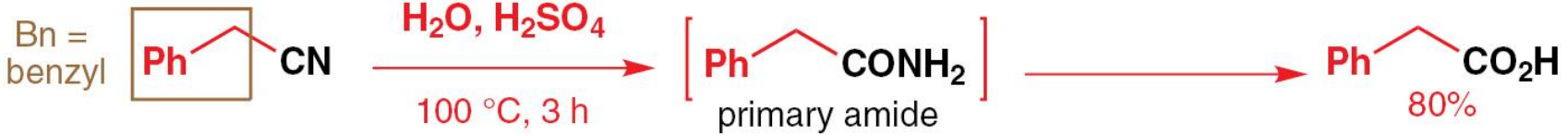


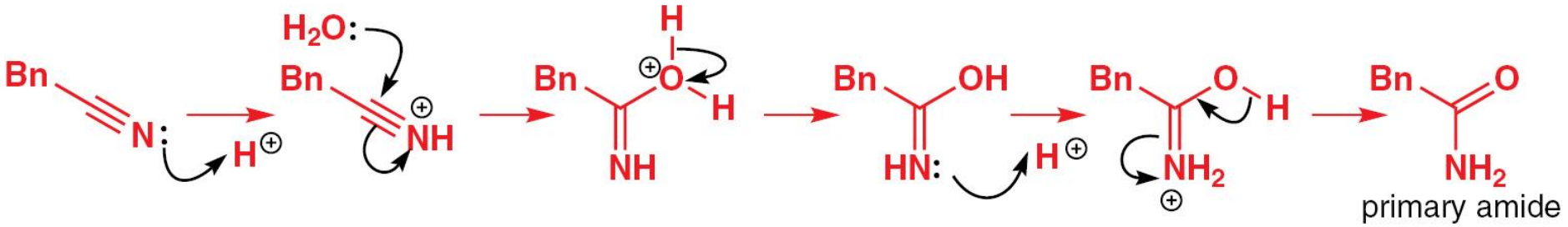


Amide dehydration-Nitrile hydrolysis

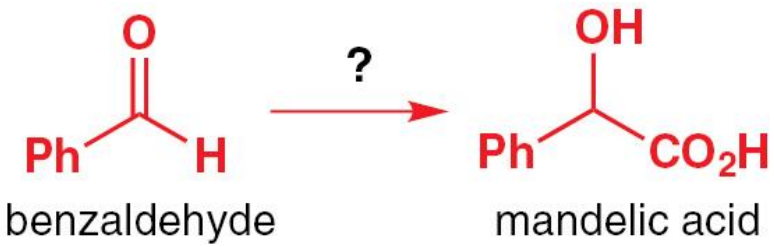






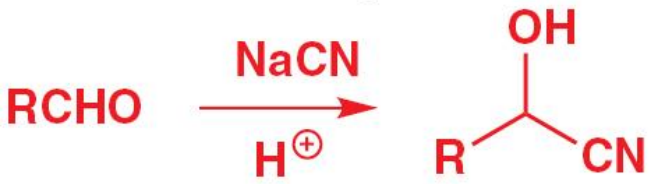


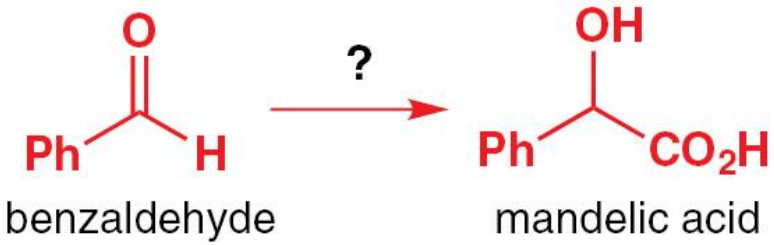
How could you synthesize mandelic acid?



How could you synthesize mandelic acid?

reminder from Chapter 6:

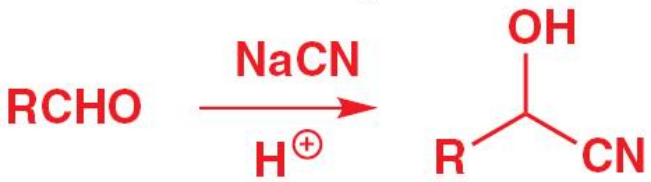


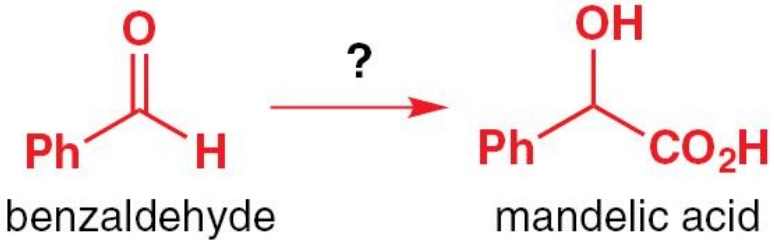


benzaldehyde mandelic acid

How could you synthesize mandelic acid?

reminder from Chapter 6:





benzaldehyde mandelic acid

synthesis of mandelic acid from benzaldehyde

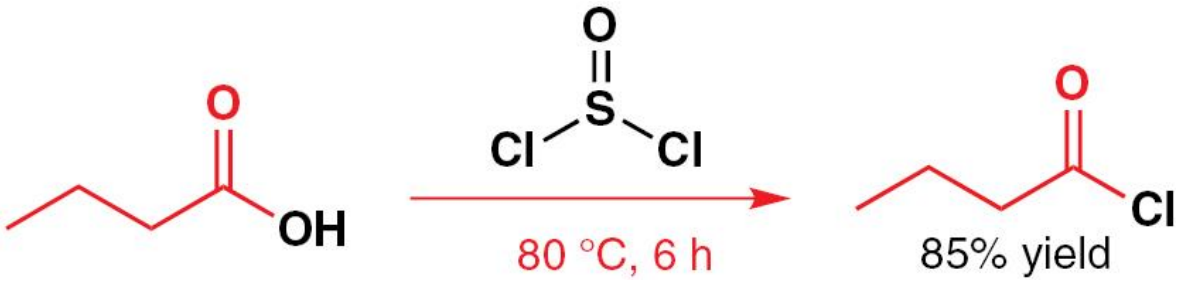


mandelic acid
50–52% yield

Making acid chlorides from carboxylic acids

- **With thionyl chloride**

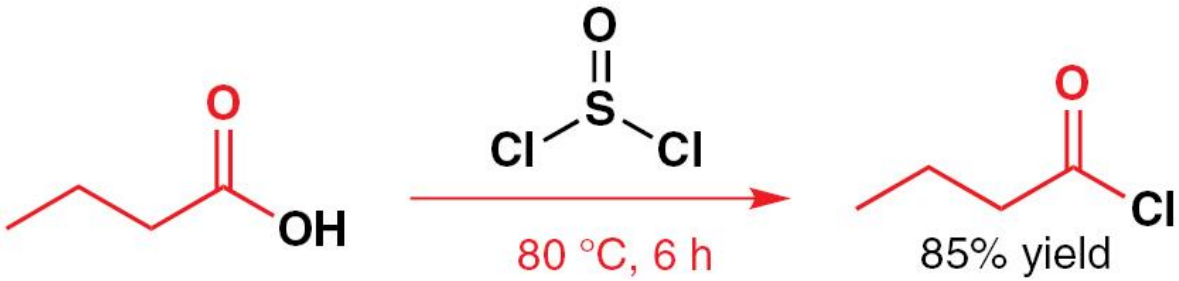
acid chlorides can be made from carboxylic acids with thionyl chloride



Making acid chlorides from carboxylic acids

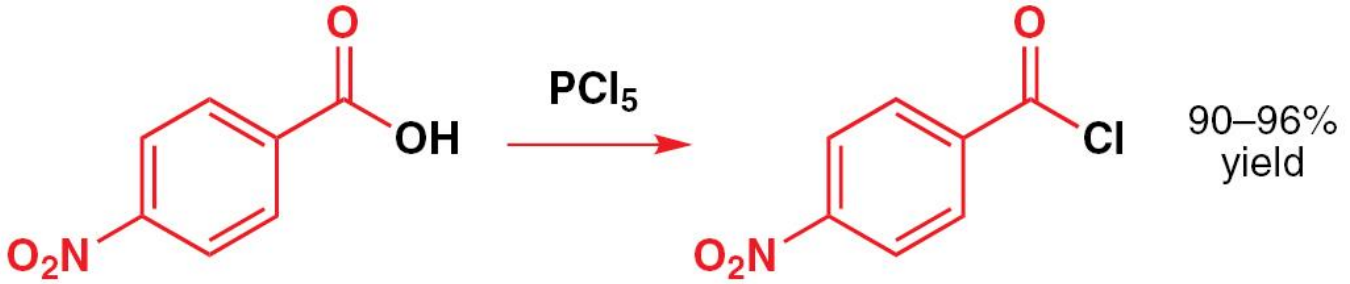
- **With thionyl chloride**

acid chlorides can be made from carboxylic acids with thionyl chloride



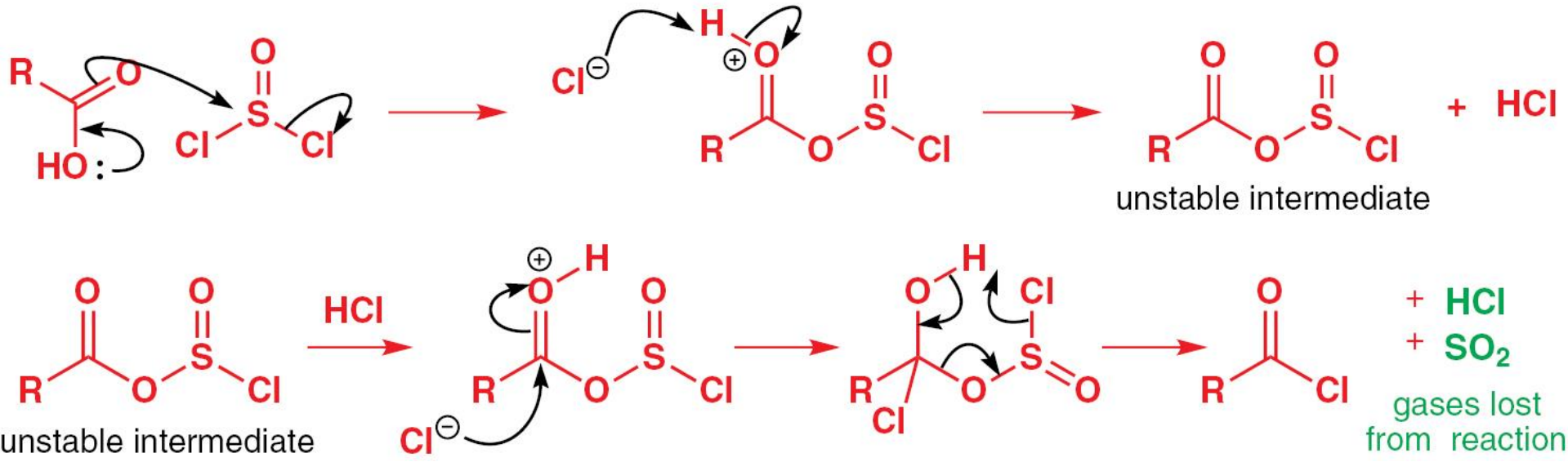
- **With phosphorus pentachloride**

acid chlorides can be made from carboxylic acids with phosphorus pentachloride



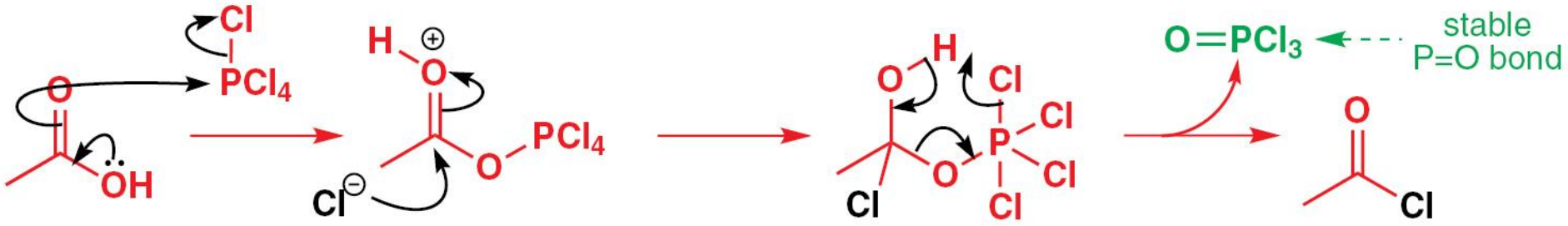
Making acid chlorides from carboxylic acids

- **With thionyl chloride: Mechanism for the reaction**

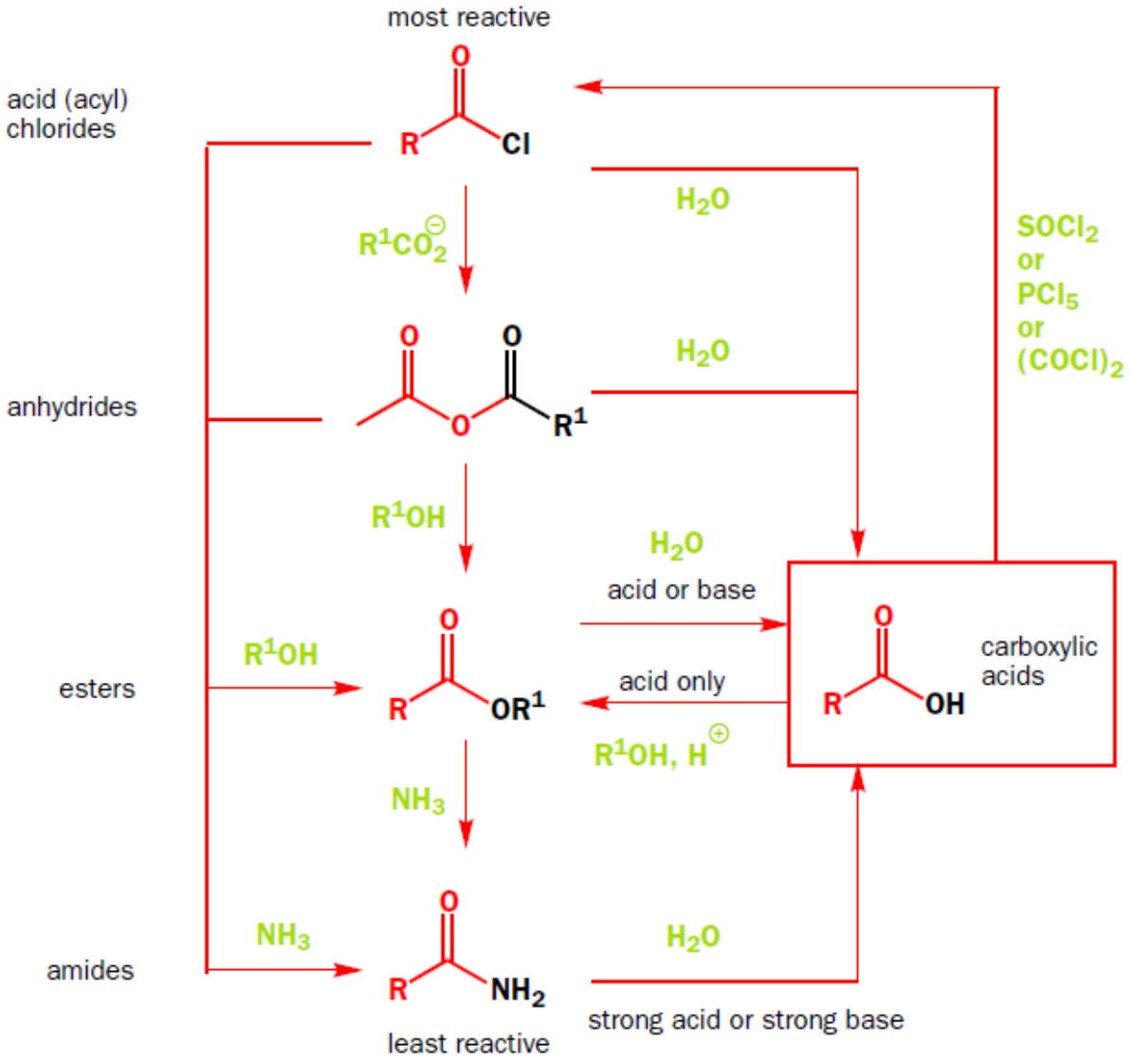


Making acid chlorides from carboxylic acids

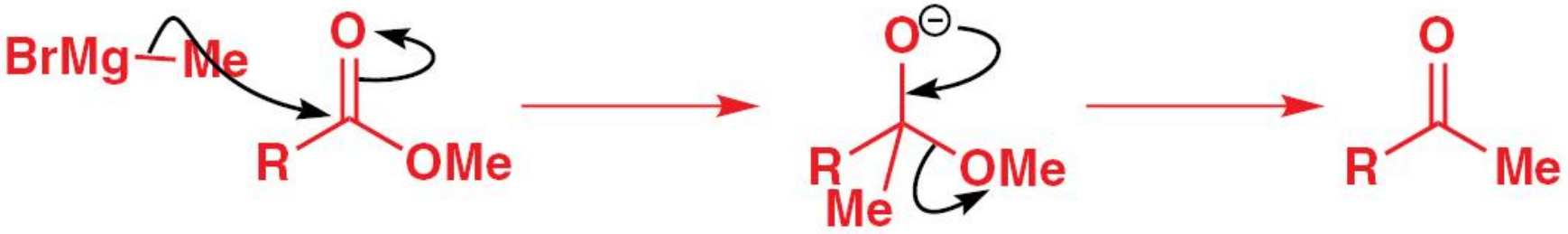
- **With phosphorus pentachloride: Mechanism for the reaction**



Reactivity Chart

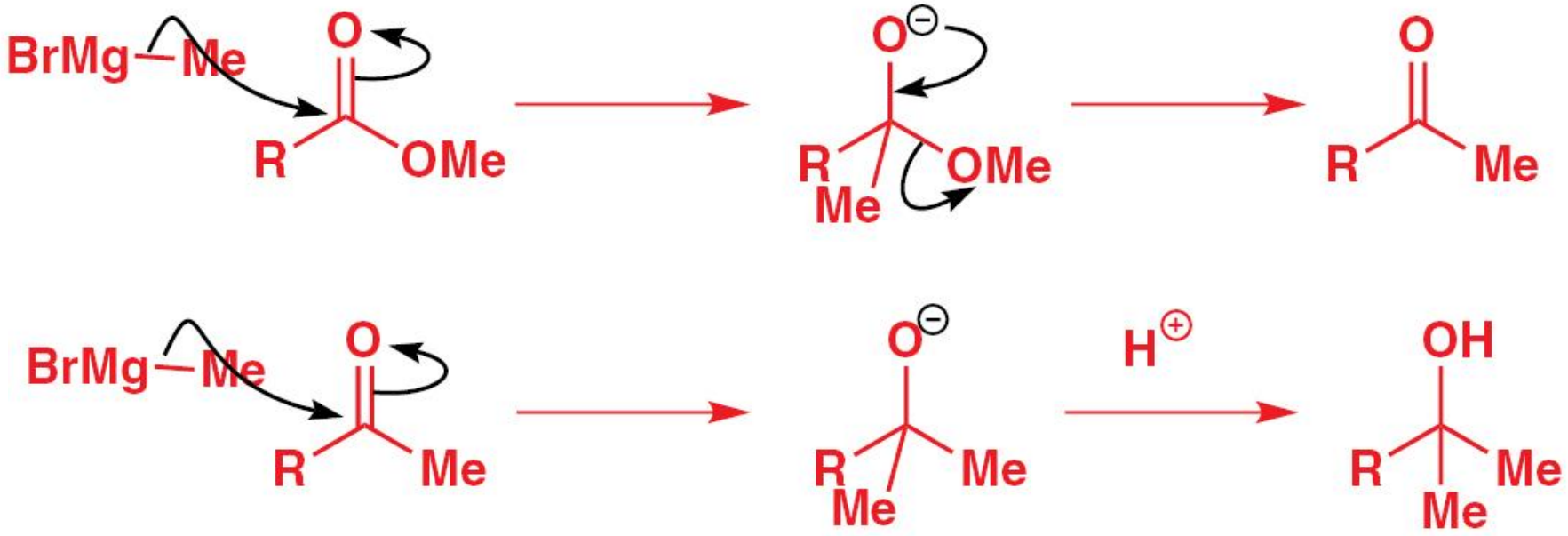


Can we make ketones from esters?



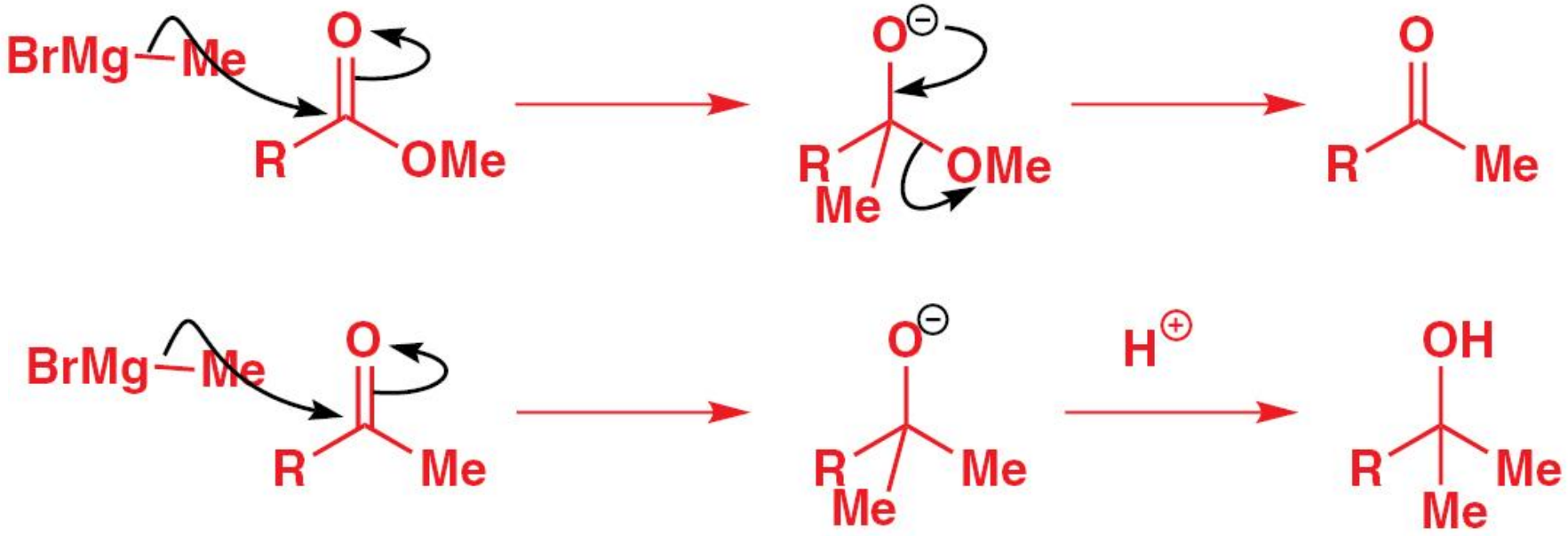
Can we make ketones from esters?

No, we make alcohols



Can we make ketones from esters?

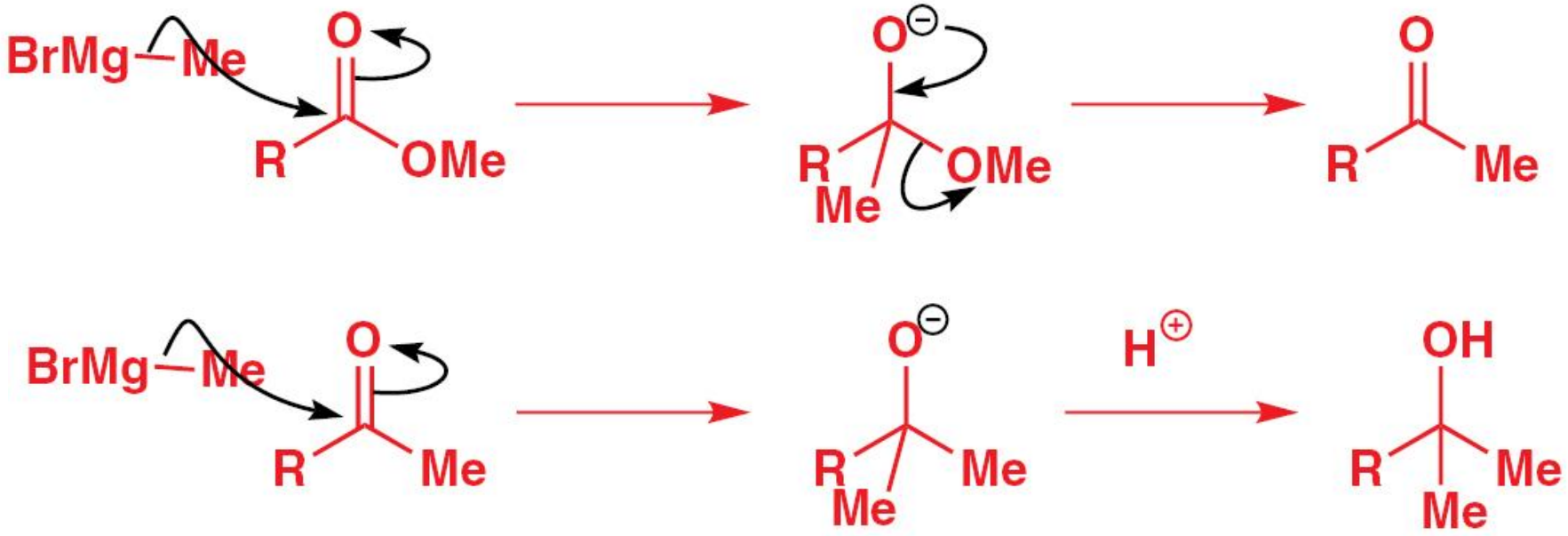
No, we make alcohols



Even if we use 1 equivalent of RMgX we would still get the ROH: Ketones are more reactive than esters

Can we make ketones from esters?

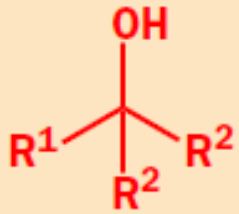
No, we make alcohols



Even if we use 1 equivalent of RMgX we would still get the ROH: Ketones are more reactive than esters

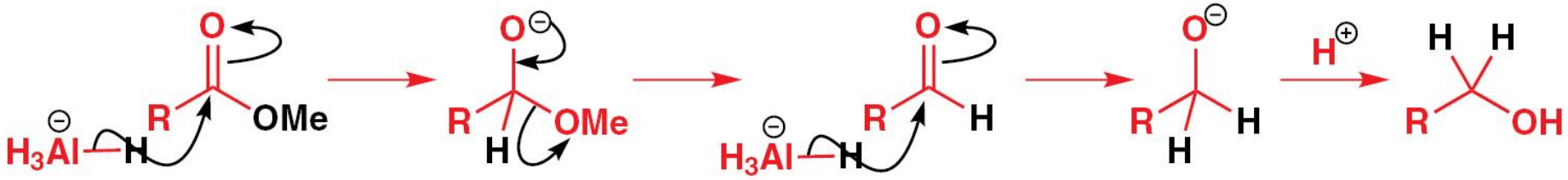
Tertiary alcohol synthesis

Tertiary alcohols with two identical R^2 groups can be made from ester plus two equivalents of organolithium or Grignard reagent.



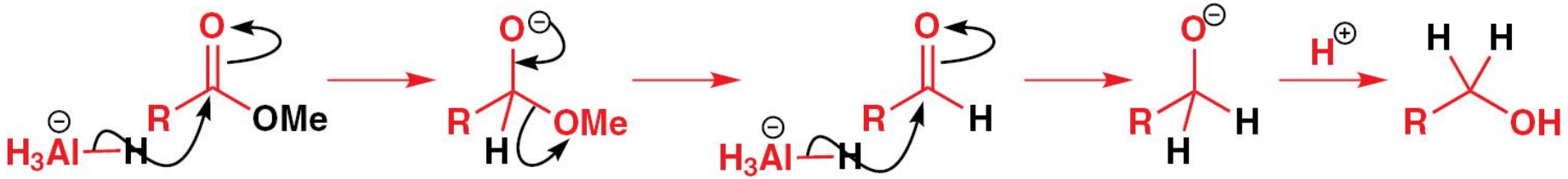
Another way to make alcohols from esters

reduction of esters by $LiAlH_4$

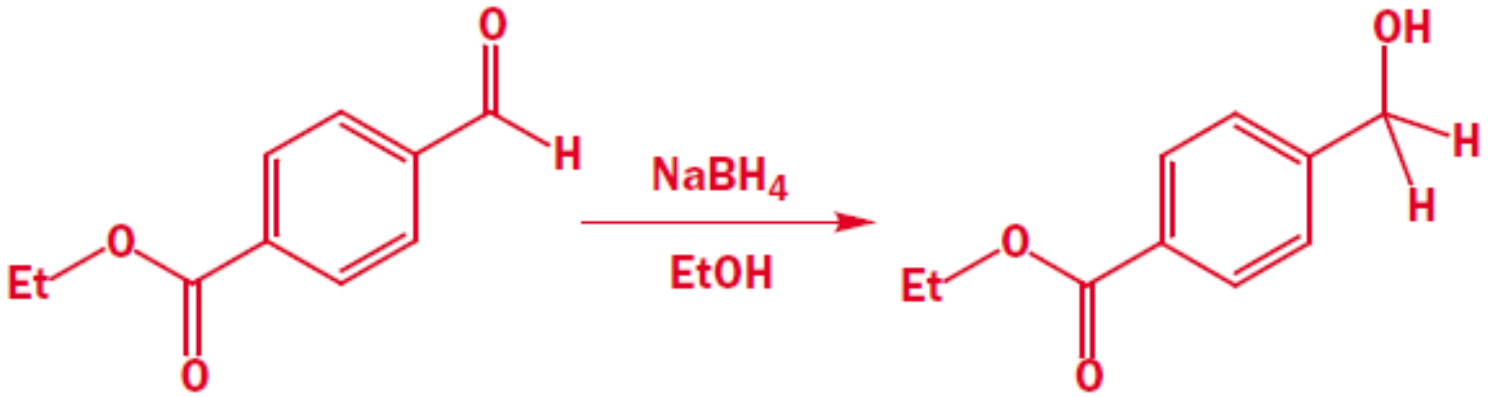


Another way to make alcohols from esters

reduction of esters by $LiAlH_4$



- **$NaBH_4$ does not reduce esters**



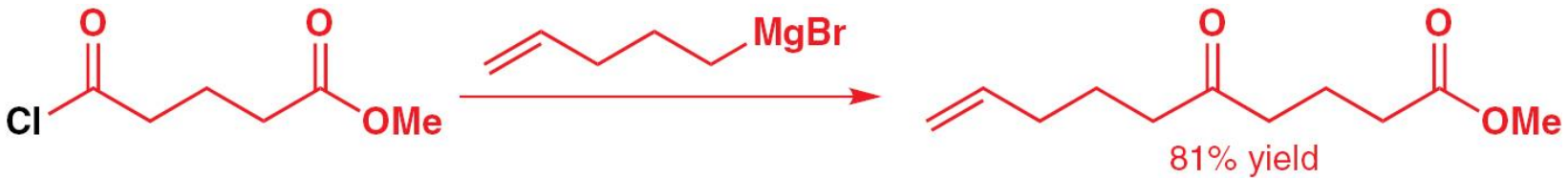
How can we make carbonyl compounds from acid derivatives?

How can we make carbonyl compounds from acid derivatives?

1. **By making the reactants more reactive**

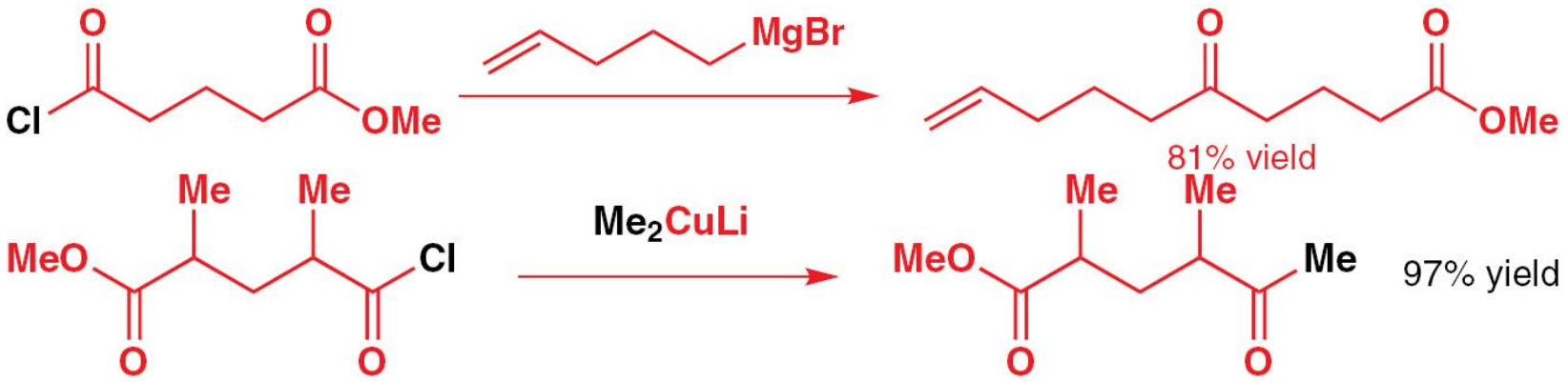
How can we make carbonyl compounds from acid derivatives?

1. **By making the reactants more reactive**



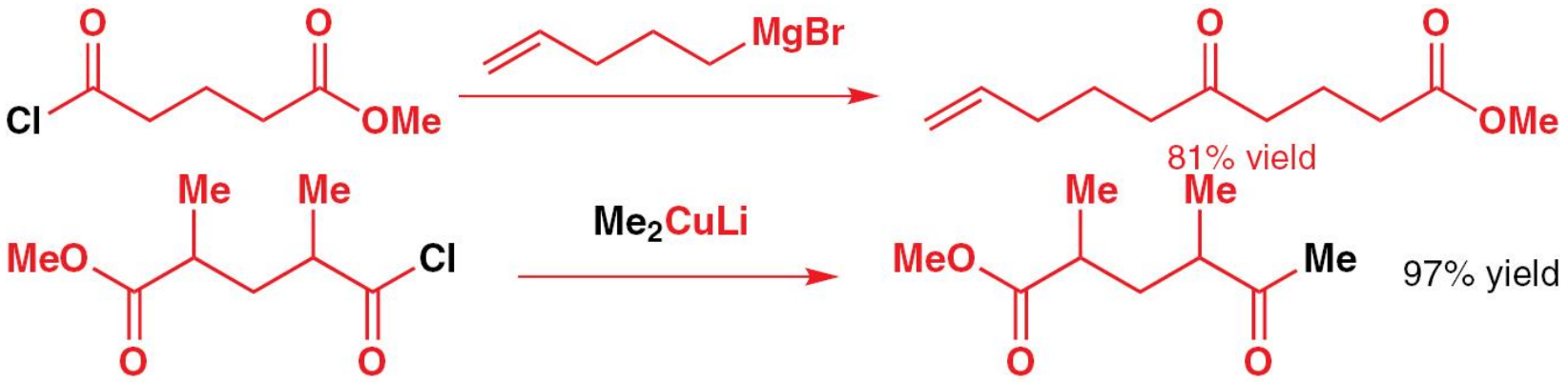
How can we make carbonyl compounds from acid derivatives?

1. **By making the reactants more reactive**



How can we make carbonyl compounds from acid derivatives?

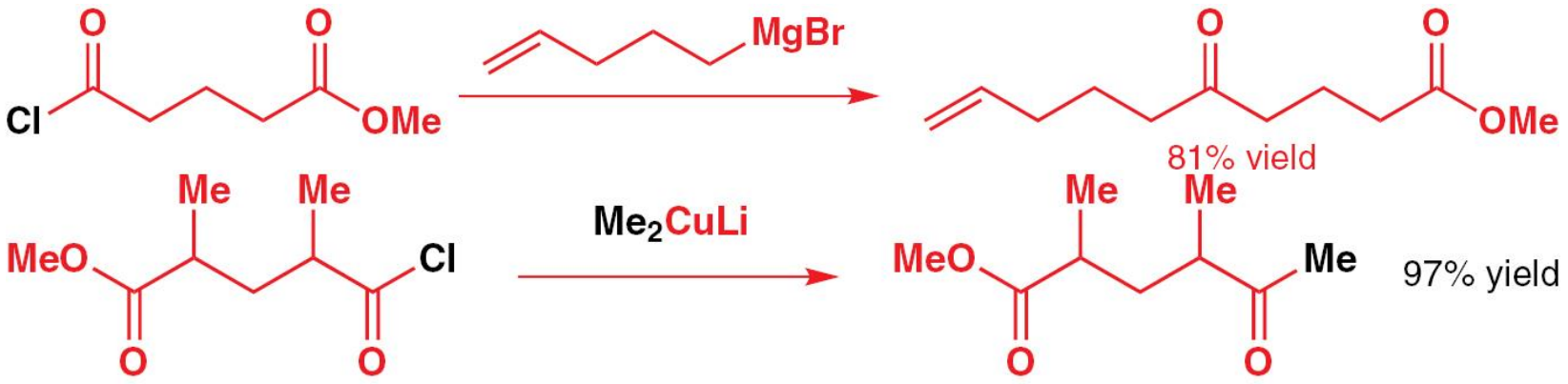
1. By making the reactants more reactive



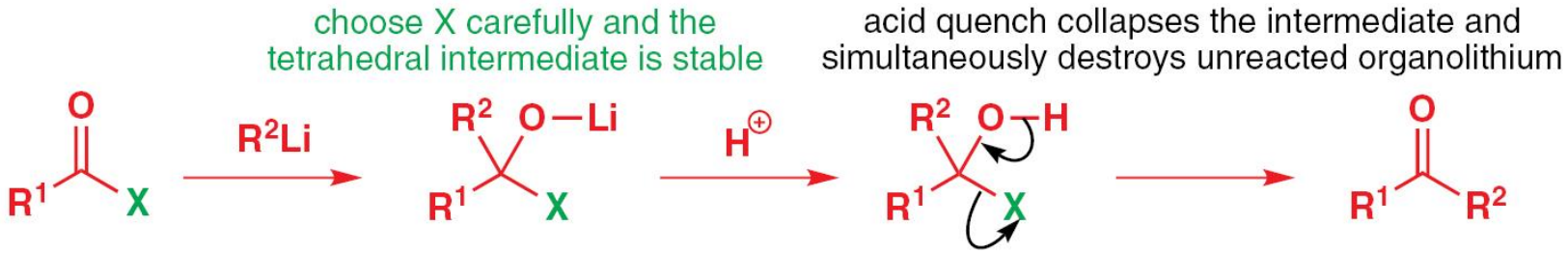
2. … Or by making the products less reactive

How can we make carbonyl compounds from acid derivatives?

1. By making the reactants more reactive



2. … Or by making the products less reactive



Examples

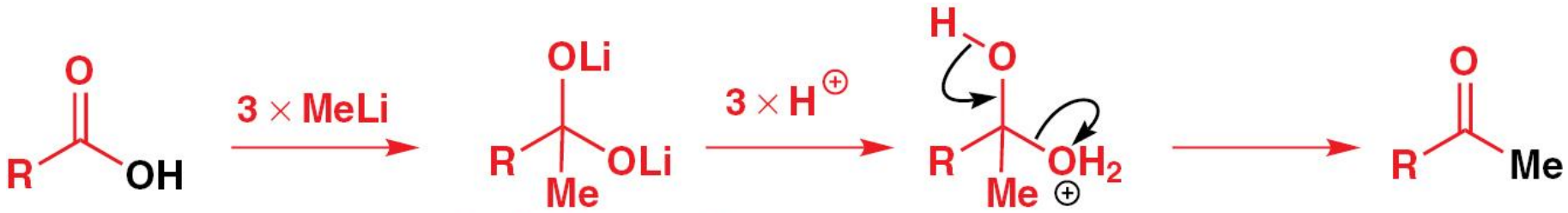


tetrahedral intermediate:
stable under anhydrous conditions

Examples



tetrahedral intermediate:
stable under anhydrous conditions

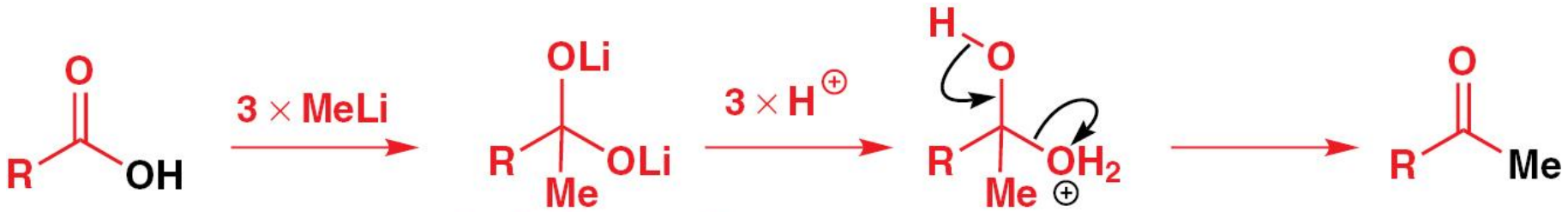


tetrahedral intermediate

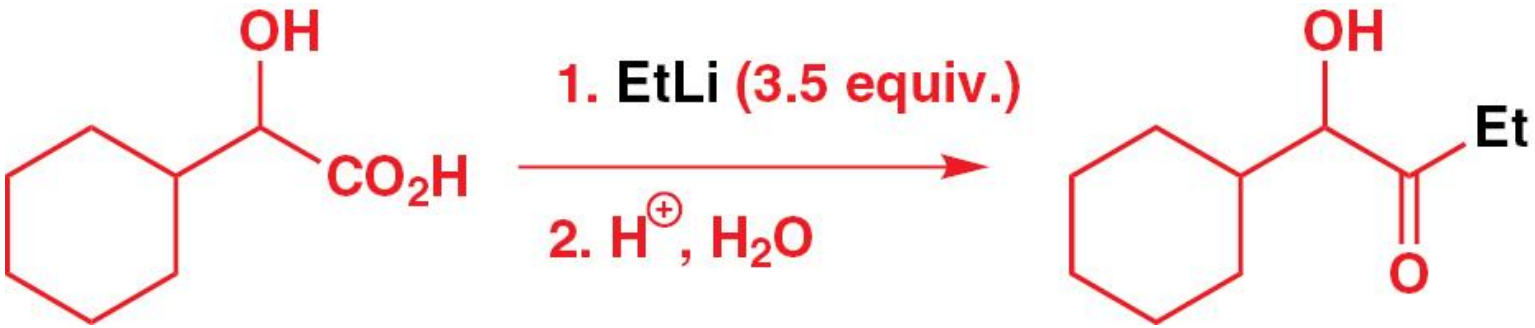
Examples



tetrahedral intermediate:
stable under anhydrous conditions



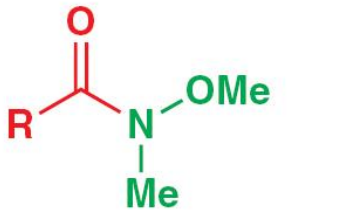
tetrahedral intermediate



65% yield

Examples

a Weinreb amide (an *N*-methoxy-*N*-methyl amide)



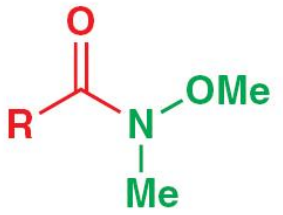
easily made



acyl chloride amine

Examples

a Weinreb amide (an *N*-methoxy-*N*-methyl amide)

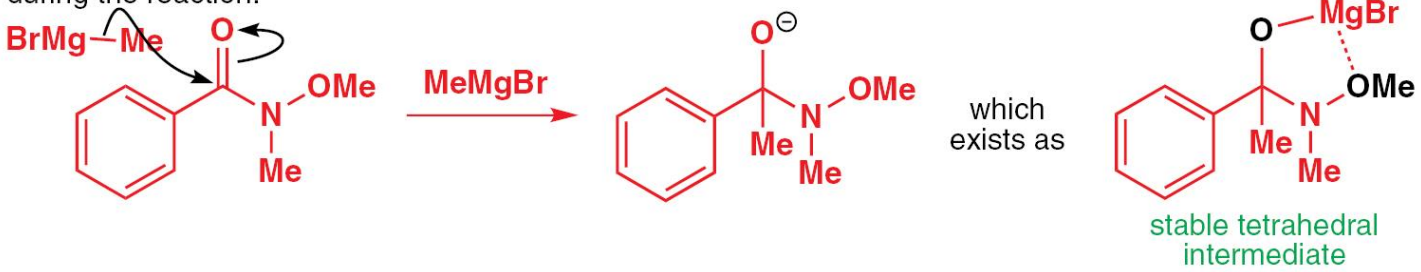


easily made

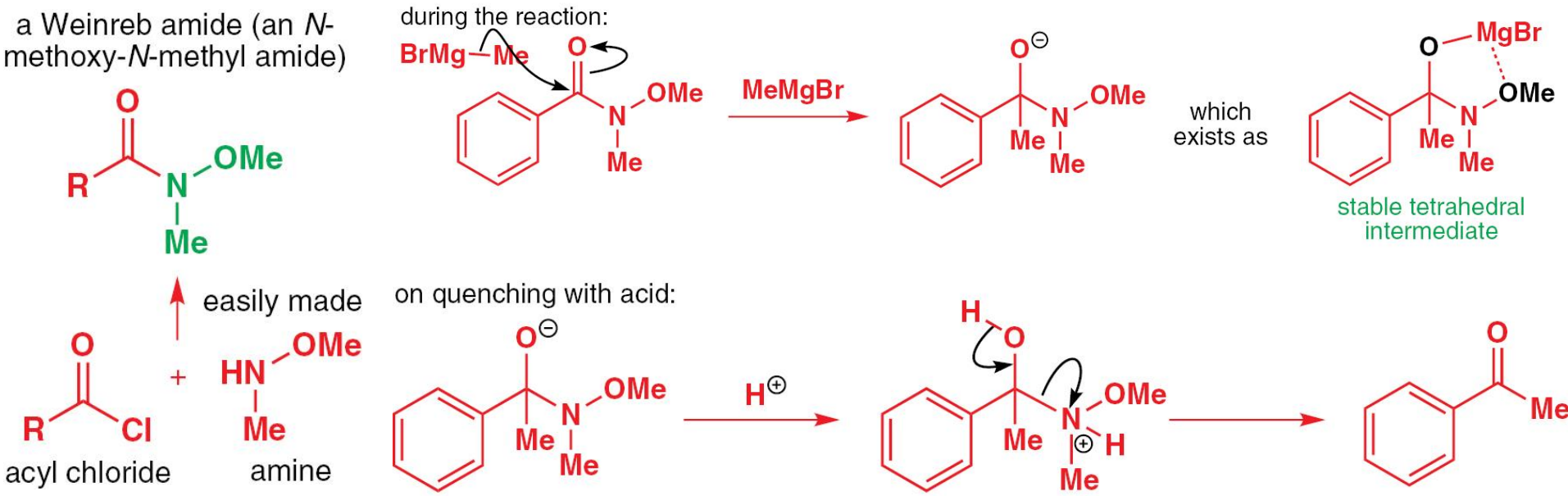


acyl chloride amine

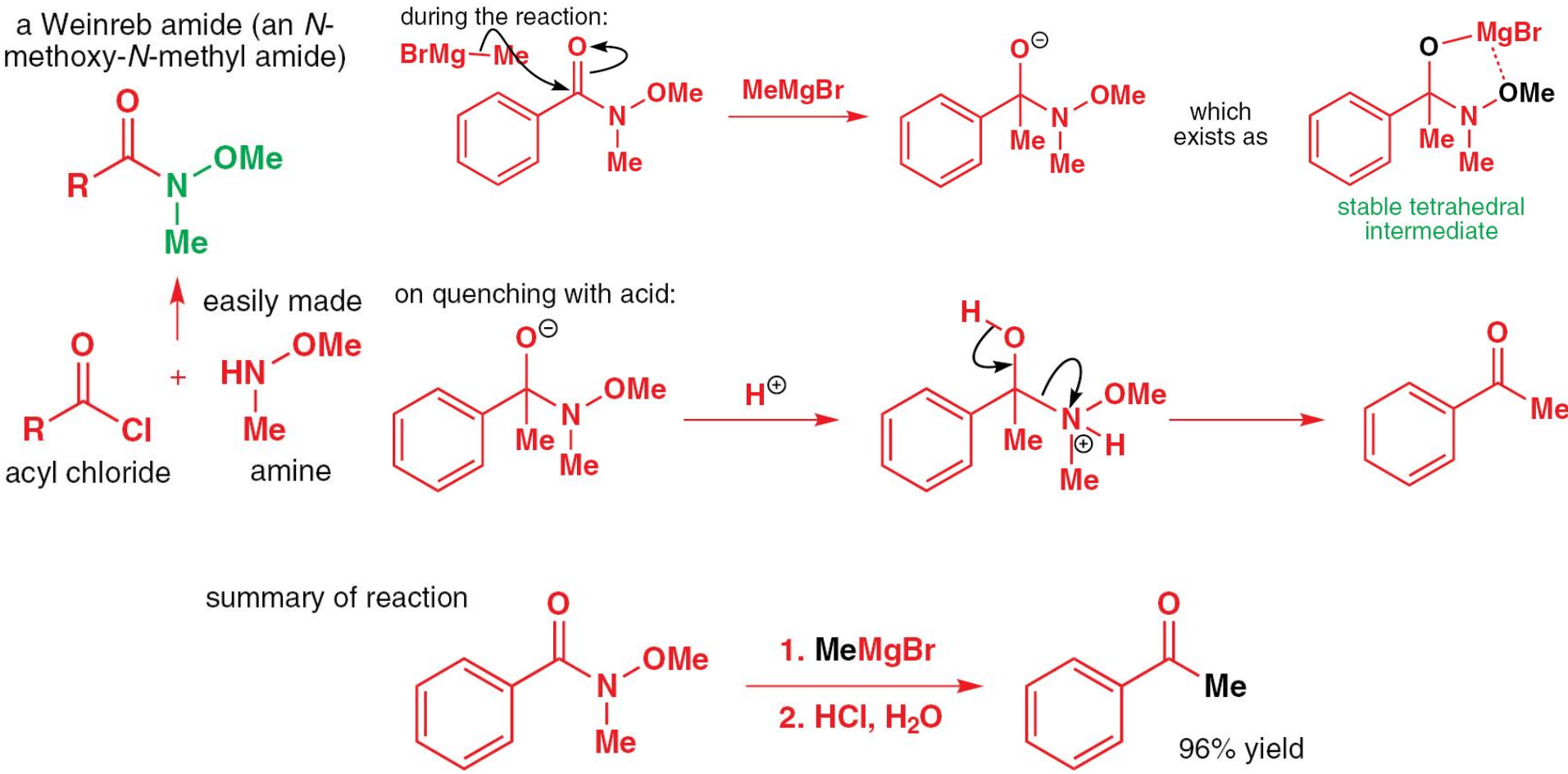
during the reaction:



Examples


a Weinreb amide (an N-methoxy-N-methyl amide)
during the reaction:
MeMgBr
which exists as
stable tetrahedral intermediate
easily made
acyl chloride
amine
on quenching with acid:

Examples


a Weinreb amide (an N-methoxy-N-methyl amide)
during the reaction:
MeMgBr
which exists as
stable tetrahedral intermediate
easily made
acyl chloride
amine
on quenching with acid:
summary of reaction
1. MeMgBr
2. HCl, H2O
96% yield

Summary

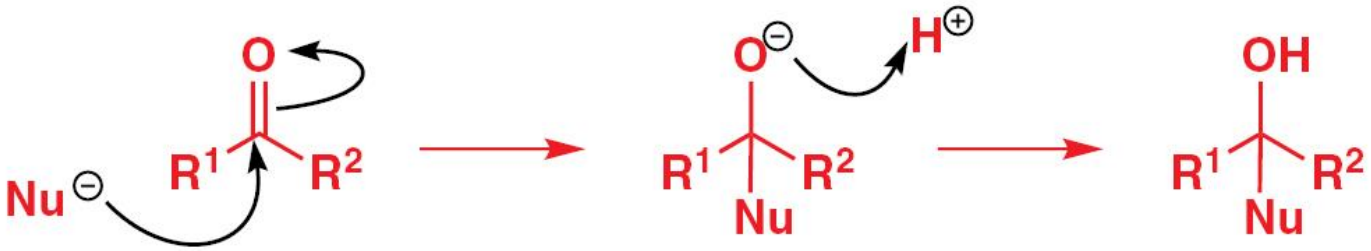
- Nucleophilic attack followed by loss of leaving group
- What makes a good nucleophile
- What makes a good leaving group
- There is always a tetrahedral intermediate
- How to make acid derivatives
- Reactivity of acid derivatives
- How to make ketones from acids
- How to reduce esters to alcohols

Nucleophilic Substitution at the C=O with loss of the Carbonyl Oxygen

- Replacement of carbonyl oxygen
- Acetal formation
- Imine formation
- Stable and unstable imines
- Reductive amination
- Strecker synthesis of amino acids
- Wittig reaction

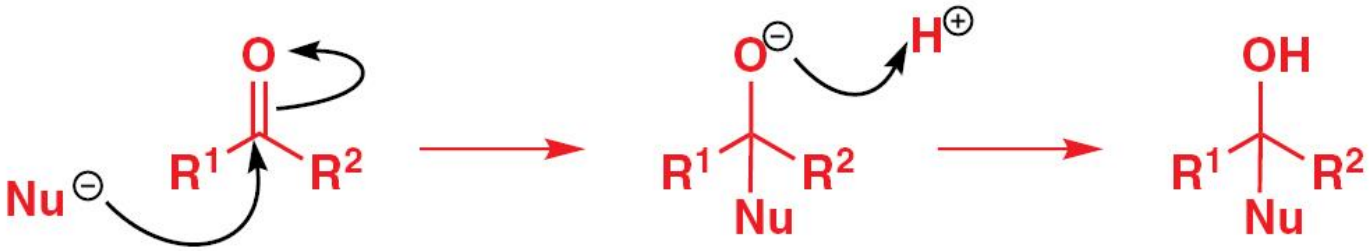
Nucleophilic addition and substitutions at the C=O group

nucleophilic addition to a carbonyl group

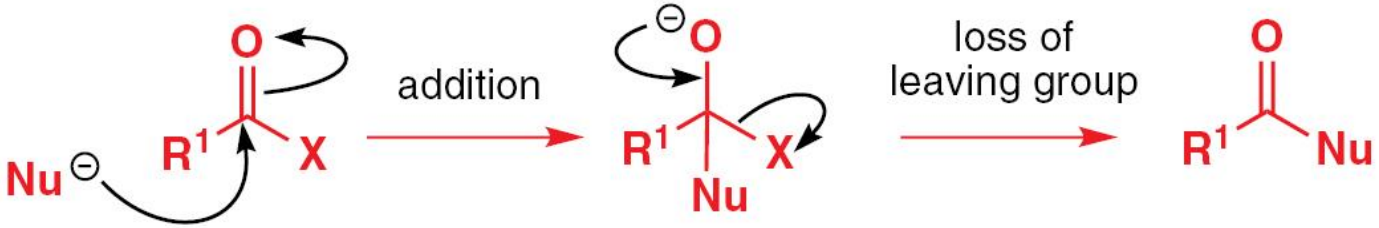


Nucleophilic addition and substitutions at the C=O group

nucleophilic addition to a carbonyl group

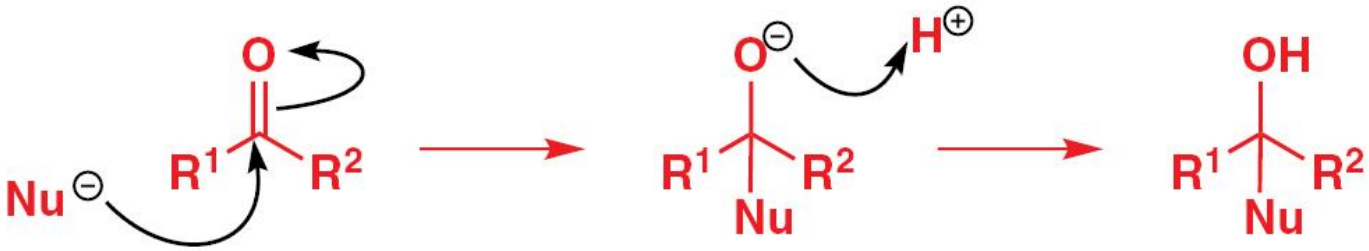


nucleophilic substitution at a carbonyl group

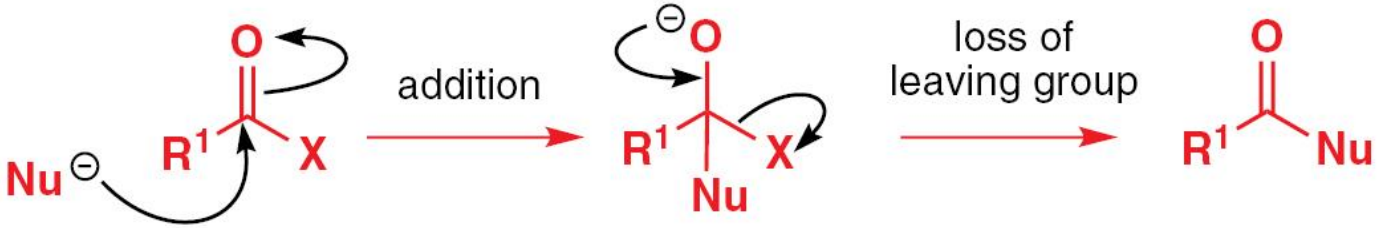


Nucleophilic addition and substitutions at the C=O group

nucleophilic addition to a carbonyl group



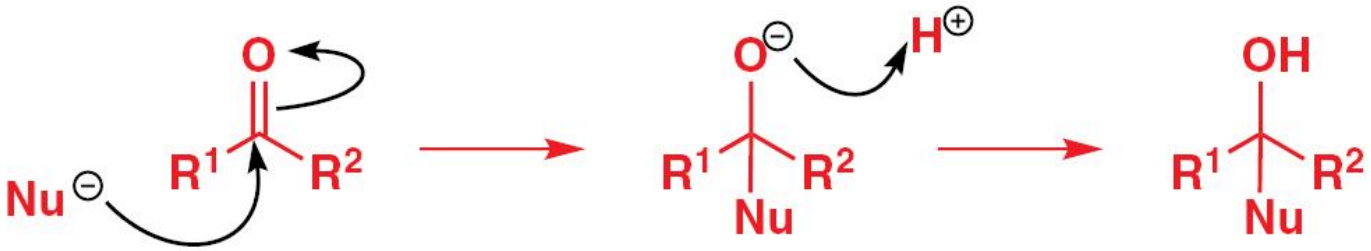
nucleophilic substitution at a carbonyl group



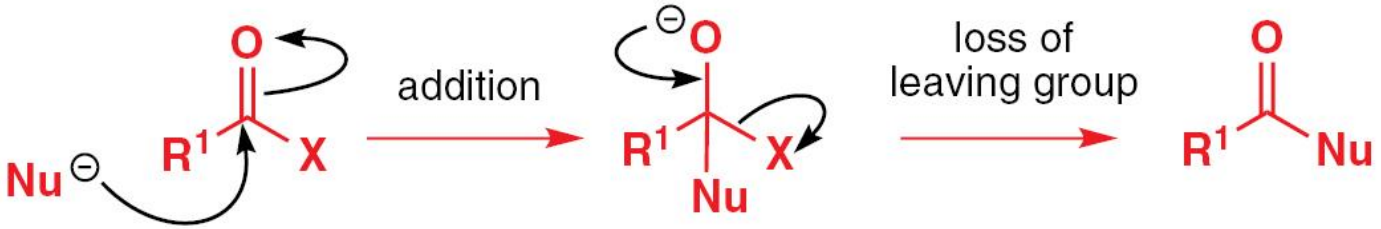
…And another kind of substitution reactions

Nucleophilic addition and substitutions at the C=O group

nucleophilic addition to a carbonyl group

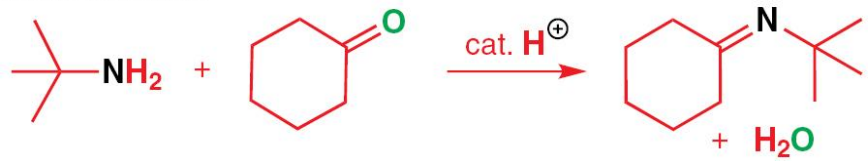


nucleophilic substitution at a carbonyl group

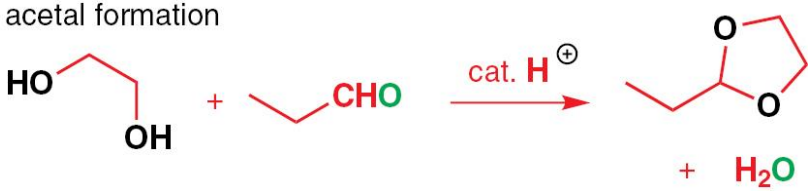


…And another kind of substitution reactions

imine formation

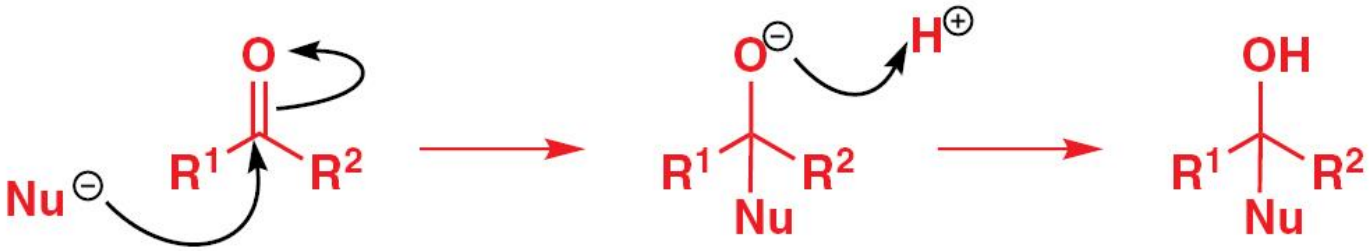


acetal formation

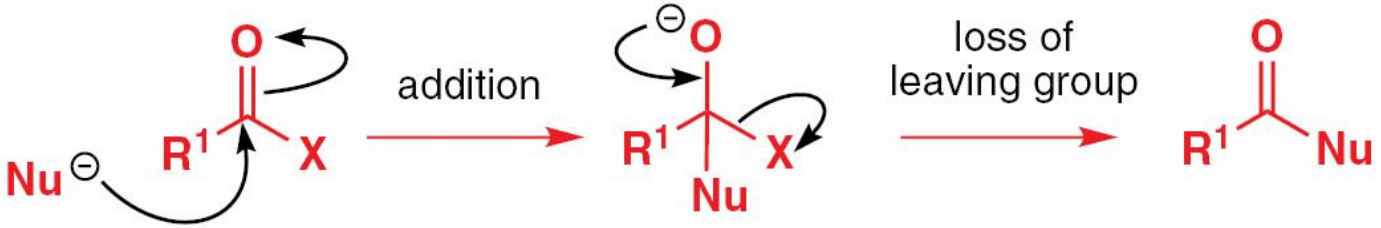


Nucleophilic addition and substitutions at the C=O group

nucleophilic addition to a carbonyl group

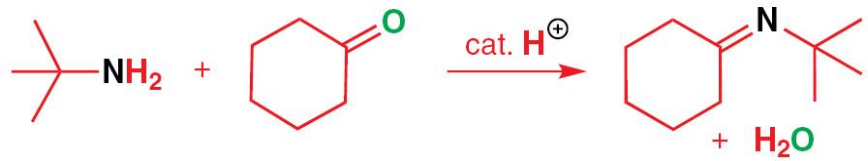


nucleophilic substitution at a carbonyl group

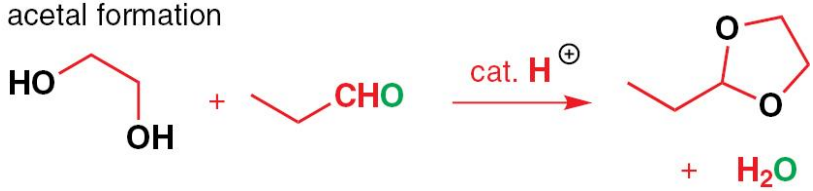


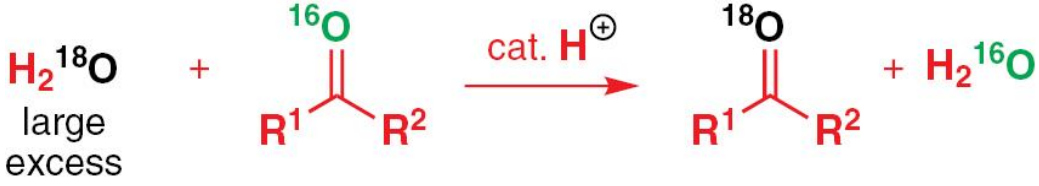
…And another kind of substitution reactions

imine formation



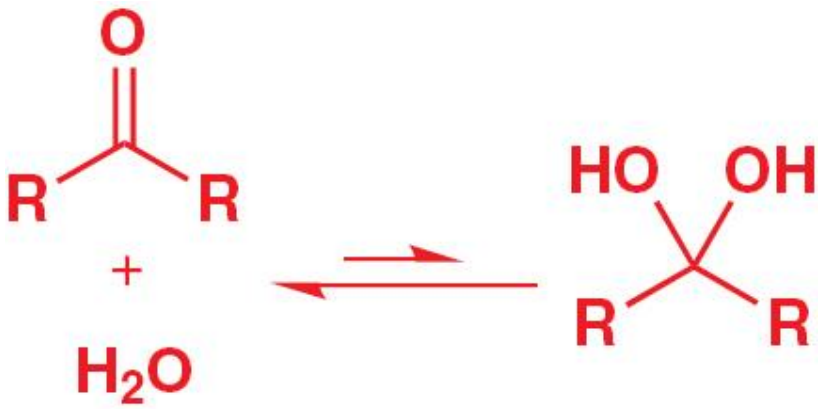
acetal formation





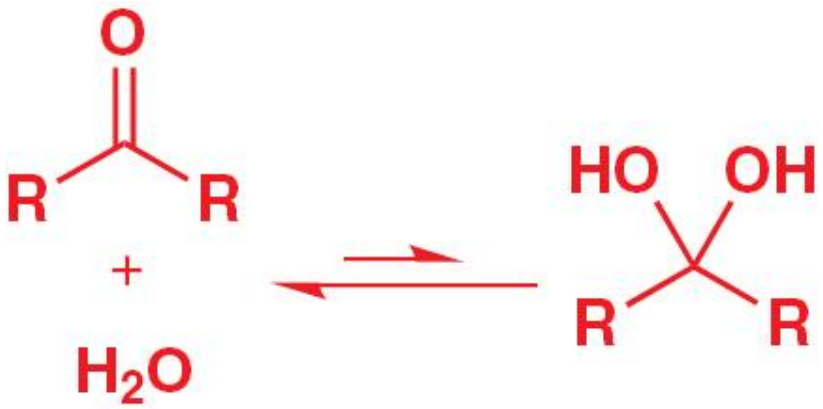
Addition of H_2O to C=O: Diol formation- Revision

Reaction



Addition of H_2O to C=O: Diol formation- Revision

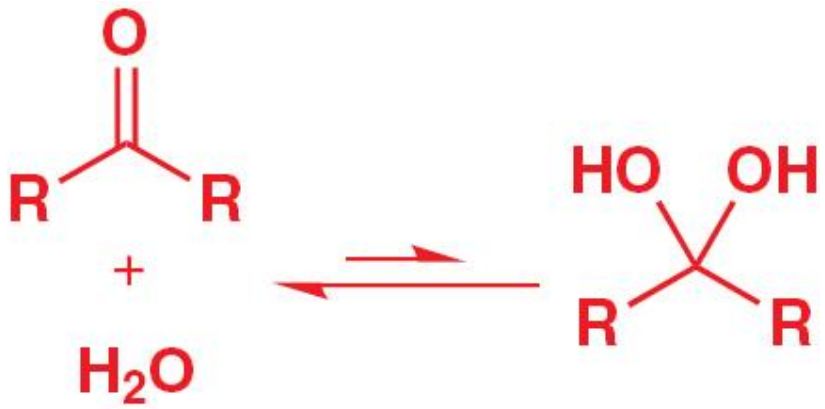
Reaction



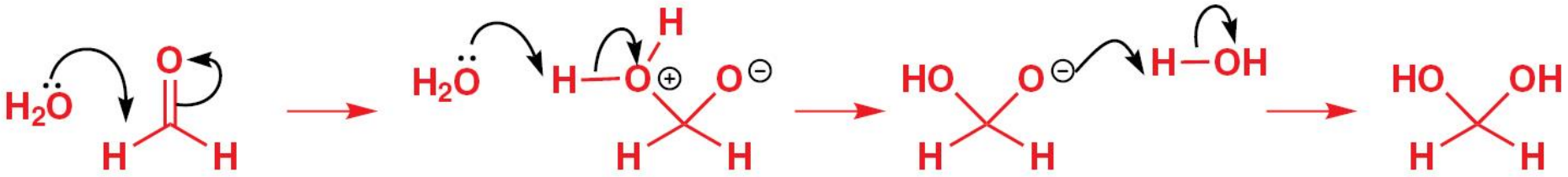
Mechanism

Addition of H_2O to C=O: Diol formation- Revision

Reaction



significant concentrations of hydrate are generally formed only from aldehydes



Mechanism

Addition of Alcohols ROH to C=O: Hemiacetal formation-Revision

Reaction



Addition of Alcohols ROH to C=O: Hemiacetal formation-Revision

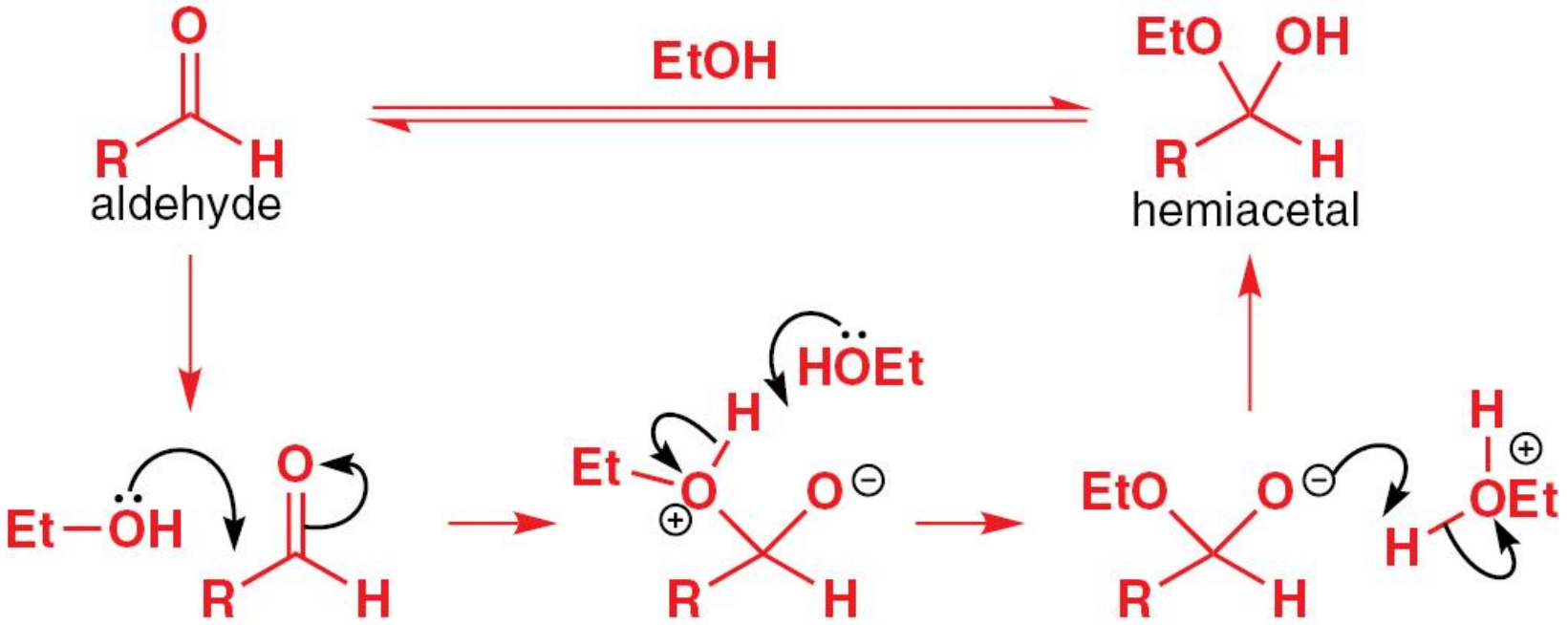
Reaction



Mechanism

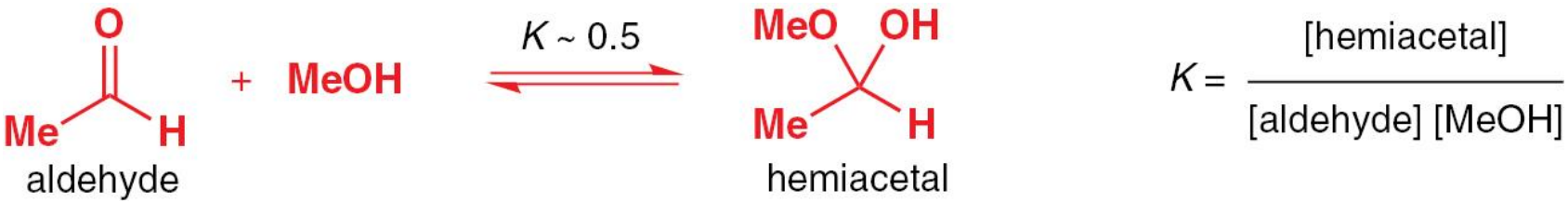
Addition of Alcohols ROH to C=O: Hemiacetal formation-Revision

Reaction

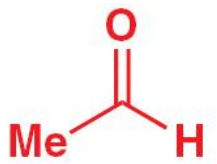


Mechanism

Hemiacetal formation-Revision



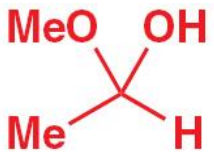
Hemiacetal formation-Revision



aldehyde

+ MeOH

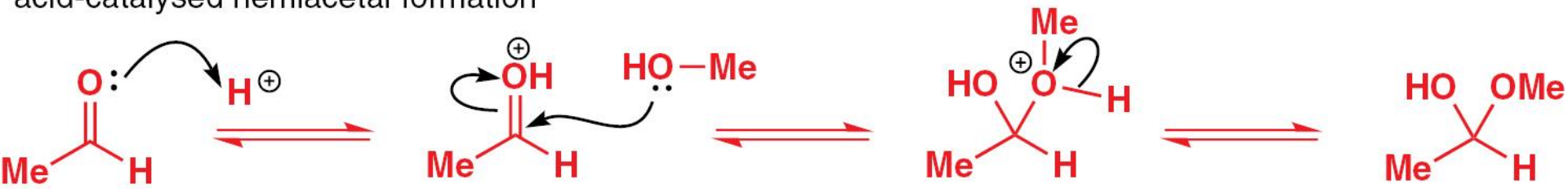
$K \sim 0.5$



hemiacetal

$$K = \frac{[\text{hemiacetal}]}{[\text{aldehyde}]\,[\text{MeOH}]}$$

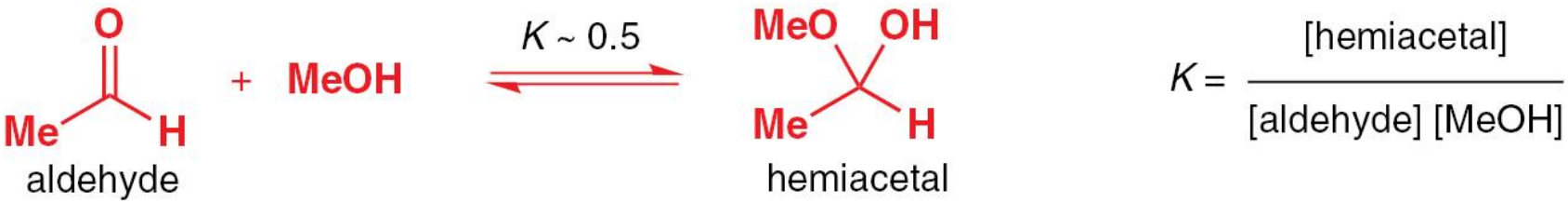
acid-catalysed hemiacetal formation



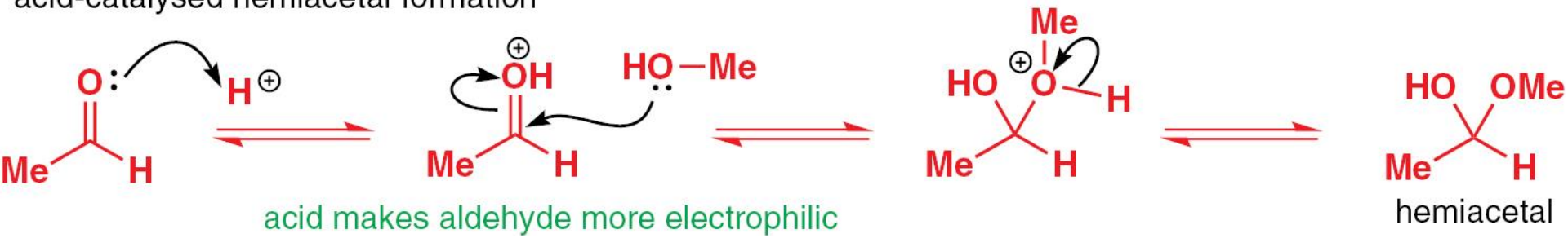
acid makes aldehyde more electrophilic

hemiacetal

Hemiacetal formation-Revision

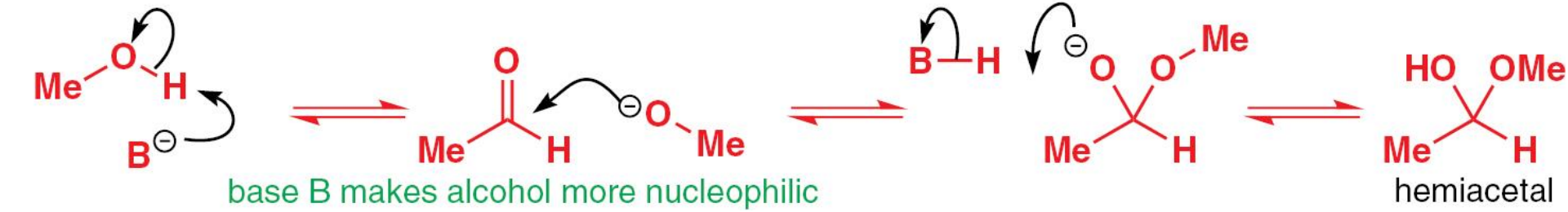


acid-catalysed hemiacetal formation



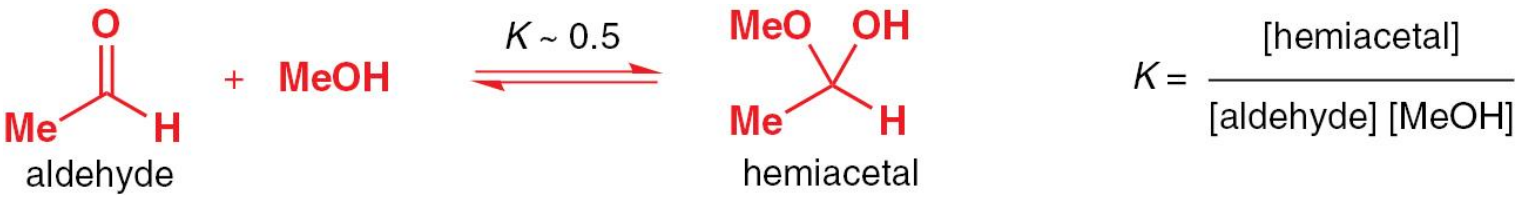
acid makes aldehyde more electrophilic

base-catalysed hemiacetal formation

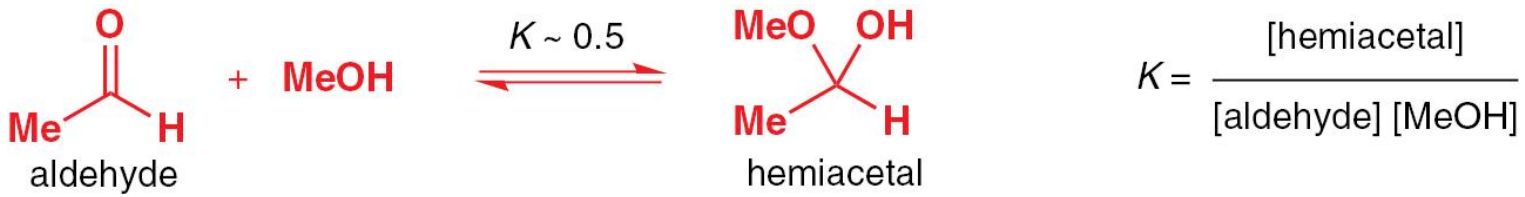


base B makes alcohol more nucleophilic

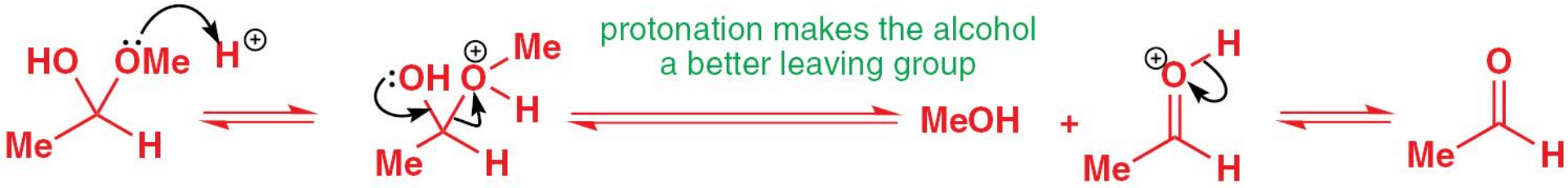
Decomposition of hemiacetals



Decomposition of hemiacetals

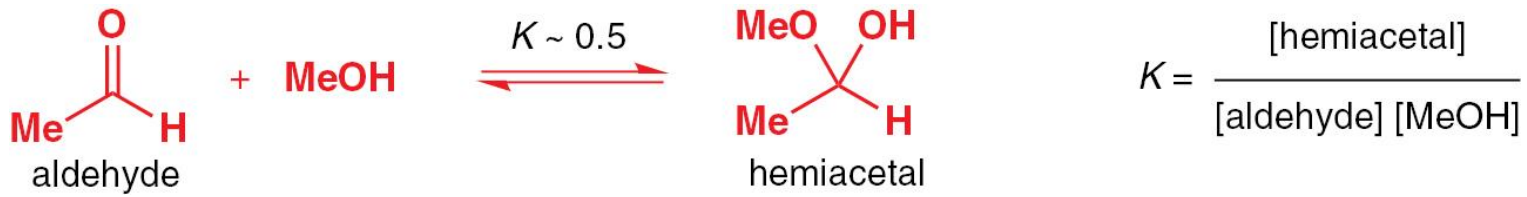


acid-catalysed hemiacetal decomposition

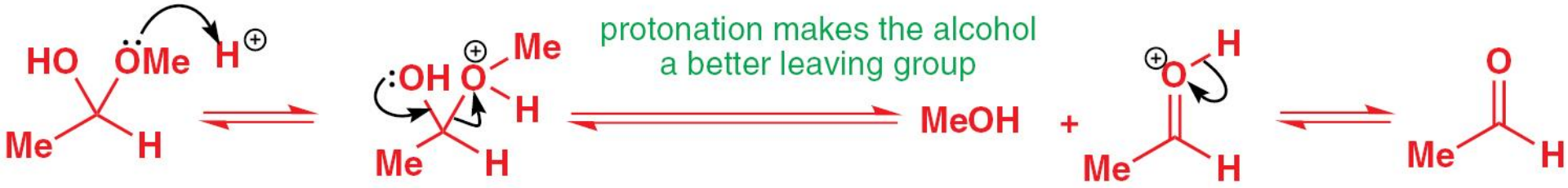


'

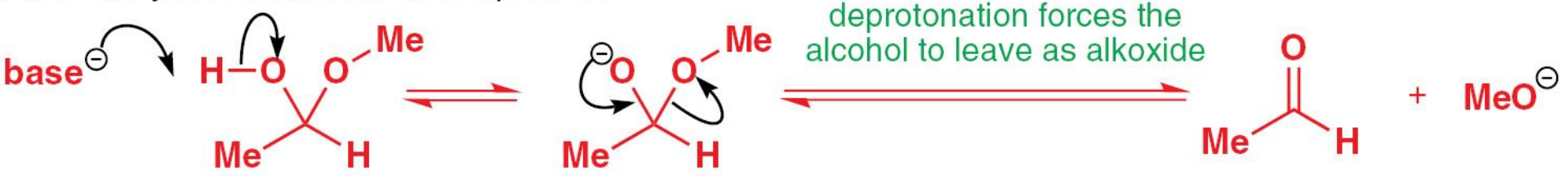
Decomposition of hemiacetals



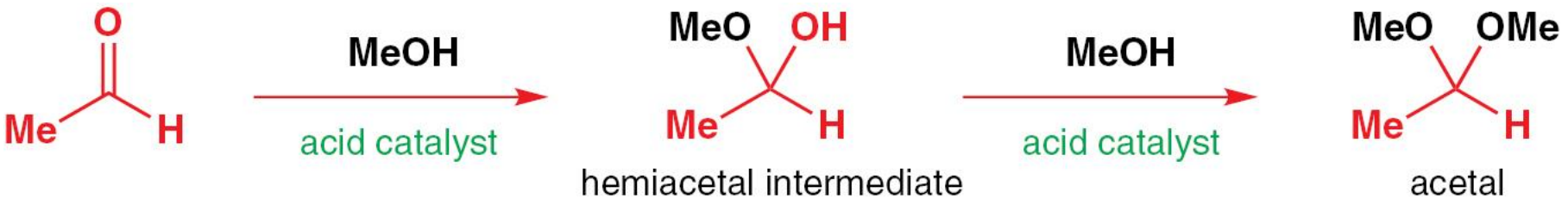
acid-catalysed hemiacetal decomposition



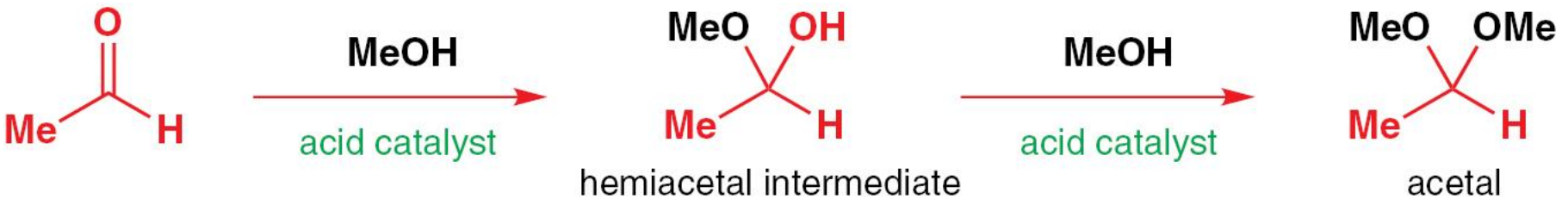
base-catalysed hemiacetal decomposition



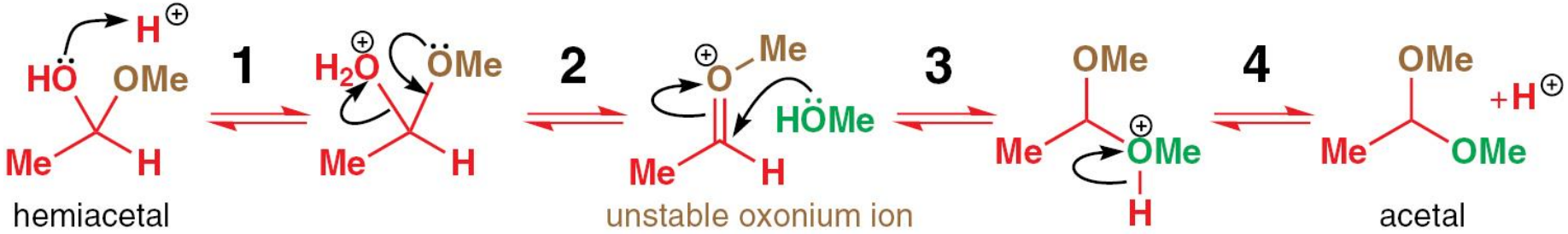
Acetal formation



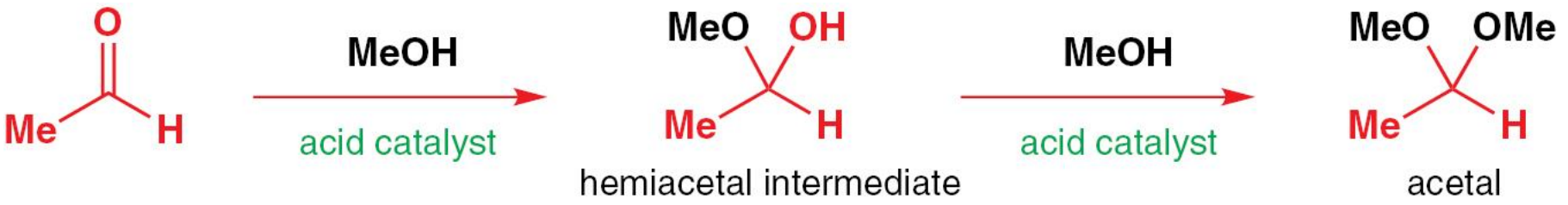
Acetal formation



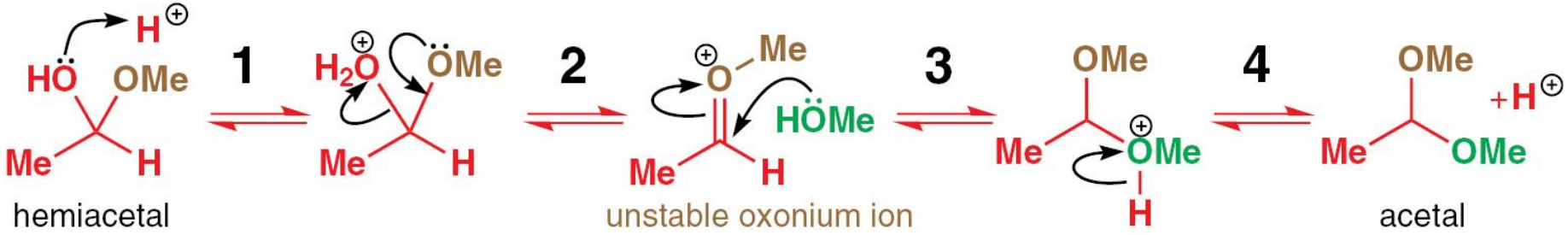
acid-catalysed acetal formation from hemiacetal

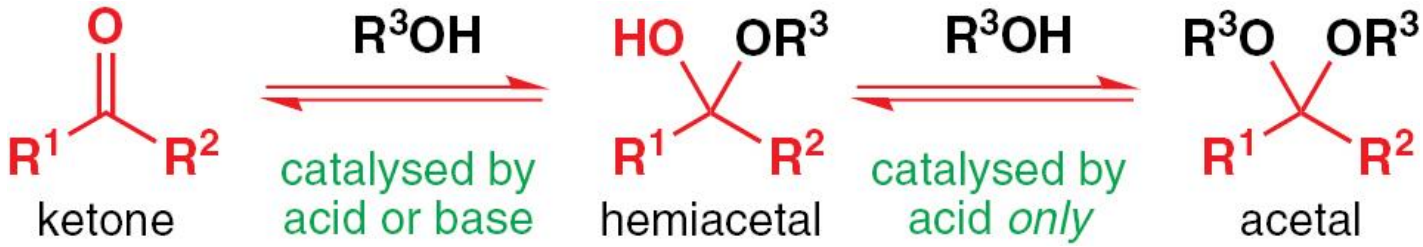


Acetal formation

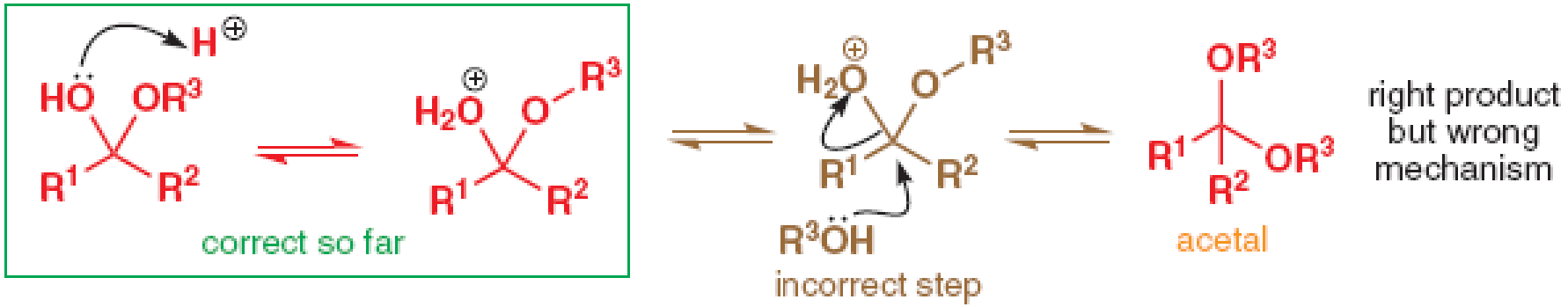


acid-catalysed acetal formation from hemiacetal

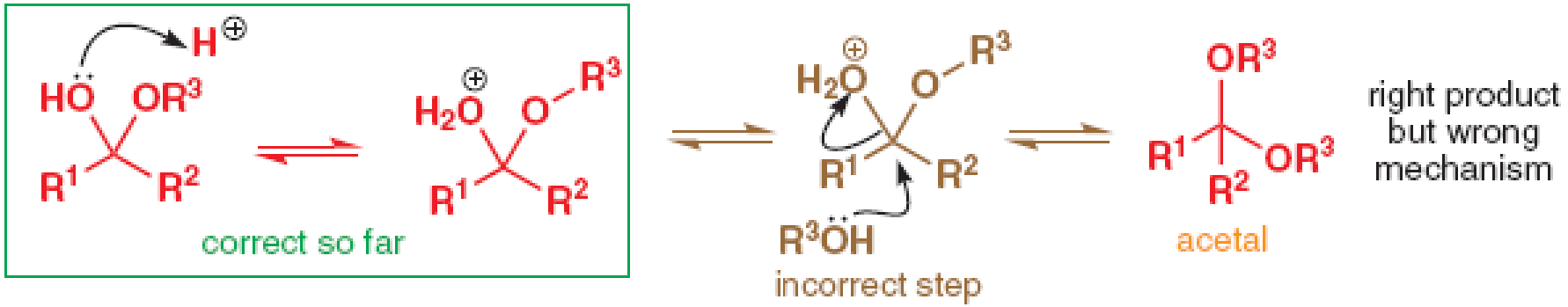


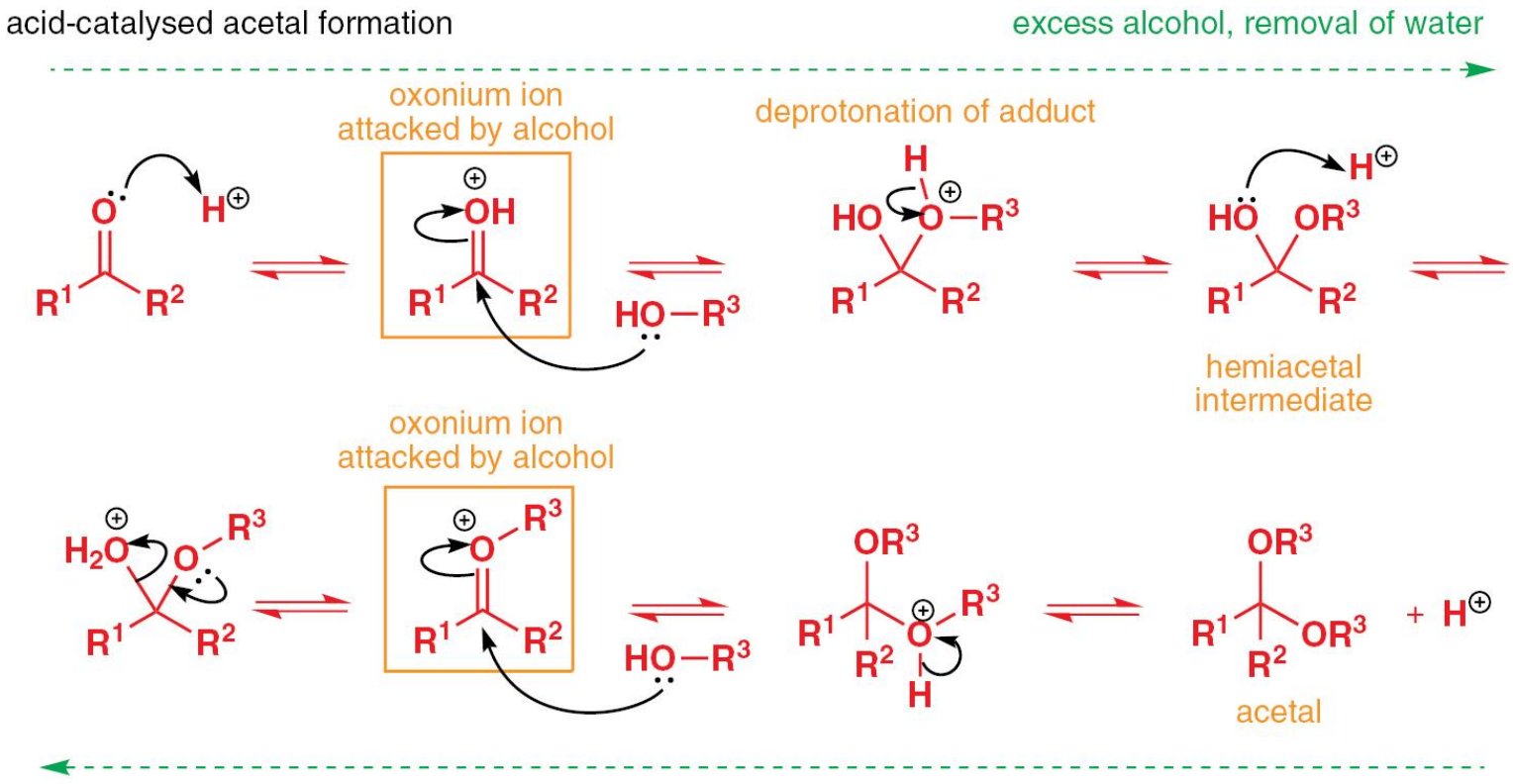


Acetal formation-Mechanism



Acetal formation-Mechanism

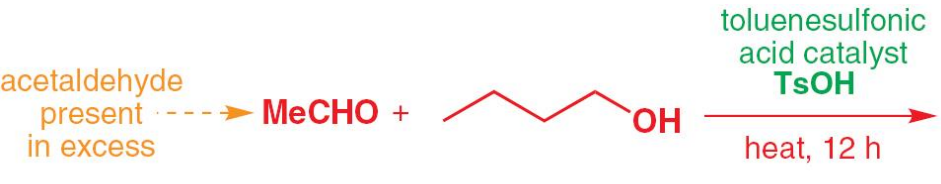




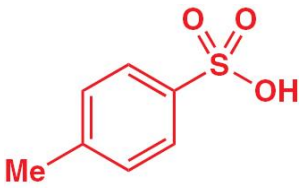
Acetal formation and decomposition-Examples



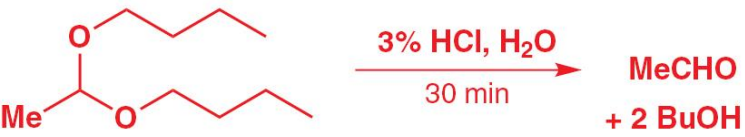
Acetal formation and decomposition-Examples



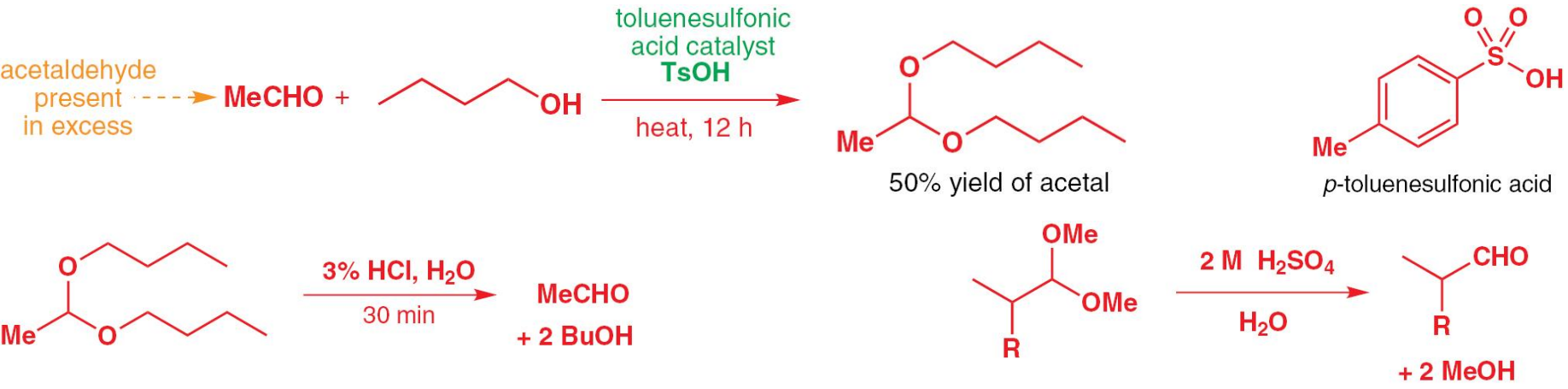
50% yield of acetal



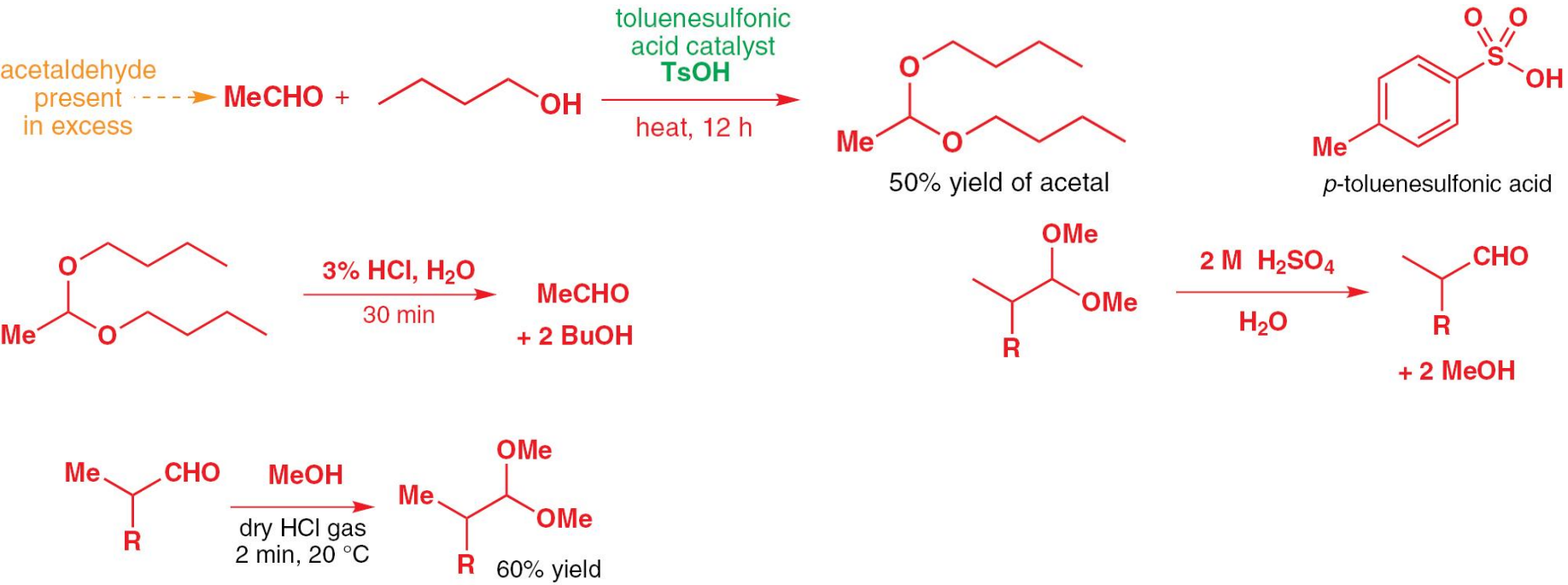
p-toluenesulfonic acid



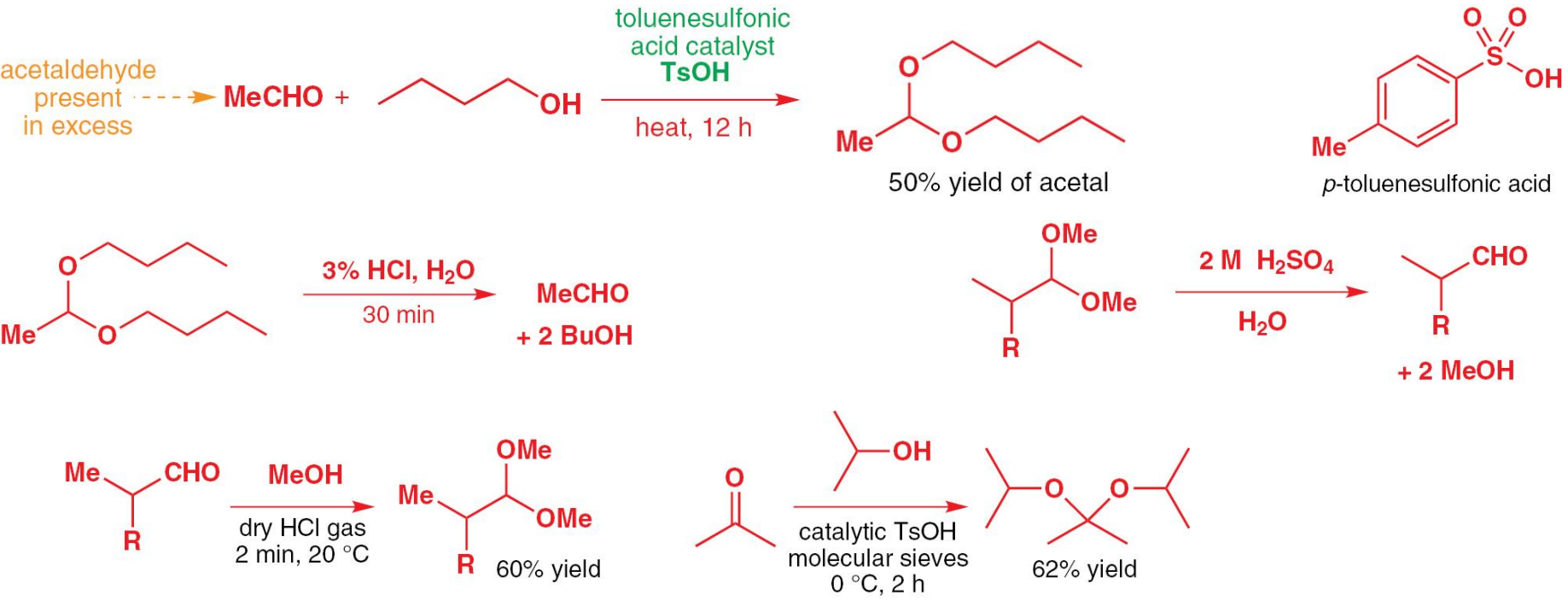
Acetal formation and decomposition-Examples



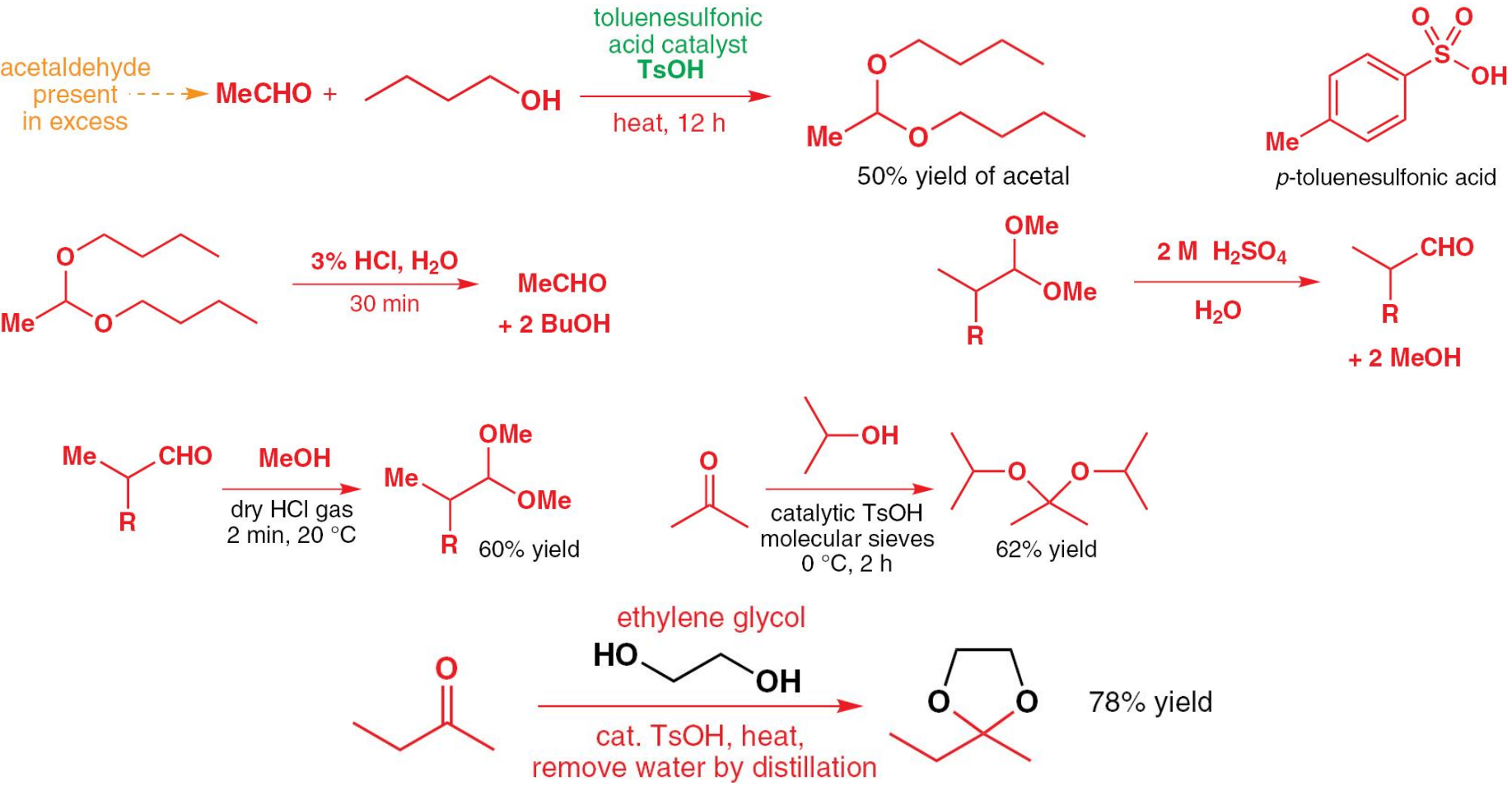
Acetal formation and decomposition-Examples



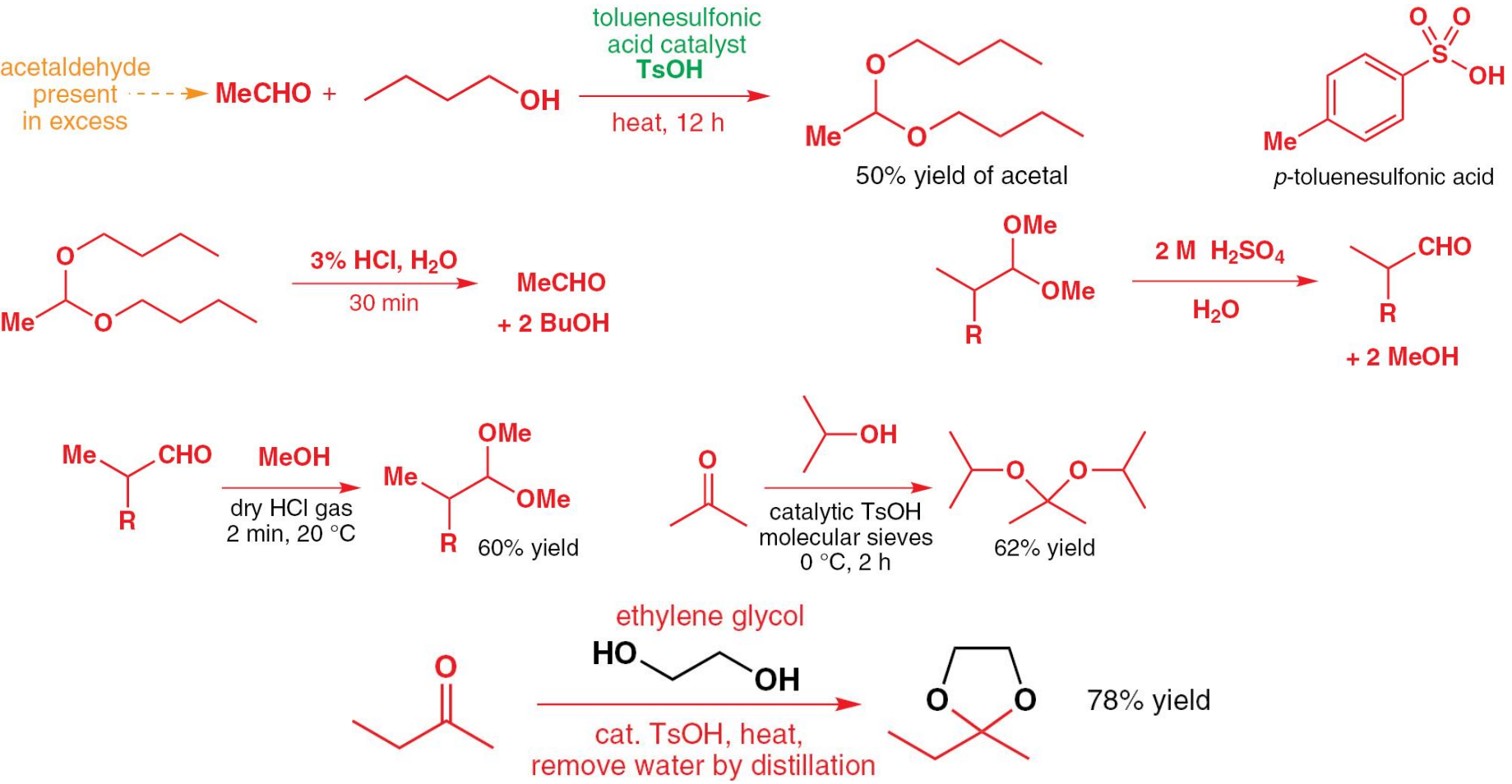
Acetal formation and decomposition-Examples



Acetal formation and decomposition-Examples

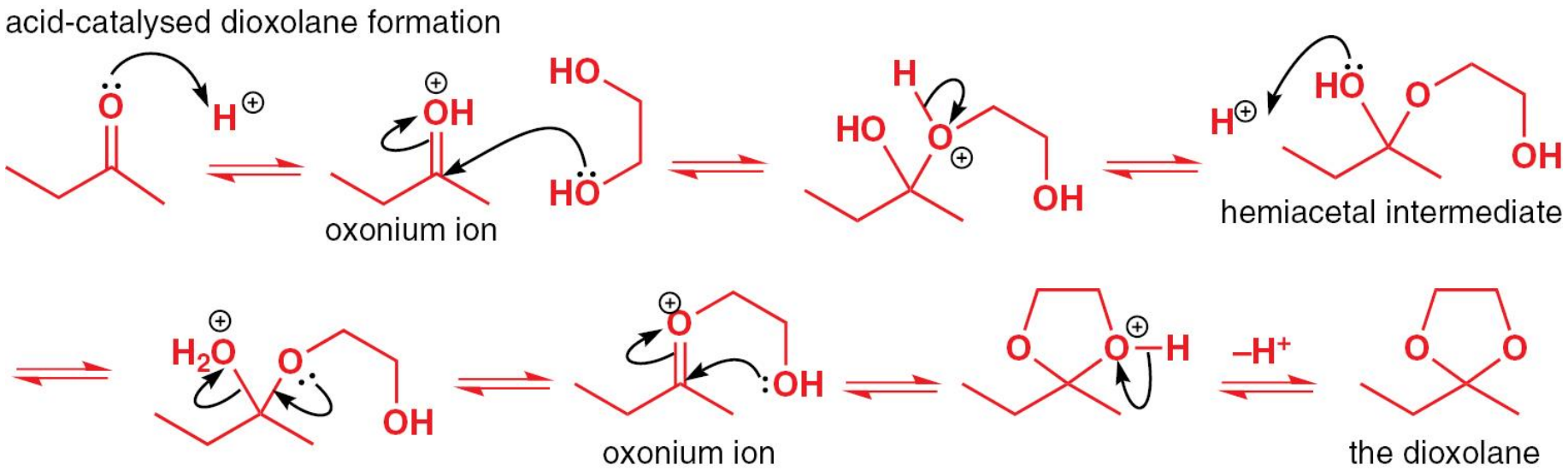


Acetal formation and decomposition-Examples

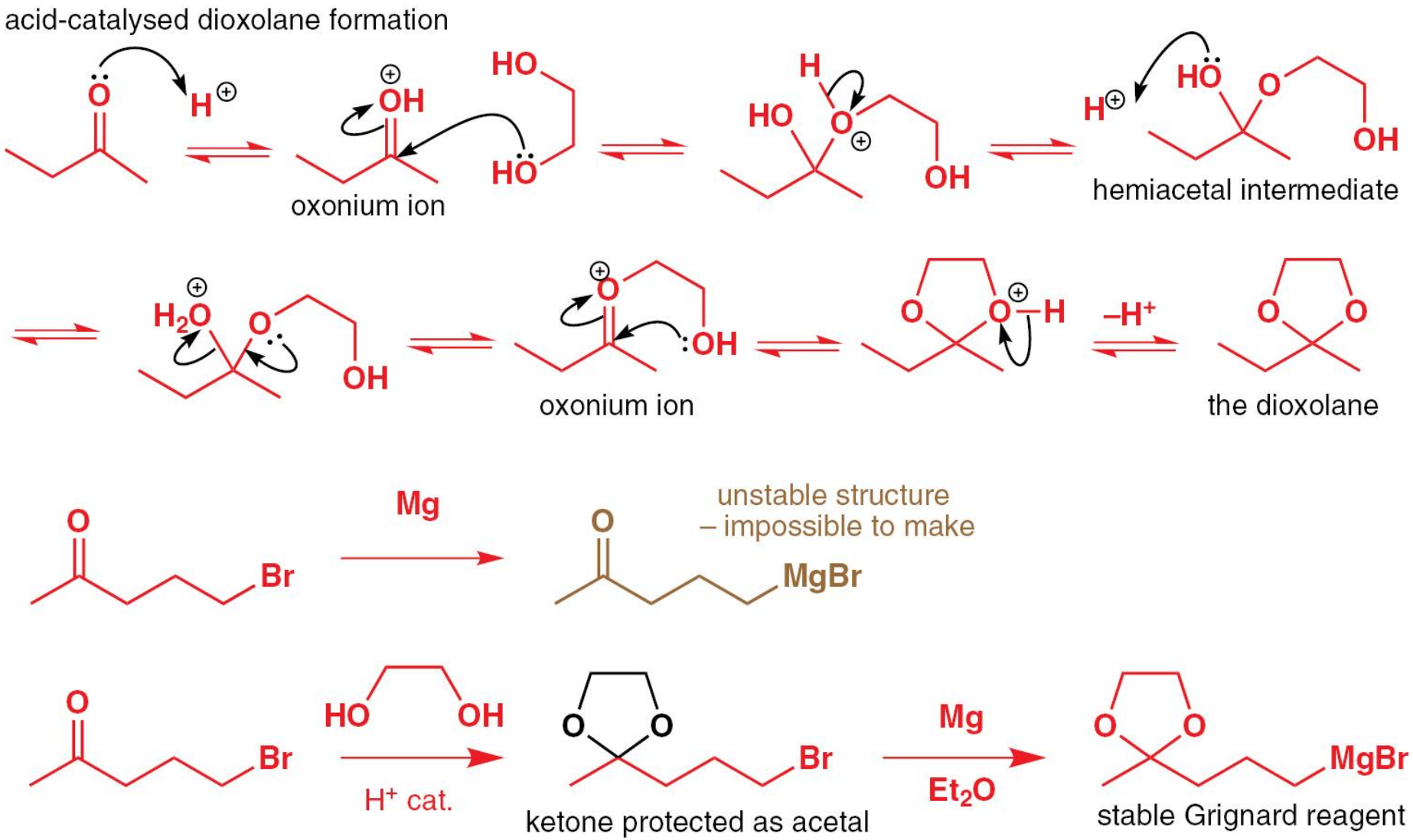


Mechanism?

Acetals as protecting groups

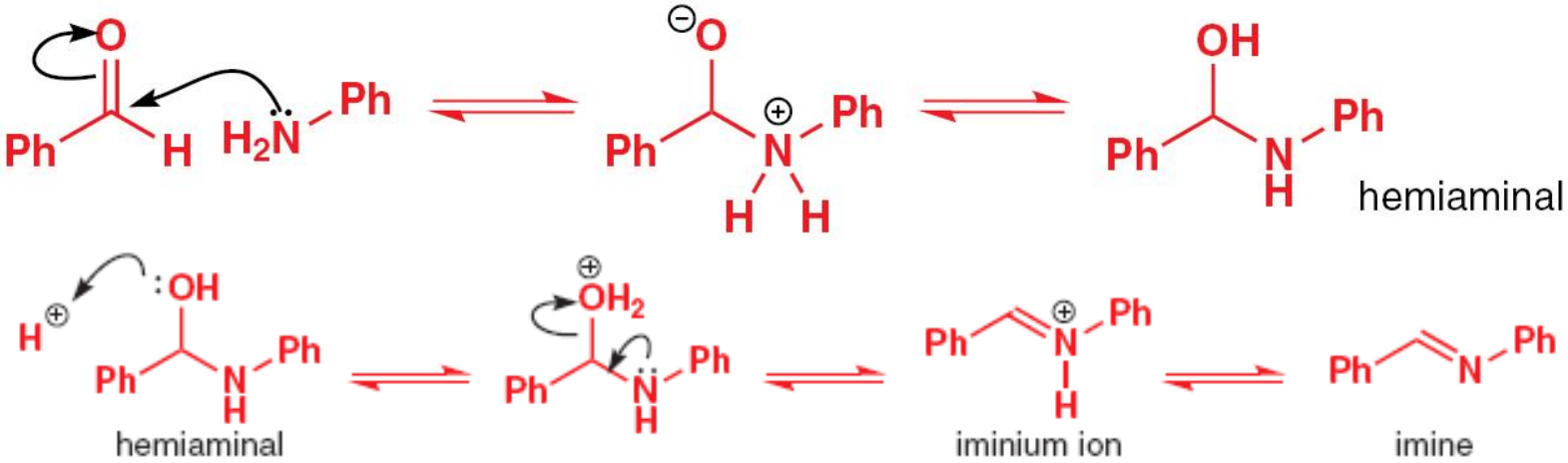


Acetals as protecting groups



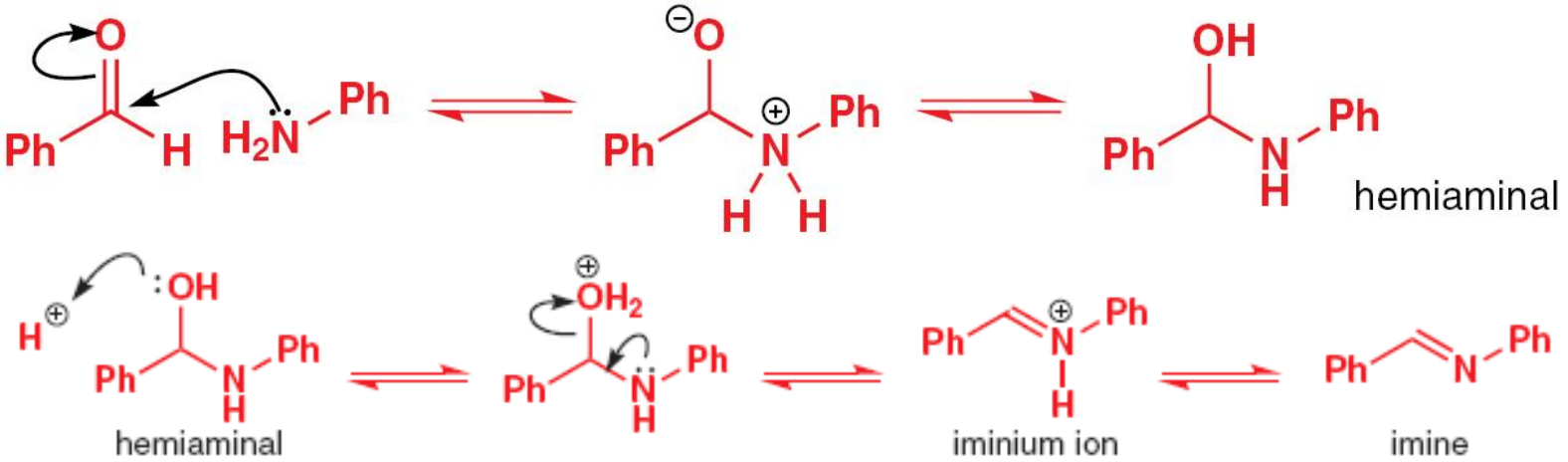
Reaction with primary amines: Imine formation

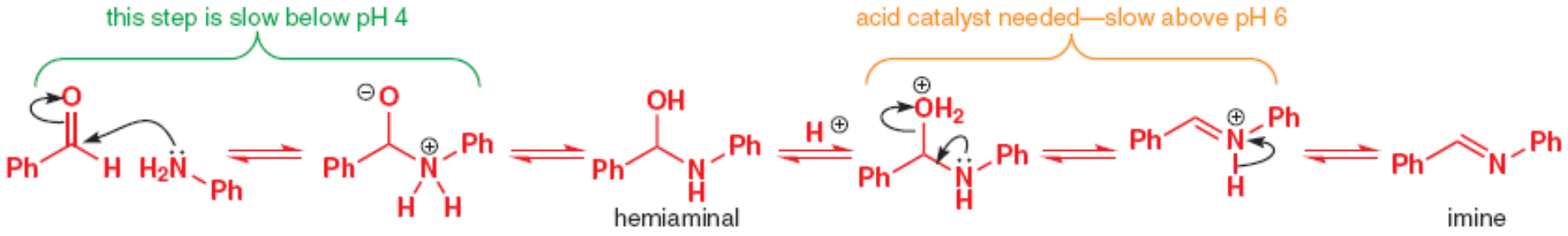
First step in imine formation:
the amine attacks the carbonyl group to form the hemiaminal intermediate:



Reaction with primary amines: Imine formation

First step in imine formation:
the amine attacks the carbonyl group to form the hemiaminal intermediate:





Imine formation-Examples



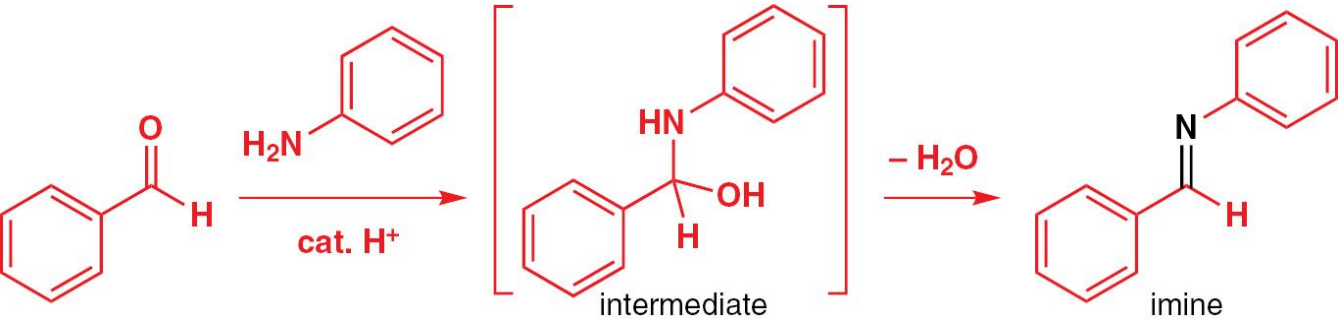
Imine formation-Examples





Imine formation-Examples

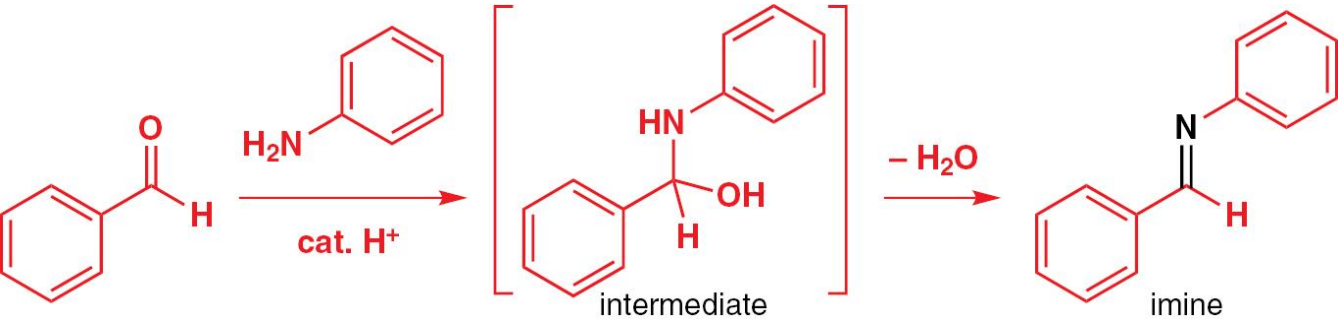






Imine formation-Examples

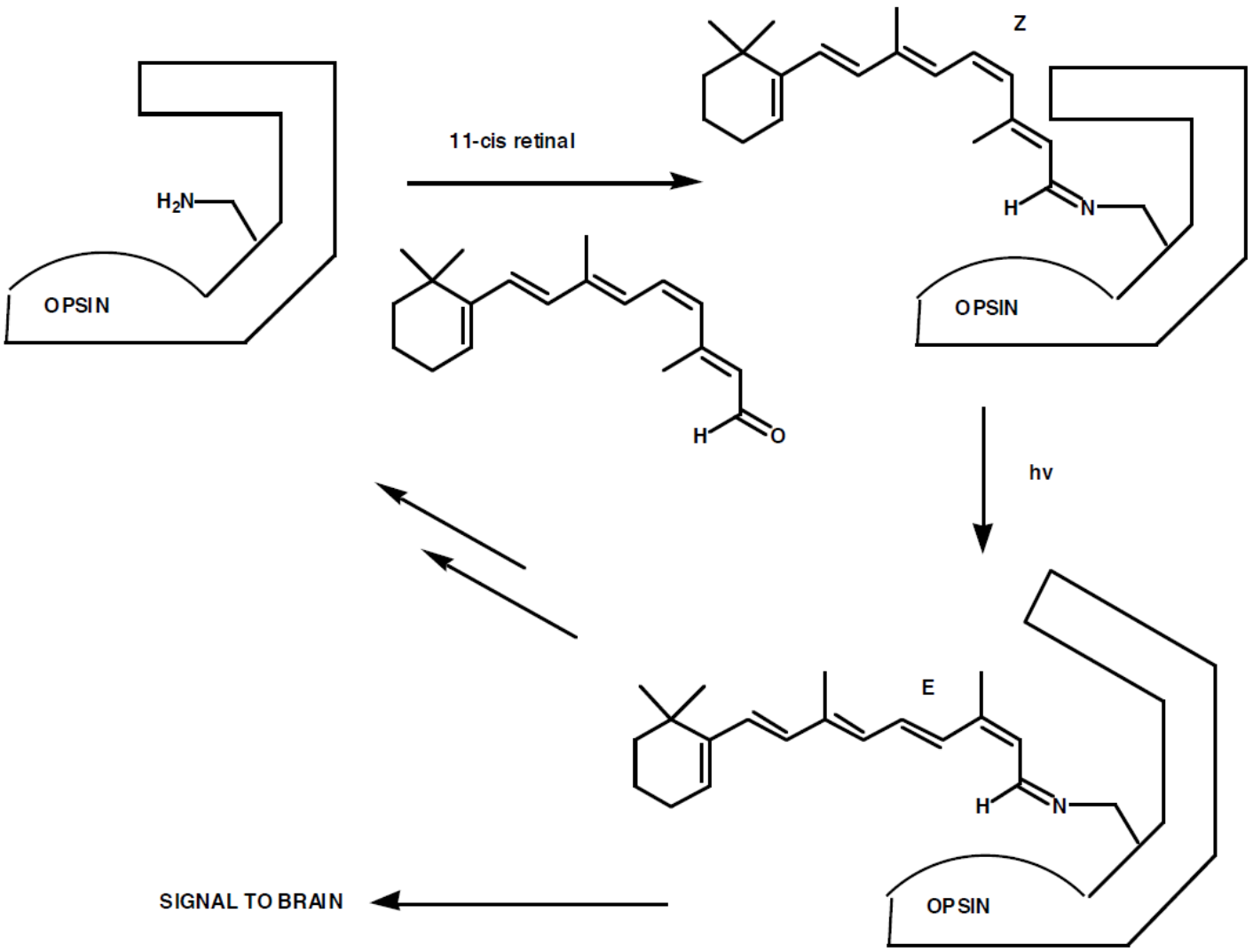




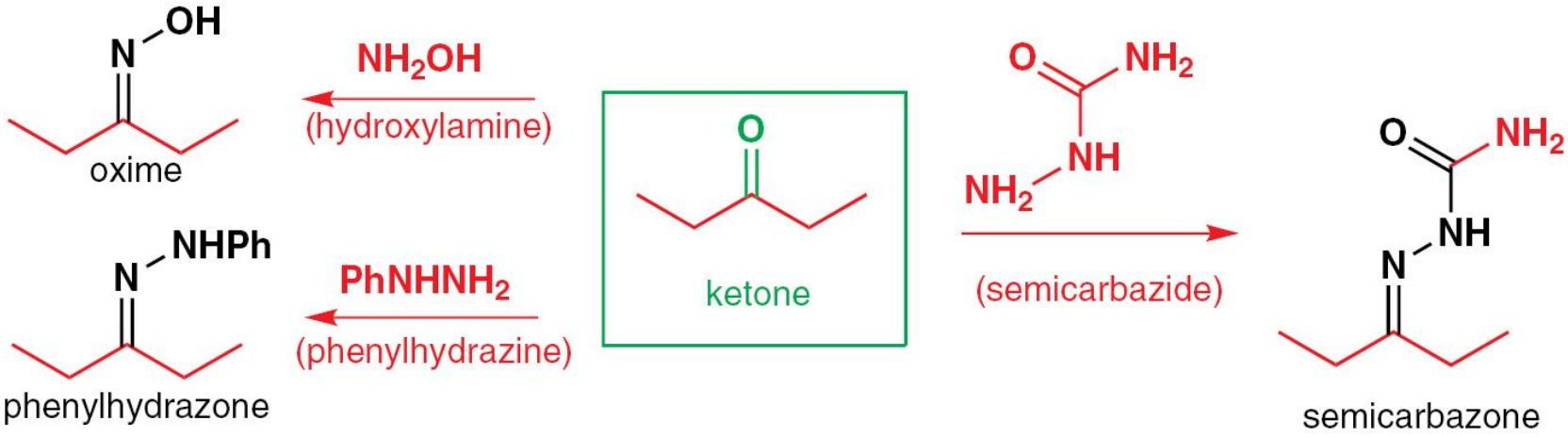


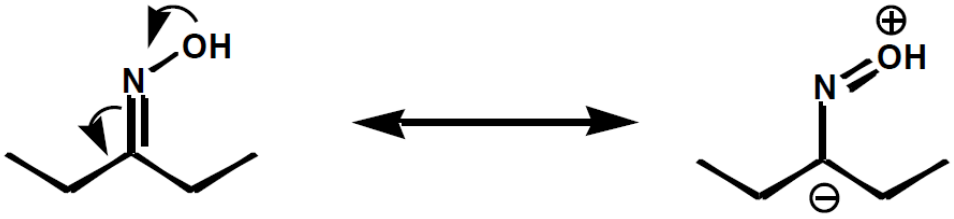


Imines and the Chemistry of vision

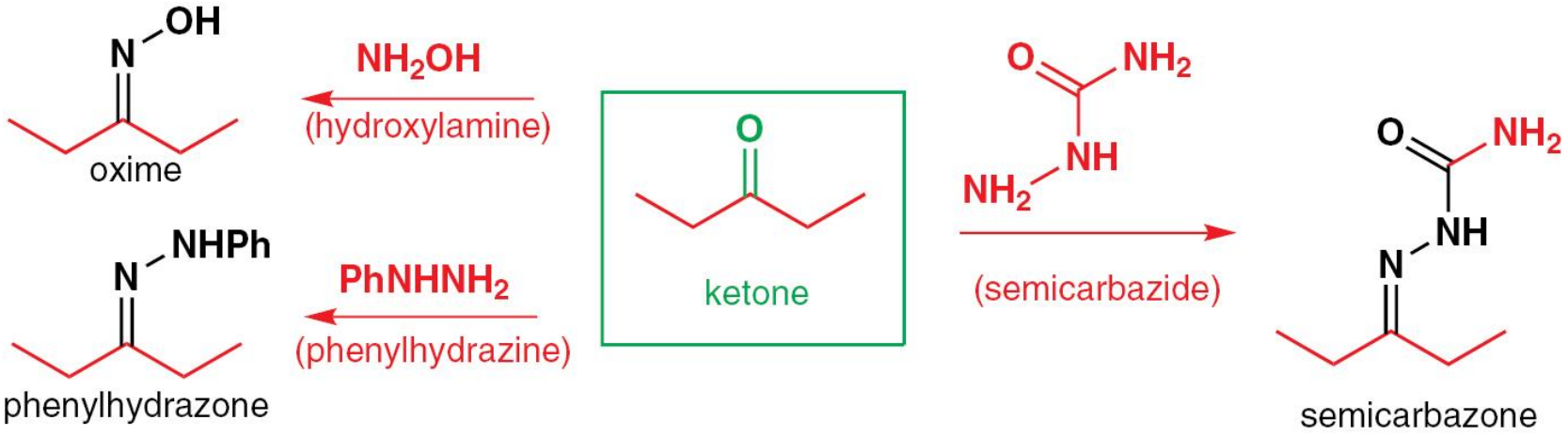


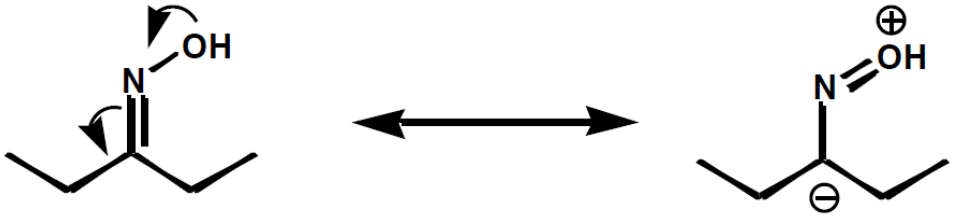
Other amino-derivatives: Oximes, Hydrazones, Semicarbazones

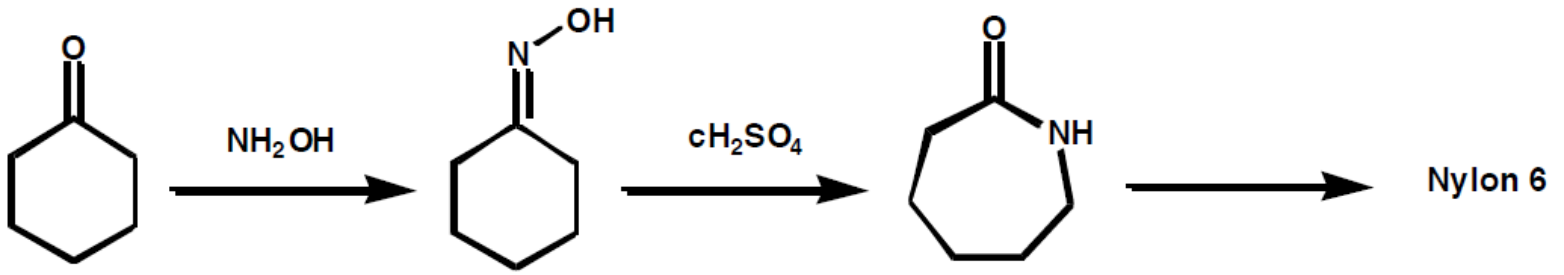




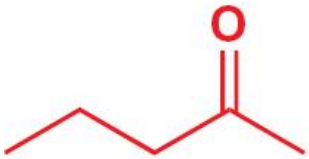
Other amino-derivatives: Oximes, Hydrazones, Semicarbazones







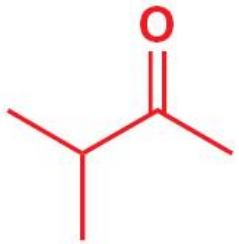
Semicarbazones and Phenylhydrazones



b.p. 102 °C

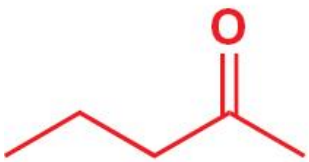


b.p. 102 °C



b.p. 106 °C

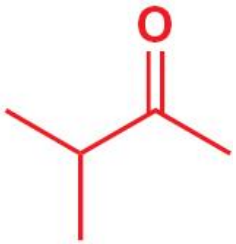
Semicarbazones and Phenylhydrazones



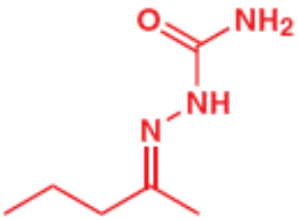
b.p. 102 °C



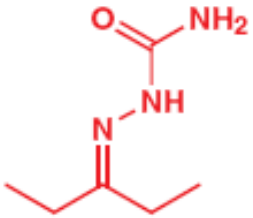
b.p. 102 °C



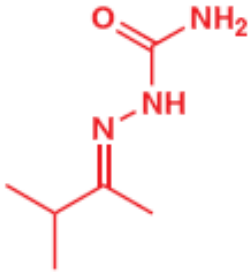
b.p. 106 °C



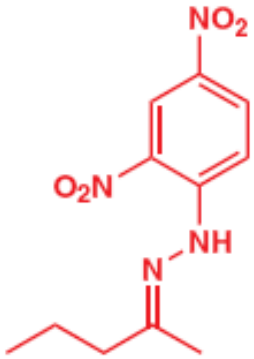
m.p. 112 °C



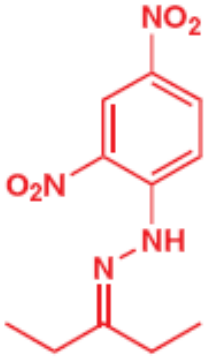
m.p. 139 °C



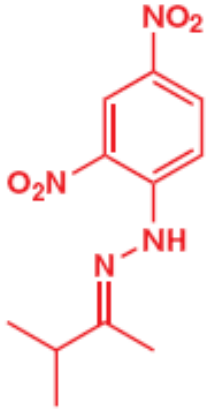
m.p. 157 °C



m.p. 143 °C

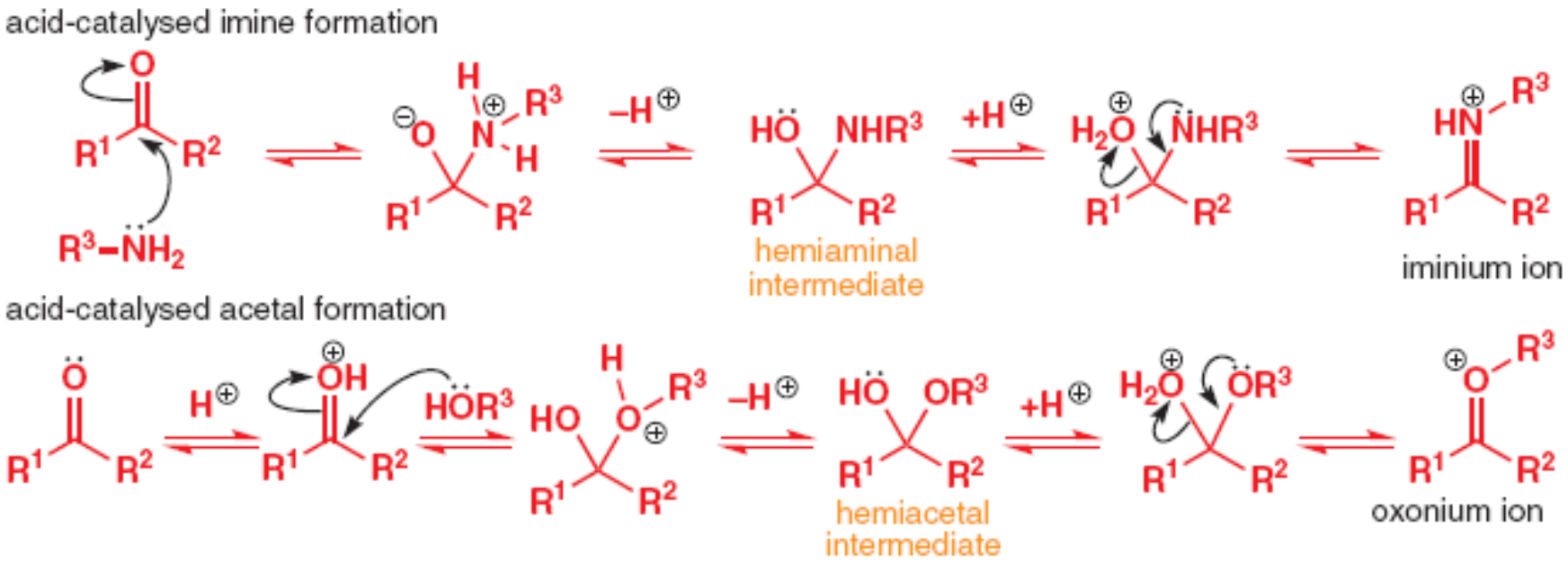


m.p. 156 °C

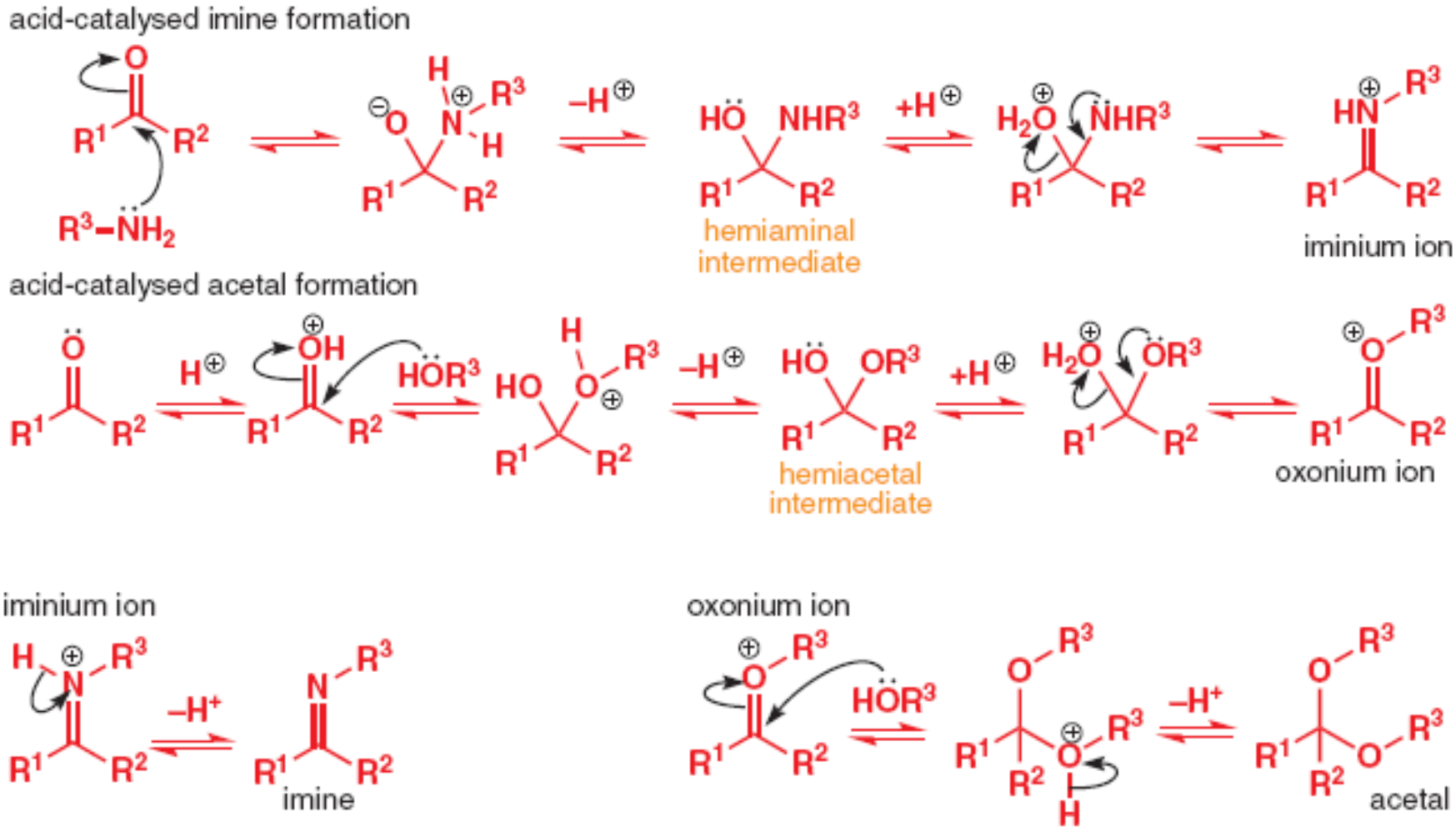


m.p. 125 °C

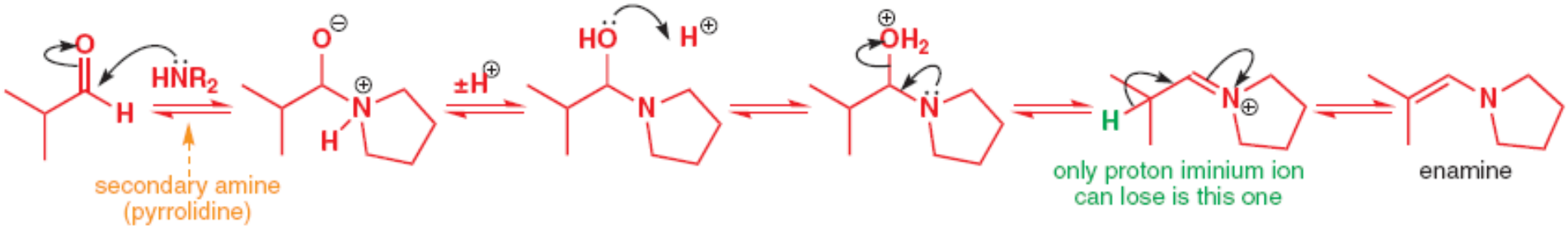
Comparison of Imine and acetal formation



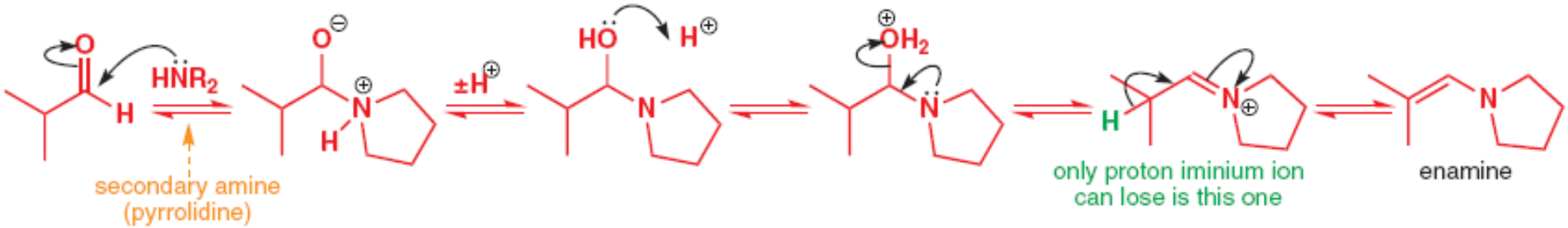
Comparison of Imine and acetal formation



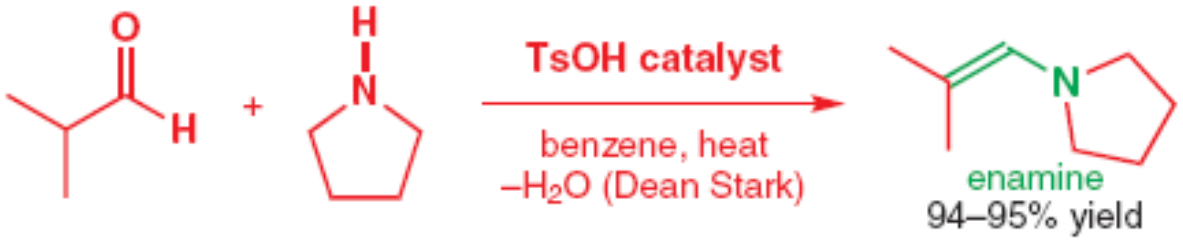
Reaction with secondary amines: Enamine formation

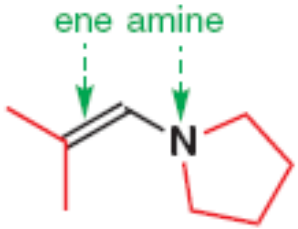


Reaction with secondary amines: Enamine formation

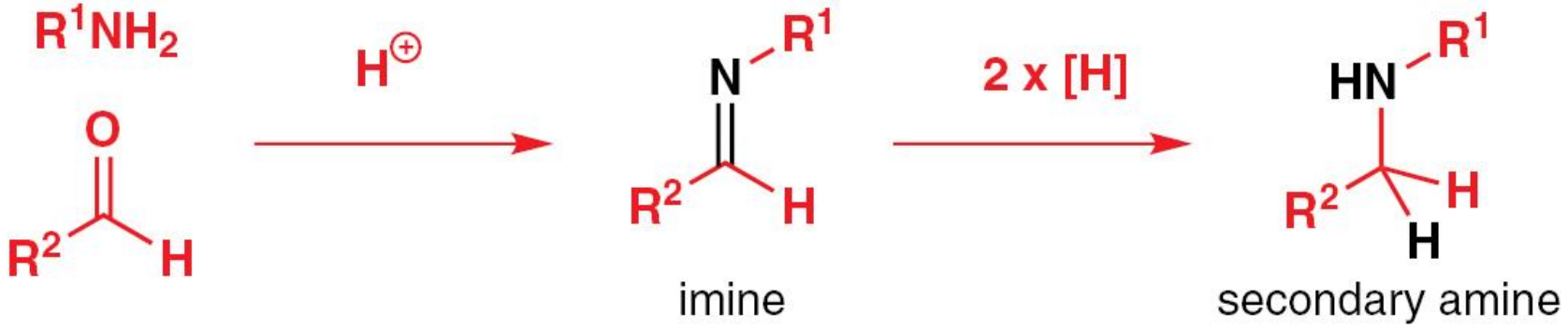


Example

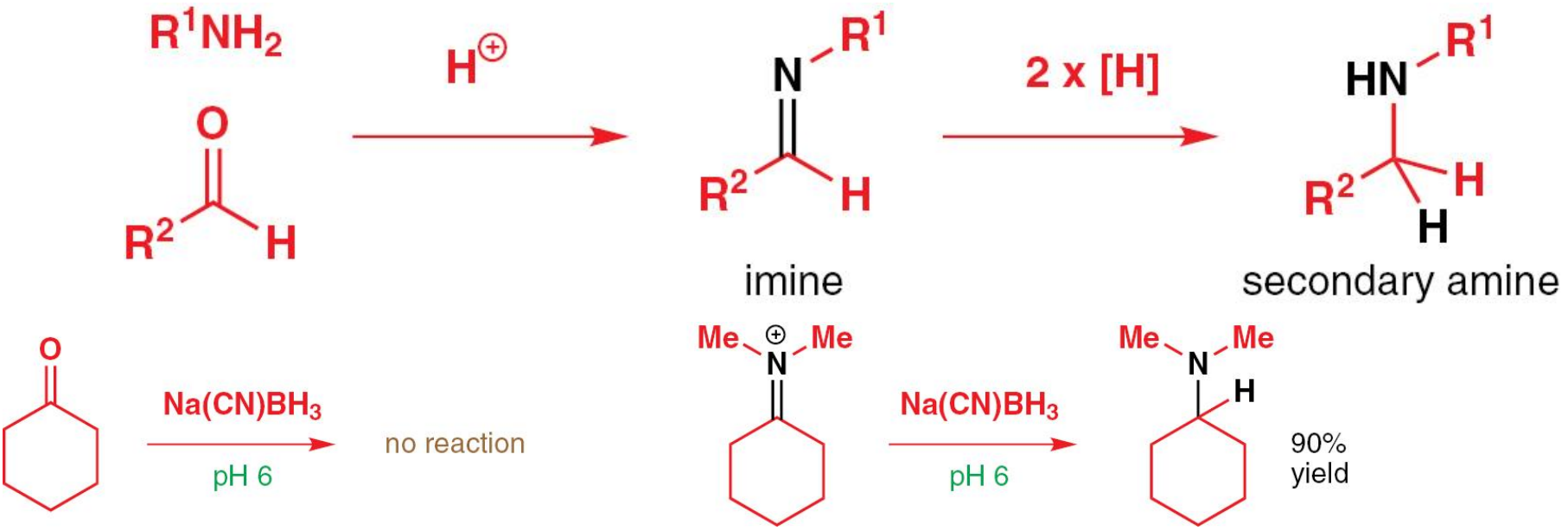




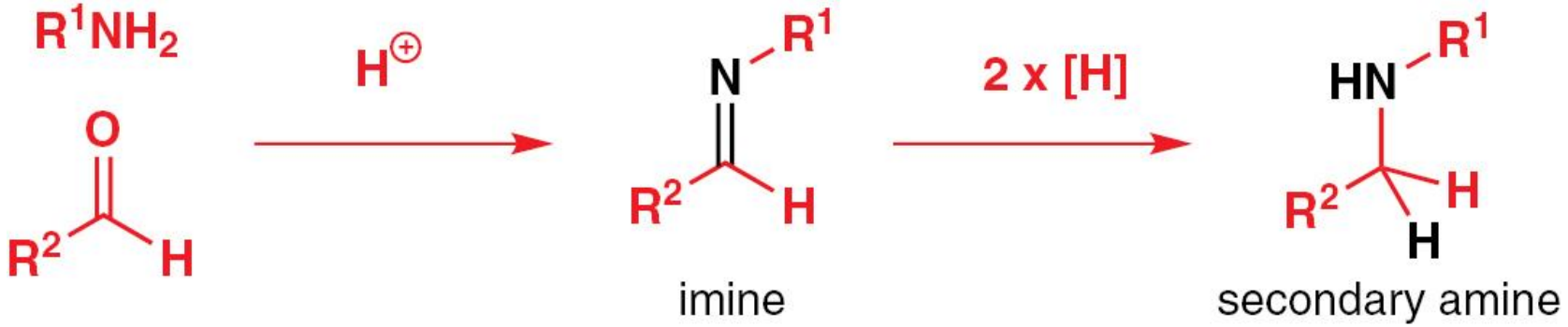
Reductive Amination

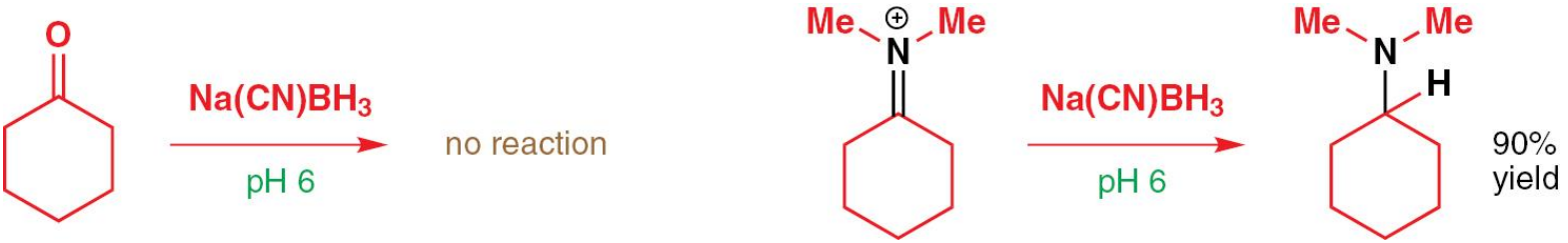


Reductive Amination

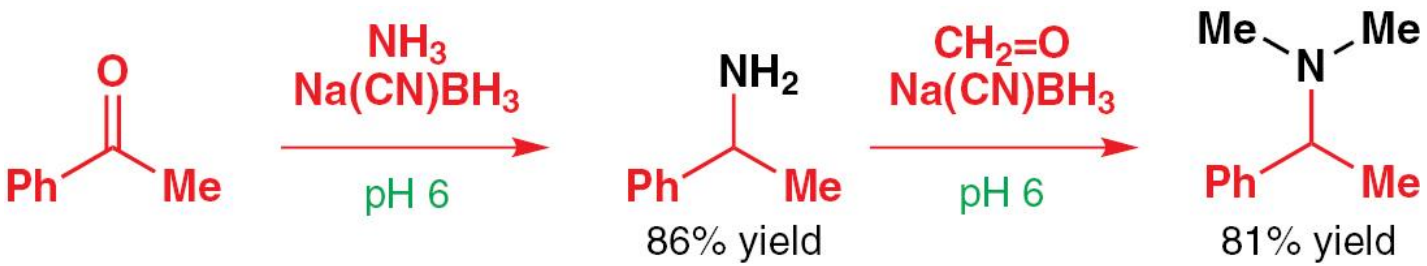


Reductive Amination

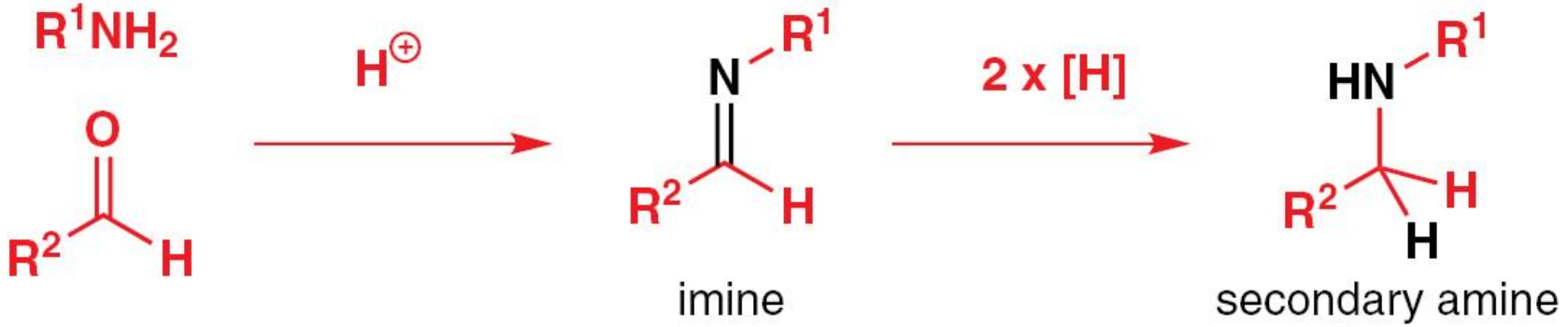


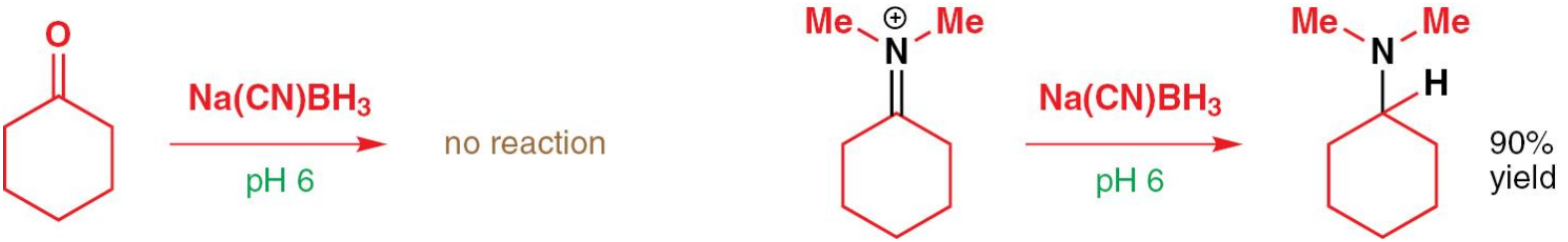


Examples

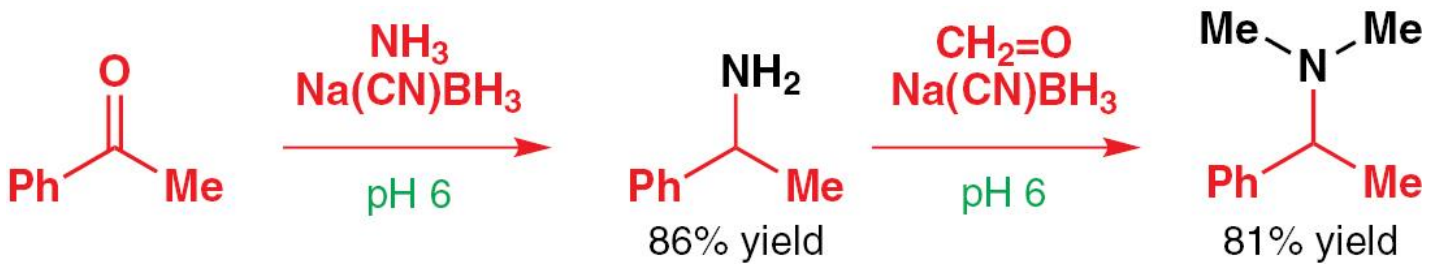


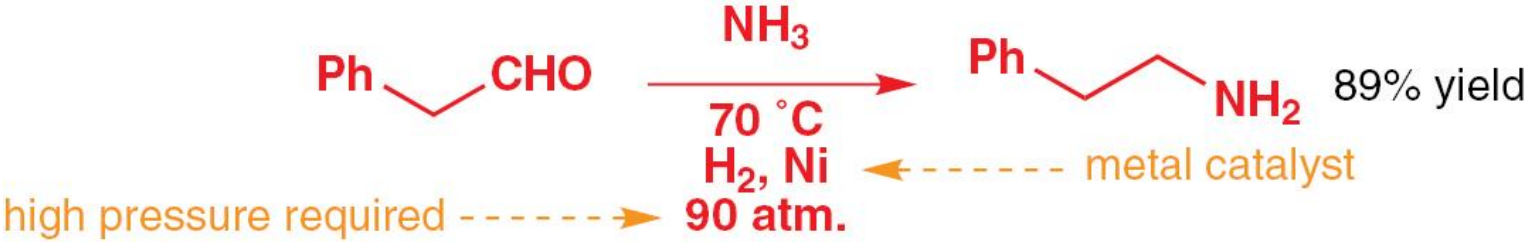
Reductive Amination





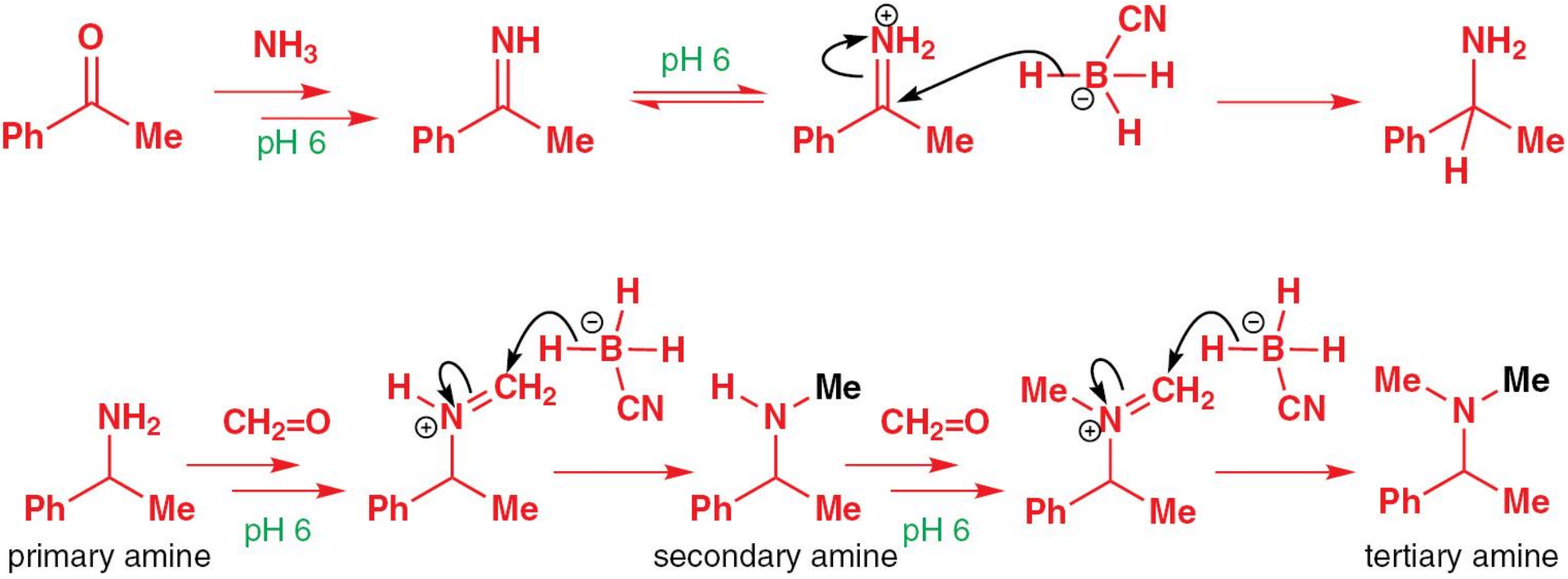
Examples





Reductive Amination

Mechanism



Summary

- Replacement of carbonyl oxygen
- Acetal formation
- Imine formation
- Stable and unstable imines
- Reductive amination